Federated Learning (FL) Lab

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1 Our simulation platform

2 Hands-on algorithms

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本节路线与听讲方法

1. 预备：FedAvg 与客户端漂移
2. 模拟平台 PFLlib
3. 经典算法：MOON · FedDyn · KT-pFL · FedMA
4. 进阶算法：Sageflow · FLChain
5. 实验室研究：效率 · 安全 · 异构

听每个算法时问三件事

1. 它针对哪类问题？
2. 它改了 FedAvg 的哪一步？
3. 它付出了什么代价？

预备

问题从哪来

FedAvg 与客户端漂移

回顾：FedAvg 的一轮只有四步

Algorithm 1 `FederatedAveraging`. The K clients are indexed by k; B is the local minibatch size, E is the number of local epochs, and η is the learning rate.

Server executes:

initialize w_0

for each round $t = 1, 2, \dots$ **do**

$m \leftarrow \max(C \cdot K, 1)$

$S_t \leftarrow$ (random set of m clients)

for each client $k \in S_t$ **in parallel do**

$w_{t+1}^k \leftarrow \text{ClientUpdate}(k, w_t)$

$w_{t+1} \leftarrow \sum_{k=1}^{K} \frac{n_k}{n} w_{t+1}^k$

ClientUpdate(k, w): *// Run on client k*

$\mathcal{B} \leftarrow$ (split $\mathcal{P}_k$ into batches of size B)

for each local epoch i from 1 to E **do**

for batch $b \in \mathcal{B}$ **do**

$w \leftarrow w - \eta \nabla \ell(w; b)$

return w to server

① 下发

服务器把全局模型 w_t 发给选中的客户端

② 本地训练

每个客户端在自己数据上训练 E 个 epoch

MOON • FedDyn

③ 上传

客户端把训练后的模型发回

KT-pFL • pFedEff

④ 聚合

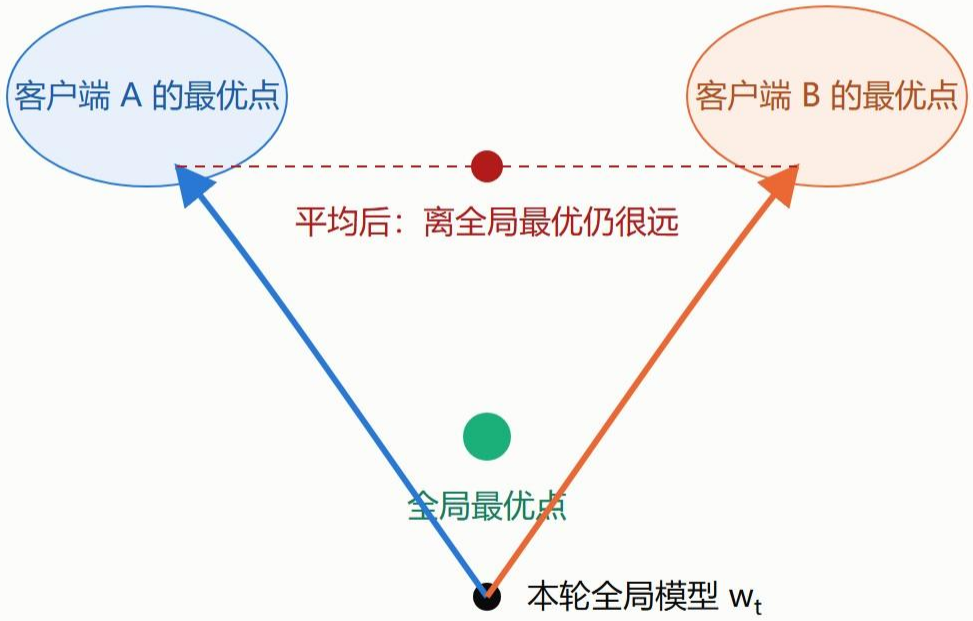
按样本数加权平均 $\Sigma\, n_k/n \cdot w^k$

FedMA • Sageflow

今天每个算法都只是在其中某一步上"动手脚"。

McMahan et al., Communication-Efficient Learning of Deep Networks from Decentralized Data, AISTATS 2017

核心困难：数据不同分布时，本地模型会"跑偏"

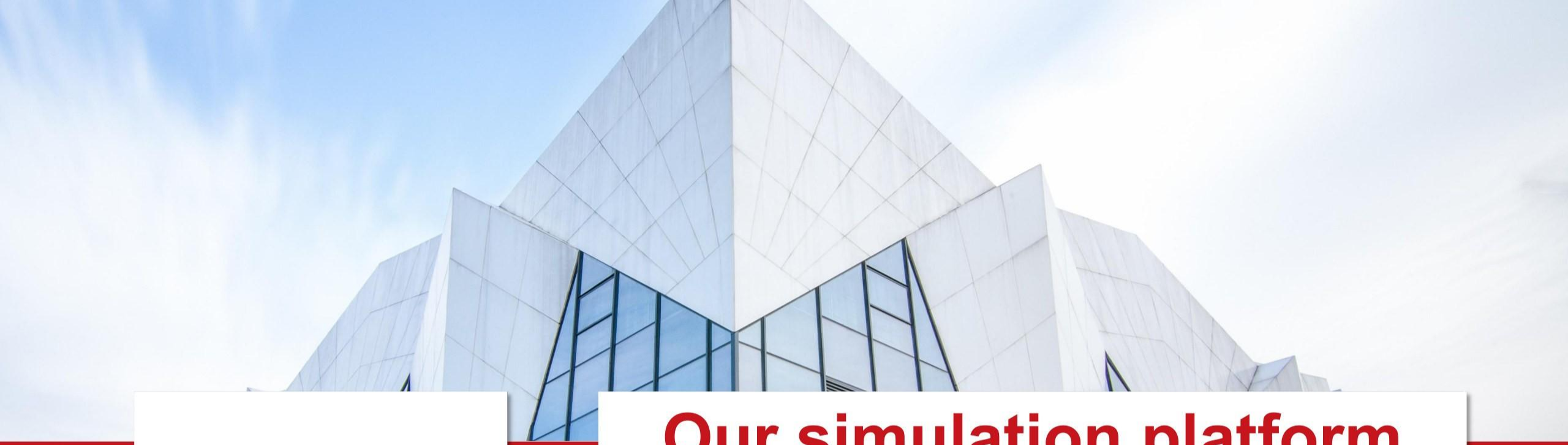


- A 只有猫狗图片，B 只有车船图片
- 各自训练多步后，模型分别走向**本地最优**
- 两个方向不一致，平均后可能两边都不好
- 数据越偏、本地步数越多，偏得越远

客户端漂移（client drift）：本地目标 ≠ 全局目标。
把自己的损失降到最低，不等于把大家的总损失降到最低。

一张地图：FedAvg 的三类问题与今天的算法

问题	表现	今天的算法（改哪一步）
统计异构 Non-IID	本地模型跑偏，平均后变差	MOON、FedDyn（② 本地训练） FedMA（④ 聚合：先对齐再平均） KT-pFL、FedALA（个性化：每家一个模型）
系统异构 设备快慢、掉线	等慢设备拖慢全局；不等又丢数据	Sageflow（④ 按过时程度分组聚合） FedVB（只传训练过的分支）
安全 恶意客户端、服务器	投毒、后门、篡改全局模型	Sageflow（④ 熵过滤 + 损失加权） Siren（客户端报警）· FLChain（区块链）
通信代价	每轮传整个模型太贵	pFedEff（③ 剪枝/掩码压缩上传）



01

Our simulation platform

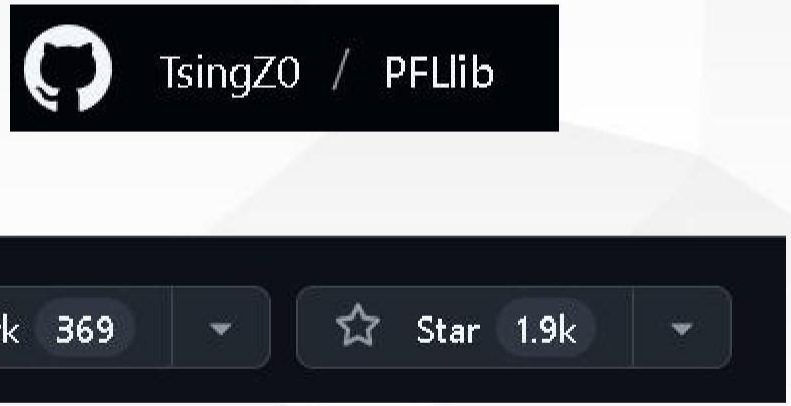
- Overview
- Dataset generation
- System introduction
- FedAvg example

SJTU

Our simulation platform

Overview

- Currently, 1.9k stars, 369 forks
- 39 traditional FL (tFL) and personalized FL (pFL) algorithms, 3 scenarios, and 24 datasets.
- Record the GPU memory usage for the model
- Differential privacy



PFLlib：实验室开源的联邦学习算法库

- github.com/TsingZ0/PFLlib · 约 2.2k stars（2026-10）
- **39 个算法**：9 个传统联邦（tFL）+ 30 个个性化联邦（pFL）
- **3 种数据场景、24 个数据集**
- 单张 RTX 3090 可模拟 500 个客户端
- 记录显存占用；支持差分隐私

Jianqing Zhang et al., PFLlib: A Beginner-Friendly and Comprehensive Personalized Federated Learning Library and Benchmark, JMLR 26(50), 2025

tFL：一个全局模型

FedAvg
FedProx · FedDyn
SCAFFOLD
MOON · FedLC
FedGen · FedNTD …

pFL：每家一个模型

Per-FedAvg · Ditto
APFL · FedALA
FedPer · FedRep
FedProto · FedKD
FedBN …

Our simulation platform

Baselines

Traditional FL (tFL)

Basic tFL

- FedAvg — Communication-Efficient Learning of Deep Networks from Decentralized Data *AISTATS 2017*

Update-correction-based tFL

- SCAFFOLD - SCAFFOLD: Stochastic Controlled Averaging for Federated Learning *ICML 2020*

Regularization-based tFL

- FedProx — Federated Optimization in Heterogeneous Networks *MLsys 2020*
- FedDyn — Federated Learning Based on Dynamic Regularization *ICLR 2021*

Model-splitting-based tFL

- MOON — Model-Contrastive Federated Learning *CVPR 2021*
- FedLC — Federated Learning With Label Distribution Skew via Logits Calibration *ICML 2022*

Knowledge-distillation-based tFL

- FedGen — Data-Free Knowledge Distillation for Heterogeneous Federated Learning *ICML 2021*
- FedNTD — Preservation of the Global Knowledge by Not-True Distillation in Federated Learning *NeurIPS 2022*

Heuristically-search-based tFL

- FedCross - FedCross: Towards Accurate Federated Learning via Multi-Model Cross-Aggregation *ICDE 2024*

Our simulation platform

Baselines

> Personalized FL (pFL)

Meta-learning-based pFL

- Per-FedAvg — Personalized Federated Learning with Theoretical Guarantees: A Model-Agnostic Meta-Learning Approach *NeurIPS 2020*

Regularization-based pFL

- pFedMe — Personalized Federated Learning with Moreau Envelopes *NeurIPS 2020*
- Ditto — Ditto: Fair and robust federated learning through personalization *ICML 2021*

Personalized-aggregation-based pFL

- APFL — Adaptive Personalized Federated Learning *2020*
- FedFomo — Personalized Federated Learning with First Order Model Optimization *ICLR 2021*
- FedAMP — Personalized Cross-Silo Federated Learning on non-IID Data *AAAI 2021*
- FedPHP — FedPHP: Federated Personalization with Inherited Private Models *ECML PKDD 2021*
- APPLE — Adapt to Adaptation: Learning Personalization for Cross-Silo Federated Learning *IJCAI 2022*
- FedALA — FedALA: Adaptive Local Aggregation for Personalized Federated Learning *AAAI 2023*

Model-splitting-based pFL

- FedPer — Federated Learning with Personalization Layers *2019*
- LG-FedAvg — Think Locally, Act Globally: Federated Learning with Local and Global Representations *2020*
- FedRep — Exploiting Shared Representations for Personalized Federated Learning *ICML 2021*
- FedRoD — On Bridging Generic and Personalized Federated Learning for Image Classification *ICLR 2022*

Our simulation platform

Datasets

Datasets and scenarios (updating)

We support 3 types of scenarios with various datasets and move the common dataset splitting code into `./dataset/utils` for easy extension. If you need another data set, just write another code to download it and then use the utils.

label skew scenario

For the ***label skew*** scenario, we introduce **16** famous datasets:

- MNIST
- EMNIST
- FEMNIST
- Fashion-MNIST
- Cifar10
- Cifar100
- AG News
- Sogou News
- Tiny-ImageNet
- Country211
- Flowers102
- GTSRB
- Shakespeare
- Stanford Cars
- COVIDx
- kvasir

Our simulation platform

Datasets

The datasets can be easily split into **IID** and **non-IID** versions. In the **non-IID** scenario, we distinguish between two types of distribution:

1. **Pathological non-IID**: In this case, each client only holds a subset of the labels, for example, just 2 out of 10 labels from the MNIST dataset, even though the overall dataset contains all 10 labels. This leads to a highly skewed distribution of data across clients.
2. **Practical non-IID**: Here, we model the data distribution using a Dirichlet distribution, which results in a more realistic and less extreme imbalance. For more details on this, refer to this paper.

Additionally, we offer a `balance` option, where data amount is evenly distributed across all clients.

feature shift scenario

For the ***feature shift*** scenario, we utilize **3** widely used datasets in Domain Adaptation:

- **Amazon Review** (raw data can be fetched from this link)
- **Digit5** (raw data available here)
- **DomainNet**

real-world scenario

For the ***real-world*** scenario, we introduce **5** naturally separated datasets:

- **Camelyon17** (5 hospitals, 2 labels)
- **iWildCam** (194 camera traps, 158 labels)
- **Omniglot** (20 clients, 50 labels)
- **HAR (Human Activity Recognition)** (30 clients, 6 labels)
- **PAMAP2** (9 clients, 12 labels)

补充：Non-IID 划分到底"偏"成什么样

划分	做法	效果
`iid`	样本随机均分	每家分布相同
`pat` 病态	每家只发固定几类（如 2 类）	极端偏斜
`dir` 实际	每个类别按 Dirichlet(α) 抽一组比例分给各家	α 越小越偏，更接近真实
`balance`	各家样本数相同	只控制数量，不控制类别

> 大作业中的 dir1.0（轻度不均）、dir0.1（严重不均）就是 Dirichlet 划分。

示意：3 个类别分给 3 个客户端的比例

	客户端 A	客户端 B	客户端 C
α 很大（接近 iid）			
类别 0	0.34	0.33	0.33
类别 1	0.32	0.35	0.33
α = 0.1（严重不均）			
类别 0	0.95	0.04	0.01
类别 1	0.02	0.90	0.08
类别 2	0.10	0.05	0.85

数值为示意。α = 0.1 时，每个类别几乎只集中在一家。

Our simulation platform

Models

Models

- for MNIST and Fashion-MNIST
  i. Mclr_Logistic(1*28*28) # convex
  ii. LeNet()
  iii. DNN(1*28*28, 100)
- for Cifar10, Cifar100 and Tiny-ImageNet
  i. Mclr_Logistic(3*32*32) # convex
  ii. FedAvgCNN()
  iii. DNN(3*32*32, 100)
  iv. ResNet18, AlexNet, MobileNet, GoogleNet, etc.
- for AG_News and Sogou_News
  - LSTM()
  - fastText() in Bag of Tricks for Efficient Text Classification
  - TextCNN() in Convolutional Neural Networks for Sentence Classification
  - TransformerModel() in Attention is all you need
- for AmazonReview
  - AmazonMLP() in Curriculum manager for source selection in multi-source domain adaptation
- for Omniglot
  - FedAvgCNN()
- for HAR and PAMAP
  - HARCNN() in Convolutional neural networks for human activity recognition using mobile sensors

Our simulation platform

Installation

Environments

Install CUDA.

Install conda latest and activate conda.

For additional configurations, refer to the `prepare.sh` script.

```
conda env create -f env_cuda_latest.yaml  # Downgrade torch via pip if needed to match the CUDA version
```

Our simulation platform

Dataset

Examples for MNIST in the *label skew* scenario

```
cd ./dataset
# Please modify train_ratio and alpha in dataset\utils\dataset_utils.py

python generate_MNIST.py iid - - # for iid and unbalanced scenario
python generate_MNIST.py iid balance - # for iid and balanced scenario
python generate_MNIST.py noniid - pat # for pathological noniid and unbalanced scenario
python generate_MNIST.py noniid - dir # for practical noniid and unbalanced scenario
python generate_MNIST.py noniid - exdir # for Extended Dirichlet strategy
```

The command line output of running `python generate_MNIST.py noniid - dir`

```
Number of classes: 10
Client 0        Size of data: 2630      Labels:  [0 1 4 5 7 8 9]
                Samples of labels:  [(0, 140), (1, 890), (4, 1), (5, 319), (7, 29), (8, 1067), (9, 184)]
--------------------------------------------------
Client 1        Size of data: 499       Labels:  [0 2 5 6 8 9]
                Samples of labels:  [(0, 5), (2, 27), (5, 19), (6, 335), (8, 6), (9, 107)]
--------------------------------------------------
Client 2        Size of data: 1630      Labels:  [0 3 6 9]
                Samples of labels:  [(0, 3), (3, 143), (6, 1461), (9, 23)]
--------------------------------------------------
```

Our simulation platform

Execution

How to start simulating (examples for FedAvg)

- Download this project to an appropriate location using git.

```
git clone https://github.com/TsingZ0/PFLlib.git
```

- Create proper environments (see Environments).
- Build evaluation scenarios (see Datasets and scenarios (updating)).
- Run evaluation:

```
cd ./system
python main.py -data MNIST -m CNN -algo FedAvg -gr 2000 -did 0 # using the MNIST dataset, the FedAvg
python main.py -data MNIST -m CNN -algo FedAvg -gr 2000 -did 0,1,2,3 # running on multiple GPUs
```

Note: It is preferable to tune algorithm-specific hyper-parameters before using any algorithm on a new machine.

两条命令跑起来

```
git clone https://github.com/TsingZ0/PFLlib.git
conda env create -f env_cuda_latest.yaml

# 1. 生成数据：Dirichlet 非独立同分布的 MNIST
cd dataset
python generate_MNIST.py noniid - dir     # 可选 iid / pat / dir / exdir；α 在 utils/dataset_utils.py

# 2. 运行：换算法只改 -algo
cd ../system
python main.py -data MNIST -m CNN -algo FedAvg -gr 2000 -did 0
python main.py -data MNIST -m CNN -algo MOON   -gr 2000 -did 0
python main.py -data MNIST -m CNN -algo FedDyn -gr 2000 -did 0
#  -data 数据集 · -m 模型 · -algo 算法 · -gr 全局轮数 · -did GPU 编号
```

同一份数据、同一个模型，只换算法名，就能公平比较不同算法。

Our simulation platform

- Top-down
  - Server: Aggregate, Send, Receive, Evaluate
    - Add
  - Client: Train, Receive, Send
    - Dataset
    - Model
    - Optimizer

Algorithm 1 `FederatedAveraging`. The K clients are indexed by k; B is the local minibatch size, E is the number of local epochs, and η is the learning rate.

Server executes:

initialize w_0

for each round $t = 1, 2, \dots$ **do**

$m \leftarrow \max(C \cdot K, 1)$

$S_t \leftarrow$ (random set of m clients)

for each client $k \in S_t$ **in parallel do**

$w_{t+1}^k \leftarrow \text{ClientUpdate}(k, w_t)$

$w_{t+1} \leftarrow \sum_{k=1}^{K} \frac{n_k}{n} w_{t+1}^k$

ClientUpdate(k, w): *// Run on client k*

$\mathcal{B} \leftarrow$ (split $\mathcal{P}_k$ into batches of size B)

for each local epoch i from 1 to E **do**

for batch $b \in \mathcal{B}$ **do**

$w \leftarrow w - \eta \nabla \ell(w; b)$

return w to server

代码结构：新算法只改"不同的那一步"

```
system/
  main.py                入口：解析参数，选服务器类
  flcore/
    servers/
      serverbase.py      选客户端、下发、接收、
                         聚合、评估
      serveravg.py       FedAvg
      serverdyn.py       FedDyn …
    clients/
      clientbase.py      读数据、接收模型、
                         训练、上传
      clientavg.py       FedAvg
      clientmoon.py      MOON …
    trainmodel/
      models.py          CNN、ResNet、文本模型…
```

算法	需要重写什么
MOON	只改客户端 `train()`：多加一个损失项
FedDyn	客户端和服务器都改：两边都有状态
FedALA	只改客户端"接收全局模型"那一步

> 服务器 = 选择、下发、接收、聚合、评估；客户端 = 接收、训练、上传。和 FedAvg 的四步一一对应。

02

Hands-on algorithms

- MOON
- FedDyn
- KT-pFL
- FedMA

MOON

MOON algorithm

- Observation
  - The global model trained on a whole dataset can learn a better representation than the local model trained on a skewed subset.
  - Propose model-contrastive learning (MOON), which corrects the local updates by maximizing the agreement of representation learned by the current local model and the representation learned by the global model.

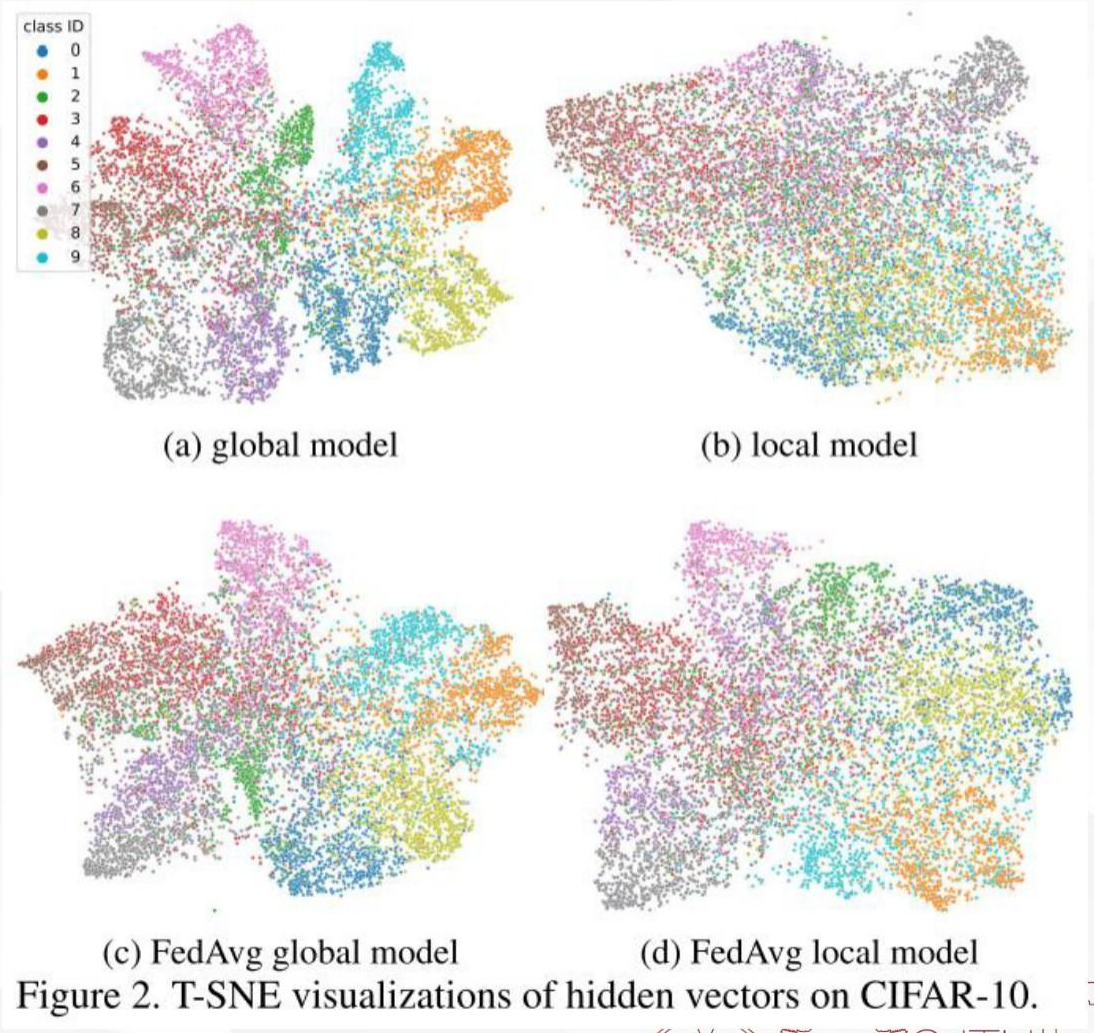


Figure 2. T-SNE visualizations of hidden vectors on CIFAR-10.

MOON algorithm

- Contrastive learning
  - The key idea of contrastive learning is to reduce the distance between the representations of different augmented views of the same image (i.e., positive pairs), and increase the distance between the representations of augmented views of different images (i.e., negative pairs)

$$l_{i,j} = -\log \frac{\exp(\text{sim}(x_i, x_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{I}_{[k \neq i]} \exp(\text{sim}(x_i, x_k)/\tau)}$$

MOON ② 预备知识：对比学习

- 同一张图做两种增强（裁剪、调色），表示要**拉近**：正样本对
- 不同图片的表示要**推远**：负样本对

$$l_{i,j} = -\log \frac{\exp(\text{sim}(x_i, x_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{I}_{[k \neq i]} \exp(\text{sim}(x_i, x_k)/\tau)}$$

符号	含义
sim	余弦相似度，越大越像
分子	正样本对的相似度
分母	与所有其他样本的相似度之和
τ	温度，越小对"像不像"越敏感

> 本质是 softmax 交叉熵："正样本"是正确答案。正样本相对越像，损失越小。

SimCLR (Chen et al., ICML 2020)

MOON algorithm

- MOON
  - Global representation $z_{glob} = R_{w^t}(x)$
  - Local representation w_i^{t-1} (i.e., $z_{prev} = R_{w_i^{t-1}}(x)$)
  - Current representation $z = R_{w_i^t}(x)$
  -

$$\ell_{con} = -\log \frac{\exp(\text{sim}(z, z_{glob})/\tau)}{\exp(\text{sim}(z, z_{glob})/\tau) + \exp(\text{sim}(z, z_{prev})/\tau)}$$

$$\ell = \ell_{sup}(w_i^t; (x, y)) + \mu \ell_{con}(w_i^t; w_i^{t-1}; w^t; x),$$

$$\min_{w_i^t} \mathbb{E}_{(x,y)\sim D^i}[\ell_{sup}(w_i^t; (x, y)) + \mu \ell_{con}(w_i^t; w_i^{t-1}; w^t; x)].$$

MOON ③ 把"两个视图"换成"两个模型"

表示	来自	角色
z	正在训练的本地模型	锚点
z_{glob}	本轮收到的全局模型	**正样本：靠近**
z_{prev}	自己上一轮的本地模型	**负样本：远离**

> 上一轮本地模型代表"按偏斜数据跑偏的方向"，推开它，就是不再往偏处走。

$$\ell_{con} = -\log \frac{\exp(\text{sim}(z, z_{glob})/\tau)}{\exp(\text{sim}(z, z_{glob})/\tau) + \exp(\text{sim}(z, z_{prev})/\tau)}$$

$$\ell = \ell_{sup}(w_i^t; (x, y)) + \mu \ell_{con}(w_i^t; w_i^{t-1}; w^t; x),$$

$$\min_{w_i^t} \mathbb{E}_{(x,y)\sim D^i}[\ell_{sup}(w_i^t; (x, y)) + \mu \ell_{con}(w_i^t; w_i^{t-1}; w^t; x)].$$

- ℓ_{sup}：普通交叉熵分类损失
- μ：对比项权重；$\mu = 0$ 就退化为 FedAvg

MOON algorithm

MOON

Algorithm 1: The MOON framework

Input: number of communication rounds T, number of parties N, number of local epochs E, temperature τ, learning rate η, hyper-parameter μ

Output: The final model w^T

Server executes:
initialize w^0
for $t = 0, 1, ..., T-1$ **do**
  for $i = 1, 2, ..., N$ **in parallel do**
    send the global model w^t to P_i
    $w_i^t \leftarrow$ **PartyLocalTraining**(i, w^t)
  $w^{t+1} \leftarrow \sum_{k=1}^{N} \frac{|\mathcal{D}^i|}{|\mathcal{D}|} w_k^t$
return w^T

PartyLocalTraining(i, w^t):
$w_i^t \leftarrow w^t$
for epoch $i = 1, 2, ..., E$ **do**
  for each batch $\mathbf{b} = \{x, y\}$ of $\mathcal{D}^i$ **do**
    $\ell_{sup} \leftarrow CrossEntropyLoss(F_{w_i^t}(x), y)$
    $z \leftarrow R_{w_i^t}(x)$
    $z_{glob} \leftarrow R_{w^t}(x)$
    $z_{prev} \leftarrow R_{w_i^{t-1}}(x)$
    $\ell_{con} \leftarrow -\log \frac{\exp(\text{sim}(z, z_{glob})/\tau)}{\exp(\text{sim}(z, z_{glob})/\tau) + \exp(\text{sim}(z, z_{prev})/\tau)}$
    $\ell \leftarrow \ell_{sup} + \mu \ell_{con}$
    $w_i^t \leftarrow w_i^t - \eta \nabla \ell$
return w_i^t to server

MOON 小结：和 FedProx 对比

	FedProx	MOON
约束什么	**参数**别离全局模型太远	**表示**要像全局模型、不像上一轮的自己
本地损失	交叉熵 + $(\mu/2)\|\|w - w^t\|\|^2$	交叉熵 + $\mu \cdot \ell_{con}$
服务器	与 FedAvg 相同	与 FedAvg 相同
额外开销	几乎没有	每个 batch 多 2 次前向；多存 1 个模型
通信量	不变	不变

$\mu = 0$ 时两者都退化为 FedAvg。

μ 太大：额外项压过分类损失，本地数据学不进去，需按数据集调。

FedDyn

FedDyn algorithm

- Intuition
  - Training models on local data that minimize local empirical loss appears to be meaningful, but yet, doing so is fundamentally inconsistent with minimizing the global empirical loss.
  - Dynamically modify the device objective with a penalty term so that, in the limit, when model parameters converge, they do so to stationary points of the global empirical loss.

FedDyn ① 动机：本地最优 ≠ 全局最优

统计异构 • 改 ② 本地训练 + ④ 聚合

- 每个客户端最小化自己的损失 L_k
- 但全局目标是 $\Sigma_k L_k$，两者的最优点**本质上不一致**
- FedAvg、FedProx 即使收敛，收敛点一般也**不是**全局损失的驻点

FedDyn：给本地目标加一个**每轮动态变化**的修正项，使得一旦收敛，收敛点就是全局损失的驻点（梯度为零）。

Durmus Alp Emre Acar et al., Federated Learning Based on Dynamic Regularization, ICLR 2021

一个比喻

每家都有"一贯的偏向"。

FedProx：用绳子把每家拴在全局模型附近 → 少偏一点，但仍有偏。

FedDyn：每家记住自己**上次往哪偏**，这次主动往反方向修正 → 偏向互相抵消。

FedDyn algorithm

- FedDyn

Algorithm 1: Federated Dynamic Regularizer - (FedDyn)

Input: $T, \boldsymbol{\theta}^0, \alpha > 0, \nabla L_k(\boldsymbol{\theta}_k^0) = \mathbf{0}$.

for $t = 1, 2, \ldots T$ **do**

  Sample devices $\mathcal{P}_t \subseteq [m]$ and transmit $\boldsymbol{\theta}^{t-1}$ to each selected device,

  for *each device* $k \in \mathcal{P}_t$, *and in parallel* **do**

    Set $\boldsymbol{\theta}_k^t = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} L_k(\boldsymbol{\theta}) - \langle \nabla L_k(\boldsymbol{\theta}_k^{t-1}), \boldsymbol{\theta} \rangle + \frac{\alpha}{2} \|\boldsymbol{\theta} - \boldsymbol{\theta}^{t-1}\|^2$,

    Set $\nabla L_k(\boldsymbol{\theta}_k^t) = \nabla L_k(\boldsymbol{\theta}_k^{t-1}) - \alpha \left(\boldsymbol{\theta}_k^t - \boldsymbol{\theta}^{t-1}\right)$,

    Transmit device model $\boldsymbol{\theta}_k^t$ to server,

  end for

  for *each device* $k \notin \mathcal{P}_t$, *and in parallel* **do**

    Set $\boldsymbol{\theta}_k^t = \boldsymbol{\theta}_k^{t-1}$, $\nabla L_k(\boldsymbol{\theta}_k^t) = \nabla L_k(\boldsymbol{\theta}_k^{t-1})$,

  end for

  Set $\boldsymbol{h}^t = \boldsymbol{h}^{t-1} - \alpha \frac{1}{m} \left(\sum_{k \in \mathcal{P}_t} \boldsymbol{\theta}_k^t - \boldsymbol{\theta}^{t-1}\right)$,

  Set $\boldsymbol{\theta}^t = \left(\frac{1}{|\mathcal{P}_t|} \sum_{k \in \mathcal{P}_t} \boldsymbol{\theta}_k^t\right) - \frac{1}{\alpha} \boldsymbol{h}^t$

end for

FedDyn ② 本地目标逐项拆解

$$\theta_k^t = \text{argmin}_\theta \left[L_k(\theta) - \langle \nabla L_k(\theta_k^{t-1}), \theta \rangle + (\alpha/2)\|\theta - \theta^{t-1}\|^2 \right]$$

项	含义	大白话
$L_k(\theta)$	客户端 k 的本地损失	照常学自己的数据
$-\langle \nabla L_k(\theta_k^{t-1}), \theta \rangle$	线性修正项：减去上一轮自己的梯度方向	**抵消本客户端一贯的偏向**（记账纠偏）
$(\alpha/2)\|\theta - \theta^{t-1}\|^2$	近端项：离本轮全局模型的距离	别走太远（与 FedProx 相同）

通信量与 FedAvg 相同：修正所需的"上一轮梯度"作为状态保存在客户端本地。

FedDyn algorithm

- Analysis

$$\boldsymbol{\theta}_k^t = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \left[\mathfrak{R}_k(\boldsymbol{\theta}; \boldsymbol{\theta}_k^{t-1}, \boldsymbol{\theta}^{t-1}) \triangleq L_k(\boldsymbol{\theta}) - \langle \nabla L_k(\boldsymbol{\theta}_k^{t-1}), \boldsymbol{\theta} \rangle + \frac{\alpha}{2} \|\boldsymbol{\theta} - \boldsymbol{\theta}^{t-1}\|^2 \right]$$

- The first order condition

$$\nabla L_k(\boldsymbol{\theta}_k^t) - \nabla L_k(\boldsymbol{\theta}_k^{t-1}) + \alpha(\boldsymbol{\theta}_k^t - \boldsymbol{\theta}^{t-1}) = \mathbf{0}$$

- If local device models converge, they converge to the server model, and the convergence point is a stationary point of the global loss.

if $\boldsymbol{\theta}_k^t \to \boldsymbol{\theta}_k^\infty$, it generally follows that, $\nabla L_k(\boldsymbol{\theta}_k^t) \to \nabla L_k(\boldsymbol{\theta}_k^\infty)$, and as a consequence, we have $\boldsymbol{\theta}^t \to \boldsymbol{\theta}_k^\infty$. In turn this implies that $\boldsymbol{\theta}_k^\infty \to \boldsymbol{\theta}^\infty$, i.e., is independent of k.

FedDyn 补充：状态怎么更新，为什么能收敛

① 黄框的递推式从哪来

对本地目标求梯度并令其为零（**一阶最优性条件**）：

$$\nabla L_k(\boldsymbol{\theta}_k^t) - \nabla L_k(\boldsymbol{\theta}_k^{t-1}) + \alpha(\boldsymbol{\theta}_k^t - \boldsymbol{\theta}^{t-1}) = \mathbf{0}$$

移项即得：新状态 = 旧状态 − α ×（本地模型相对全局模型的位移）。不需要重新计算梯度，也不需要上传。

② 服务器也有状态 h

先像 FedAvg 一样平均，再减去 h/α 作为修正。

③ 收敛直觉（不用推导）

1. 若训练收敛，所有本地模型都停在同一点 θ^∞，也等于服务器模型
2. 停住 → 每家的状态不再变化
3. 各家的线性修正项正好互相抵消 → $\Sigma_k \nabla L_k(\theta^\infty) = 0$
4. 即：收敛点是**全局损失的驻点**

一句话：**本地状态记账，各家偏向互相抵消。**

FedDyn ④ 与同类方法对比

方法	做法	收敛点是全局驻点?	额外通信	额外存储
FedAvg	直接平均	一般不是	—	—
FedProx	近端项：别走太远	一般不是	无	无
SCAFFOLD	控制变量纠偏	是	**多传一份控制变量**	客户端、服务器
FedDyn	近端项 + 线性修正项	是	**无**	客户端存梯度状态

代价：客户端长期保存一份与模型同大小的状态；很久才被选中一次时状态会过时。

超参数 α：太大，本地几乎不动；太小，修正不足。

KT-pFL

KT-pFL algorithm

- Intuition
  - Main idea is to allow each client to maintain a personalized soft prediction at the server that can be updated by a linear combination of all clients local soft predictions using a knowledge coefficient matrix.
  - Regardless of model structures

KT-pFL ① 换个思路：每家一个模型 个性化 · 改 ③ 上传内容

传统联邦：一个模型给所有人

$$\min_{w} \Sigma_n (D_n/D) \cdot L_n(w)$$

个性化联邦：每家一个模型

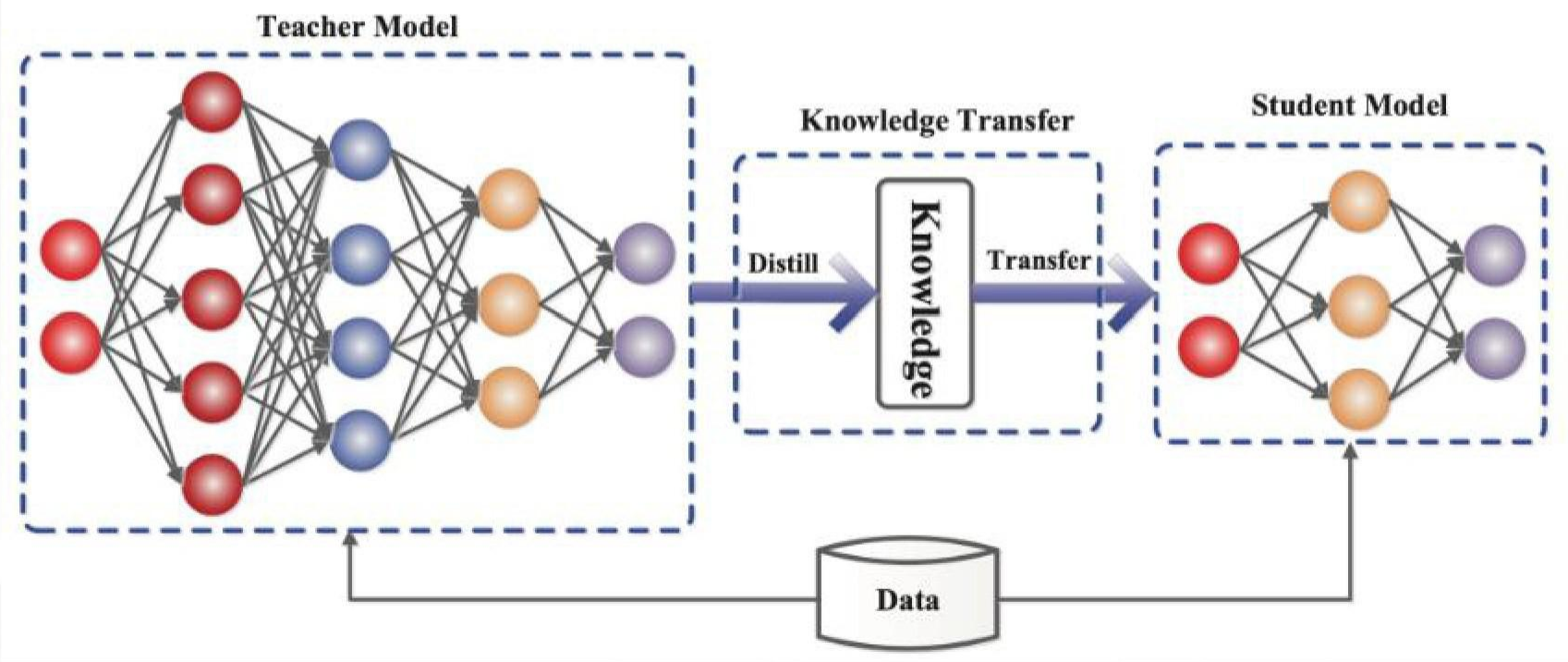
$$\min_{w^1 \ldots w^N} \Sigma_n (D_n/D) \cdot L_n(w^n)$$

Jie Zhang, Song Guo, Xiaosong Ma, Haozhao Wang, Wenchao Xu, Feijie Wu, Parameterized Knowledge Transfer for Personalized Federated Learning, NeurIPS 2021

- 各家数据不同，一个平均模型对谁都不是最好
- 完全不合作：数据太少；完全平均：抹掉个性
- KT-pFL：每家保留自己的模型，**通过知识蒸馏互相学习**
- 交换的不是参数，而是**在公共数据上的预测** → 各家模型结构可以不同

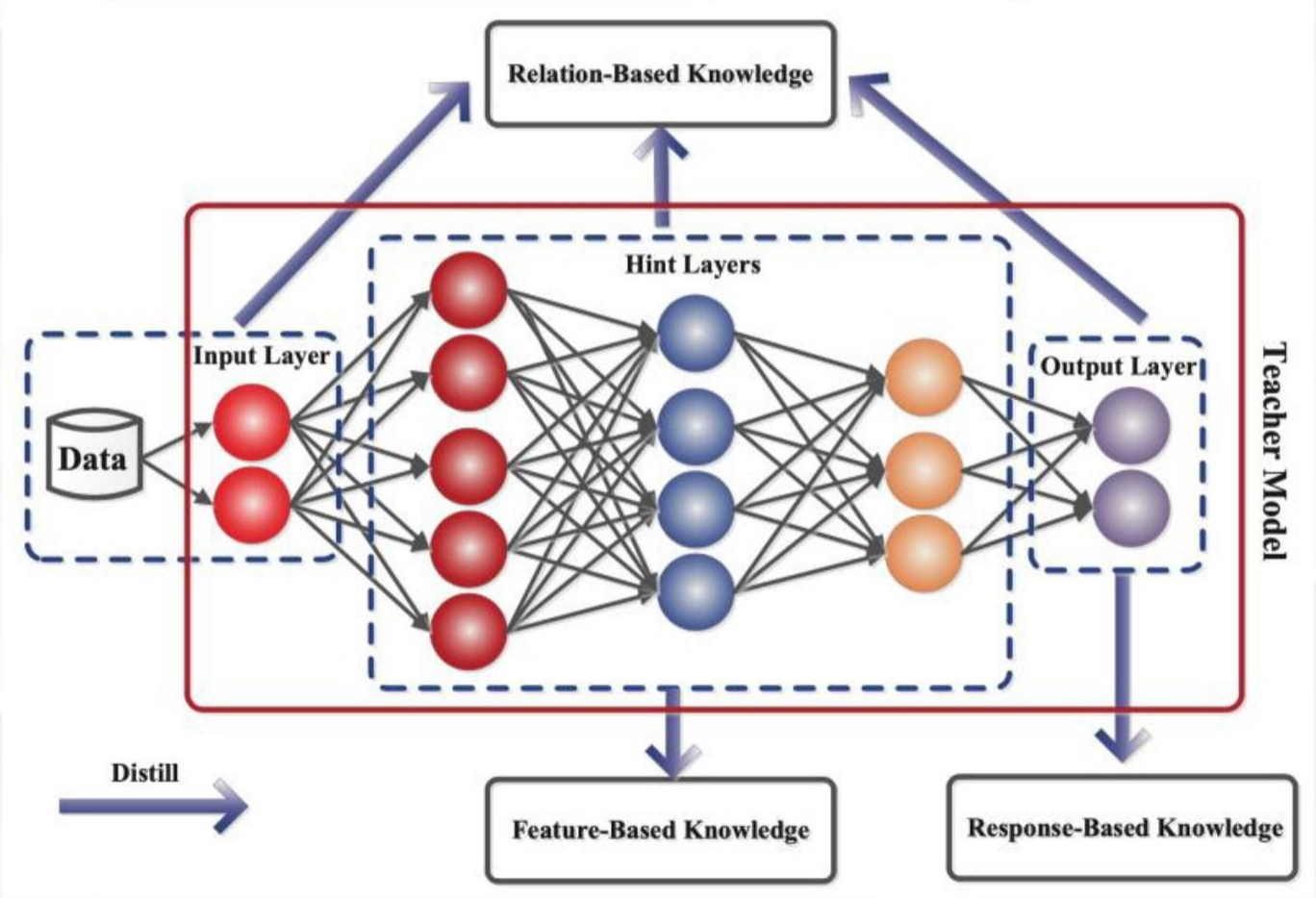
KT-pFL algorithm

- Knowledge Distillation (KD)
  - Transfer knowledge from well-learned teacher model to student model



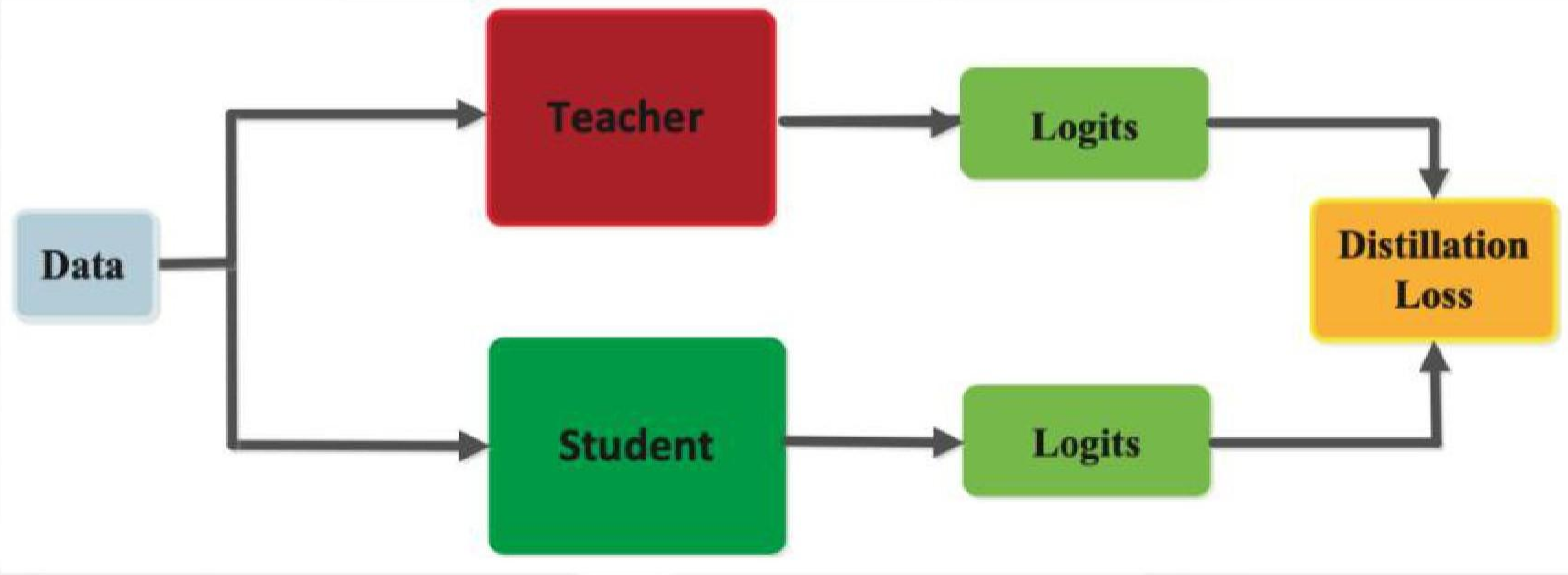
KT-pFL algorithm

- Knowledge Distillation (KD)
  - Classification



KT-pFL algorithm

- Knowledge Distillation (KD)
  - Response-based KD



KT-pFL ② 预备知识：知识蒸馏与软标签

- 老师模型输出**软标签**（概率分布），学生去拟合
- 软标签比 one-hot 信息多：
  "90% 像猫、9% 像狗" → 猫和狗相似
- 温度 T：T 越大分布越平滑，次要类别的信息越明显
- KT-pFL 用**基于响应（输出）**的蒸馏

$$s(w, x) = \exp(z_c/T) \,/\, \Sigma_{c'} \exp(z_{c'}/T)$$

标签	猫	狗	车
one-hot	1	0	0
软标签 T=1	0.90	0.09	0.01
软标签 T=4	0.53	0.30	0.17

表中数值为示意。

KT-pFL algorithm

- KT-pFL
  - Objective

$$\min_{\mathbf{w}} \mathcal{L}(\mathbf{w}) := \sum_{n=1}^{N} \frac{D_n}{D} \mathcal{L}_n(\mathbf{w}), \text{ where } \mathcal{L}_n(\mathbf{w}) = \frac{1}{D_n} \sum_{i=1}^{D_n} \mathcal{L}_{CE}(\mathbf{w}; x_i, y_i)$$

$$\min_{\mathbf{w}^1,\cdots,\mathbf{w}^N} \mathcal{L}(\mathbf{w}^1, \cdots, \mathbf{w}^N) := \sum_{n=1}^{N} \frac{D_n}{D} L_n(\mathbf{w}^n)$$

KT-pFL algorithm

- KT-pFL
  - Personalized loss function
    - Kullback–Leibler (KL) Divergence
    - c_{mn} is the knowledge coefficient which is used to estimate the contribution from client m to n.
    - $s(w^n, \hat{x})$ can be deemed to be a soft prediction of the client n

$$\mathcal{L}_{per,n}(\mathbf{w}^n) := \mathcal{L}_n(\mathbf{w}^n) + \lambda \sum_{\hat{x} \in \mathbb{D}_r} \mathcal{L}_{KL}\left(\sum_{m=1}^{N} c_{mn} \cdot s(\mathbf{w}^m, \hat{x}), s(\mathbf{w}^n, \hat{x})\right)$$

$$s(\mathbf{w}^n, \hat{x}) = \frac{\exp(z_c^n / T)}{\sum_{c=1}^{C} \exp(z_c^n / T)},$$

KT-pFL ③ 个性化损失：向"定制的老师"学习

$$\mathcal{L}_{per,n}(\mathbf{w}^n) := \mathcal{L}_n(\mathbf{w}^n) + \lambda \sum_{\hat{x}\in\mathbb{D}_r} \mathcal{L}_{KL}\left(\sum_{m=1}^{N} c_{mn} \cdot s(\mathbf{w}^m, \hat{x}), s(\mathbf{w}^n, \hat{x})\right)$$

$$s(\mathbf{w}^n, \hat{x}) = \frac{\exp(z_c^n/T)}{\sum_{c=1}^{C}\exp(z_c^n/T)},$$

符号	含义
$\mathbb{D}_r$	所有人都能拿到的**公共数据集**
$s(w^m, \hat{x})$	客户端 m 在公共样本上的软预测
c_{mn}	**知识系数**：客户端 m 对客户端 n 有多大参考价值
$\Sigma_m c_{mn} \cdot s(w^m, \hat{x})$	给客户端 n 定制的"老师"：其他人预测的加权组合

FedAvg 让每家拿到"所有人的平均"；KT-pFL 给每家一个**定制的加权组合**，数据相似的客户端互相权重大。

KT-pFL algorithm

- KT-pFL
  - Knowledge coefficient matrix

$$\mathbf{c} = \begin{Bmatrix} c_{11} & c_{12} & \cdots & c_{1N} \\ c_{21} & c_{22} & \cdots & c_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ c_{N1} & c_{N2} & \cdots & c_{NN} \end{Bmatrix}.$$

KT-pFL algorithm

- KT-pFL
  - Objective

$$\min_{\mathbf{w},\mathbf{c}} \mathcal{L}(\mathbf{w},\mathbf{c}) := \sum_{n=1}^{N} \frac{D_n}{D} \mathcal{L}_{per,n}(\mathbf{w}^n) + \rho \|\mathbf{c} - \frac{\mathbf{1}}{N}\|^2,$$

$$\mathbf{w} = [\mathbf{w}^1, \cdots, \mathbf{w}^N] \in \mathbb{R}^{\sum_{n=1}^{N} d_n}$$

KT-pFL algorithm

- KT-pFL
  - Training
    - Update w
      - Local Training
      - Distillation
    - Update c

$$\mathbf{w}^n \leftarrow \mathbf{w}^n - \eta_1 \nabla_{\mathbf{w}^n} \mathcal{L}_n(\mathbf{w}^n; \xi_n),$$

$$\mathbf{w}^n \leftarrow \mathbf{w}^n - \eta_2 \nabla_{\mathbf{w}^n} \mathcal{L}_{KL} \left(\sum_{m=1}^{N} \mathbf{c}_m^{*,T} \cdot s(\mathbf{w}^m, \xi_r), s(\mathbf{w}^n, \xi_r) \right)$$

$$\mathbf{c} \leftarrow \mathbf{c} - \eta_3 \lambda \sum_{n=1}^{N} \frac{D_n}{D} \nabla_{\mathbf{c}} \mathcal{L}_{KL} \left(\sum_{m=1}^{N} \mathbf{c}_m \cdot s(\mathbf{w}^{m,*}, \xi_r), s(\mathbf{w}^{n,*}, \xi_r) \right) - 2\eta_3 \rho (\mathbf{c} - \frac{\mathbf{1}}{N})$$

KT-pFL 补充：知识系数矩阵 c 怎么理解

示意：3 个客户端的 c（第 n 列 = 客户端 n 向谁学）

老师 m \ 学生 n	客户端 1	客户端 2	客户端 3
客户端 1	0.50	0.10	0.40
客户端 2	0.10	0.80	0.10
客户端 3	0.40	0.10	0.50

数值为示意：客户端 1、3 数据相似，互相学得多；客户端 2 数据很特别，主要靠自己。

- **c 不是人工设定的，是学出来的**（论文标题里"参数化"的含义）
- 正则项 $\rho||c - 1/N||^2$：别偏离"平均对待每个人"太远，防止只向一两家学

每轮三步

1. **本地训练**（客户端）：私有数据上更新 w^n
2. **蒸馏**（客户端）：公共数据上向"定制老师" $\Sigma_m c_{mn} s(w^m)$ 学习
3. **更新 c**（服务器）：用上传的软预测对 c 做梯度下降

KT-pFL algorithm

- KT-pFL

Algorithm 1 KT-pFL Algorithm

Input: $\mathbb{D}, \mathbb{D}_r, \eta_1, \eta_2, \eta_3$ and T
Output: $\mathbf{w} = [\mathbf{w}^1, \cdots, \mathbf{w}^N]$

Initialize $\mathbf{w}_0$ and $\mathbf{c}_0$
procedure SERVER-SIDE OPTIMIZATION
    Distribute $\mathbf{w}_0$ and $\mathbf{c}_0$ to each client
    for each communication round $t \in \{1, 2, ..., T\}$ **do**
        for each client n **in parallel do**
            $\mathbf{w}_{t+1}^n \leftarrow ClientLocalUpdate(n, \mathbf{w}_t^n, \mathbf{c}_{t,n})$
        Update knowledge coefficient matrix $\mathbf{c}$ via (7)
        Distribute $\mathbf{c}_{t+1}$ to all clients
procedure CLIENTLOCALUPDATE$(n, \mathbf{w}_t^n, \mathbf{c}_{t,n})$
    Client n receives $\mathbf{w}_t^n$ and $\mathbf{c}_n$ from the server
    for each local epoch i from 1 to E **do**
        for mini-batch $\xi_t \subseteq \mathbb{D}_n$ **do**
            Local Training: update model parameters on private data via (5)
    for each distillation step j from 1 to R **do**
        for mini-batch $\xi_{r,t} \subseteq \mathbb{D}_r$ **do**
            Distillation: update model parameters on public data via (6)
    return local parameters $\mathbf{w}_{t+1}^n$

KT-pFL algorithm

KT-pFL

- Illustration

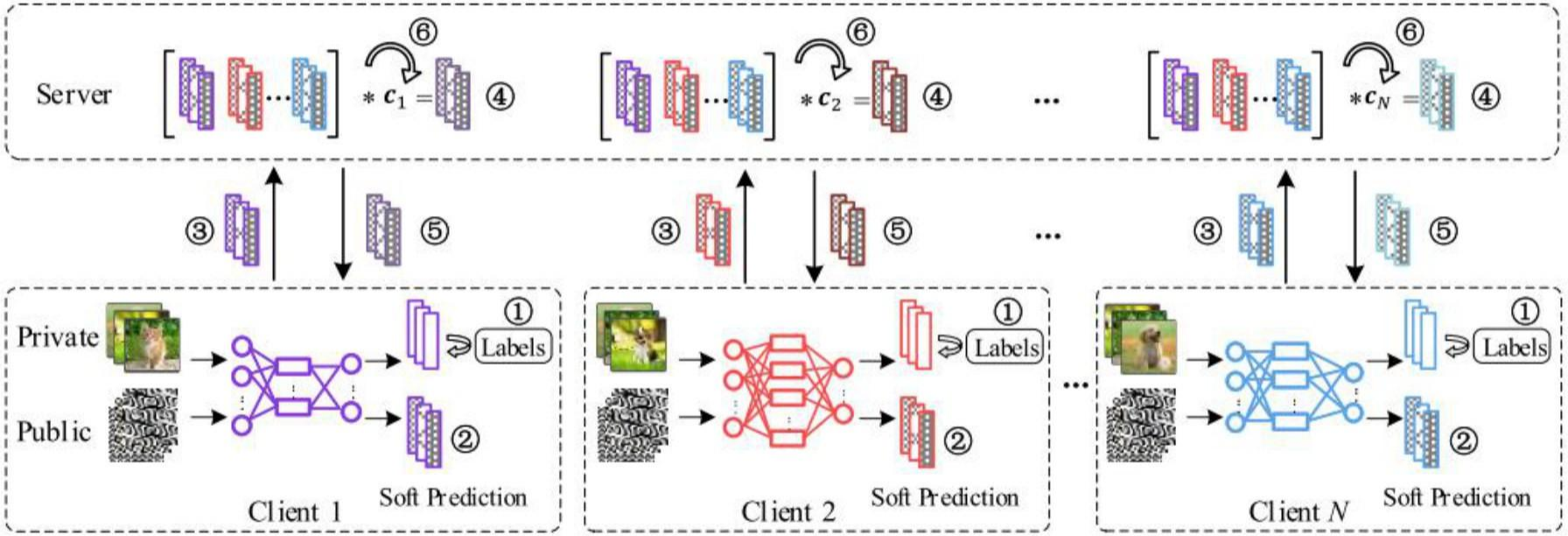


Figure 1: Illustration of the KT-pFL framework. The workflow includes 6 steps: ① local training on private data; ②, ③ each client outputs the local soft prediction on public data and sends it to the server; ④ the server calculates each client's personalized soft prediction via a linear combination of local soft predictions and knowledge coefficient matrix; ⑤ each client downloads the personalized soft prediction to perform distillation phase; ⑥ the server updates the knowledge coefficient matrix.

KT-pFL 小结：和 FedAvg 对比

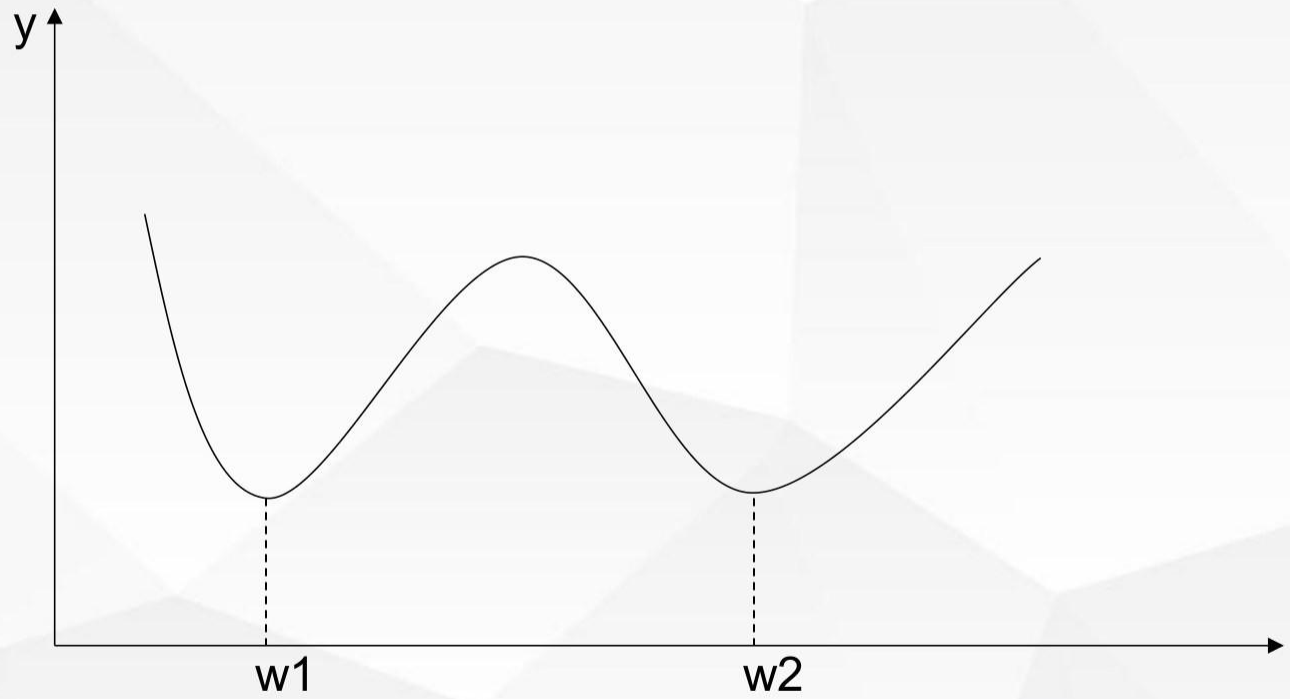
	FedAvg	KT-pFL
最终模型	一个全局模型	每家一个个性化模型
上传内容	模型参数	**公共数据上的软预测**
各家模型结构	必须相同	**可以不同**
通信量取决于	模型大小	公共数据量 × 类别数
额外需求	无	一份公共数据集；服务器维护 N×N 的 c

KT-pFL 给每家的是"定制的加权组合"，而不是所有人的平均；数据相似的客户端互相权重大。

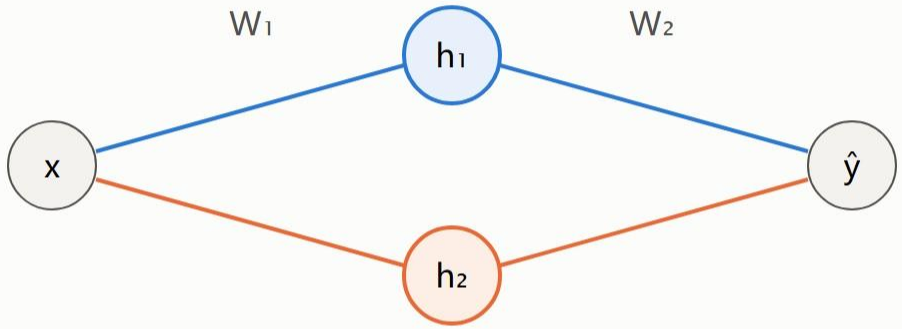
FedMA

FedMA algorithm

- Permutation invariance



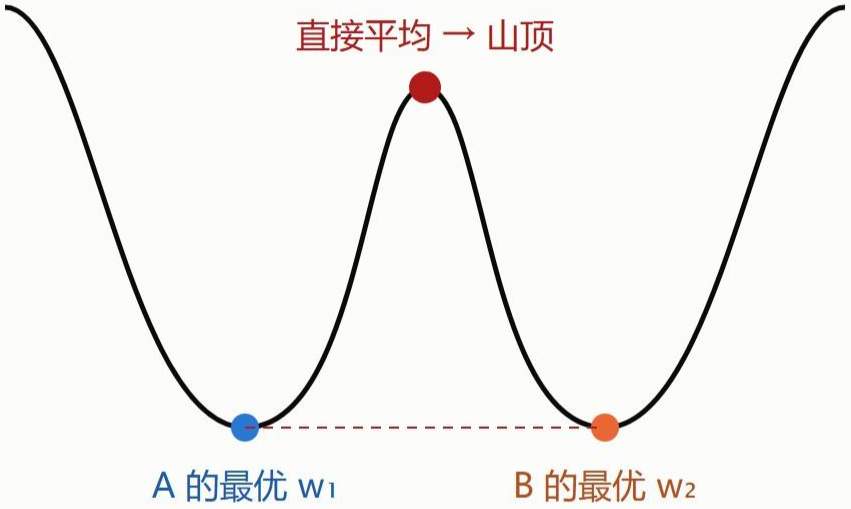
FedMA ① 排列不变性：交换两个神经元，网络功能不变 改 ④ 聚合



交换 h_1、h_2 的位置：W_1 换列、W_2 换行 → 输出不变

$$\hat{y} = \sigma(xW_1\Pi)\,\Pi^T W_2 = \sigma(xW_1)\,W_2$$

Π 为置换矩阵；σ 是**逐元素**激活，所以能与置换交换顺序。隐藏层 L 个神经元就有 L! 种等价写法。

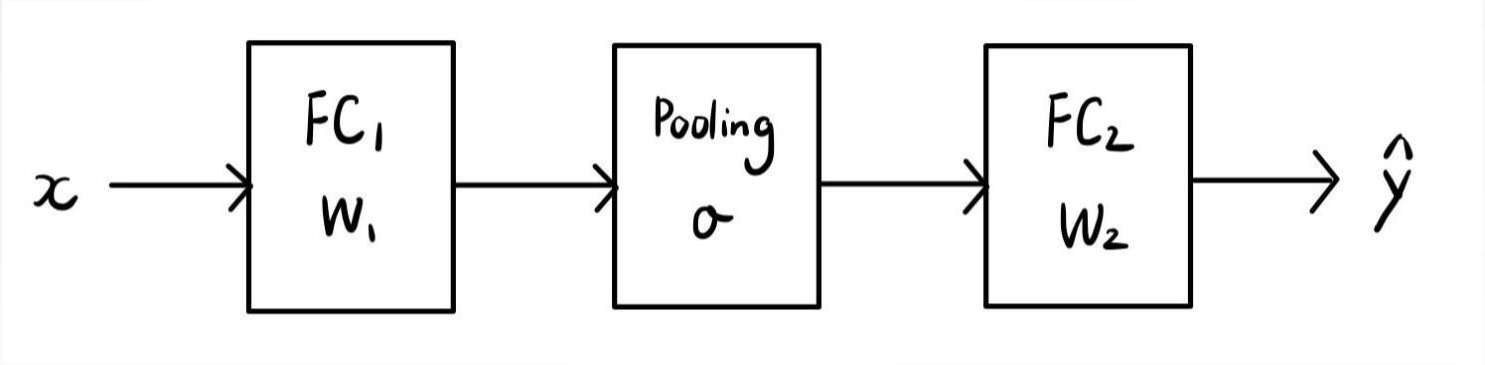


两个同样好的解，平均后反而最差

Hongyi Wang, Mikhail Yurochkin, Yuekai Sun, Dimitris Papailiopoulos, Yasaman Khazaeni, Federated Learning with Matched Averaging, ICLR 2020 · github.com/IBM/FedMA

FedMA algorithm

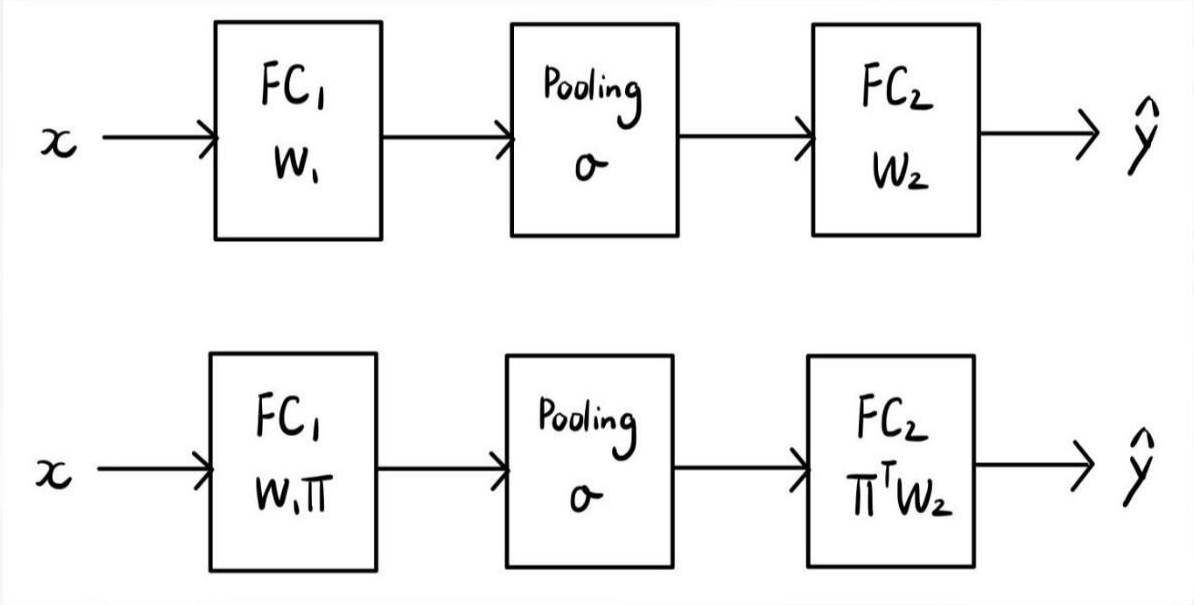
- Permutation invariance (fully-connected (FC) layer)



$$\hat{y} = \sigma(xW_1)W_2$$

FedMA algorithm

- Permutation invariance (fully-connected (FC) layer)



$$\hat{y} = \sigma(xW_1\Pi)\Pi^T W_2, \text{ where } \Pi \text{ is any } L \times L \text{ permutation matrix.}$$

FedMA algorithm

- Permutation invariance (fully-connected (FC) layer)

Client A:

$$\{W_1\Pi_j, \Pi_j^T W_2\}$$

Client B:

$$\{W_1\Pi_{j'}, \Pi_{j'}^T W_2\}$$

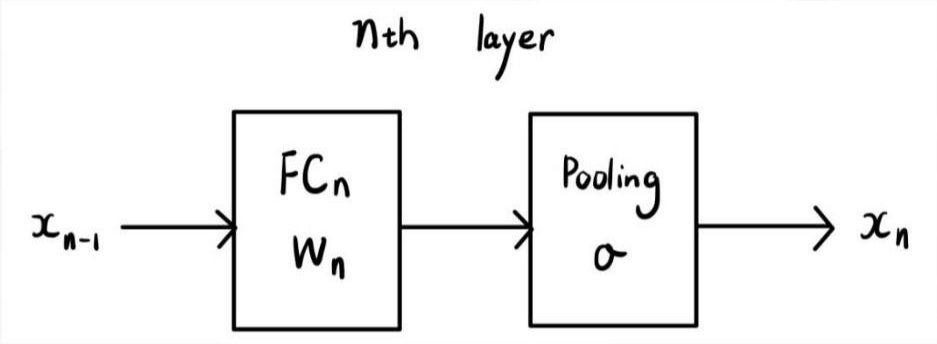
Aggregation

$$(W_1\Pi_j + W_1\Pi_{j'})/2 \neq W_1\Pi \text{ for any } \Pi$$

Solution: $(W_1\Pi_j\Pi_j^T + W_1\Pi_{j'}\Pi_{j'}^T)/2 = W_1$

FedMA algorithm

- Permutation invariance （FCs）

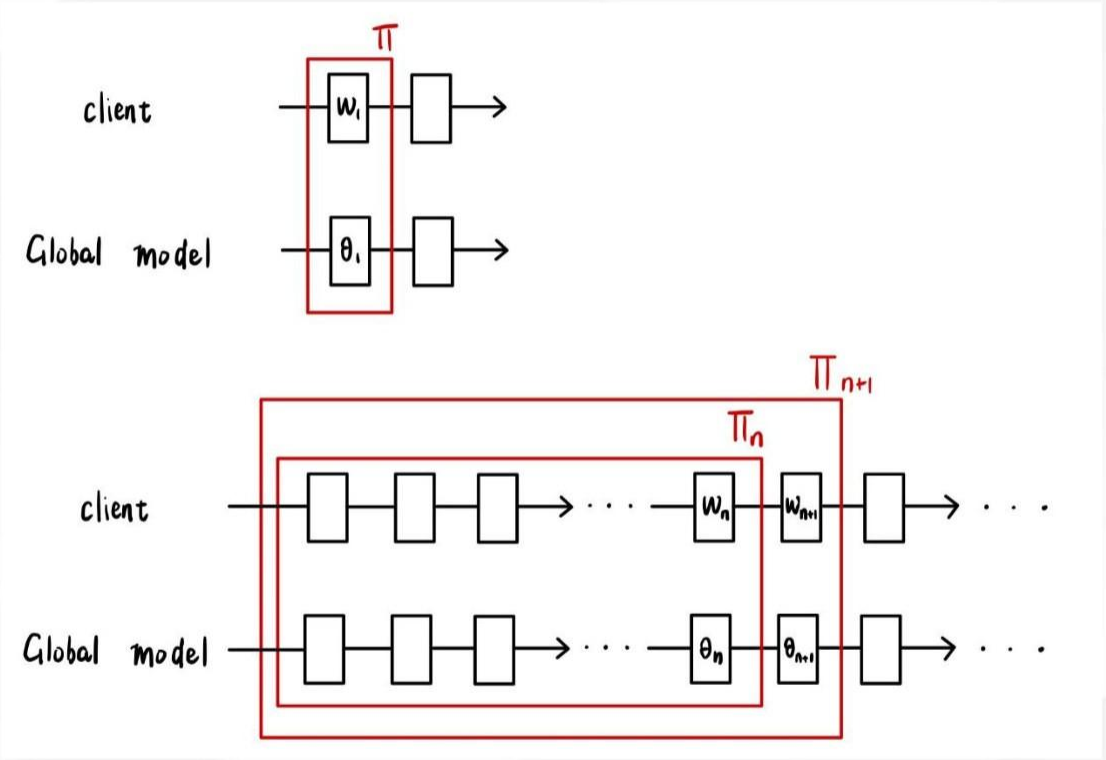


Simple FCs: $\hat{y} = \sigma(xW_1\Pi)\Pi^T W_2$

Deep FCs: $x_n = \sigma(x_{n-1}\Pi_{n-1}^T W_n \Pi_n)$

FedMA algorithm

- Permutation invariance (FCs)



$$x_n = \sigma(x_{n-1}\Pi_{n-1}^T W_n \Pi_n)$$

FedMA algorithm

- Permutation invariance （CNN)

FC : $x_n = \sigma(x_{n-1}\Pi_{n-1}^T W_n \Pi_n)$

CNN : $x_n = \sigma(\mathbf{Conv}(x_{n-1}, \Pi_{n-1}^T W_n \Pi_n))$

FedMA algorithm

- Permutation invariance (recall)

Client A: $\{W_1\Pi_j, \Pi_j^T W_2\}$

Client B: $\{W_1\Pi_{j'}, \Pi_{j'}^T W_2\}$

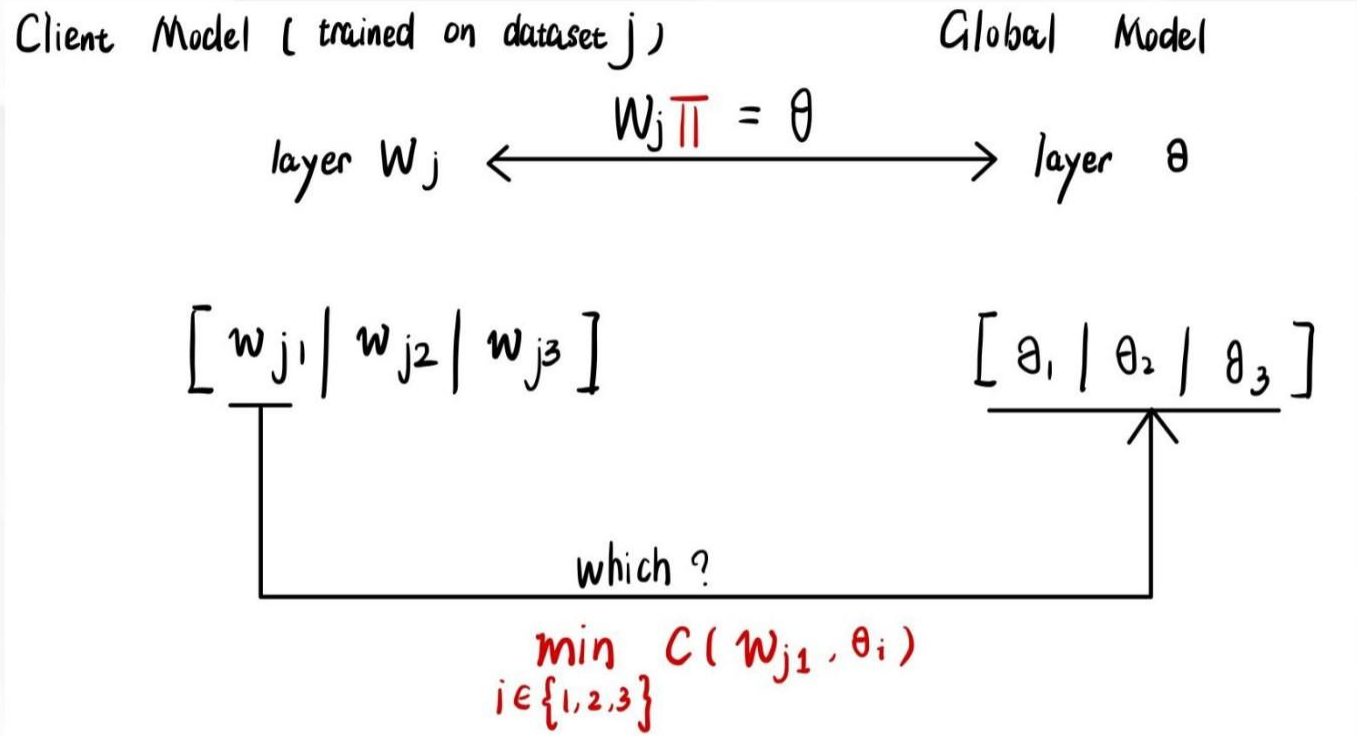
Aggregation

$$(W_1\Pi_j + W_1\Pi_{j'})/2 \neq W_1\Pi \text{ for any } \Pi$$

Solution: $(W_1\Pi_j\Pi_j^T + W_1\Pi_{j'}\Pi_{j'}^T)/2 = W_1$

FedMA algorithm

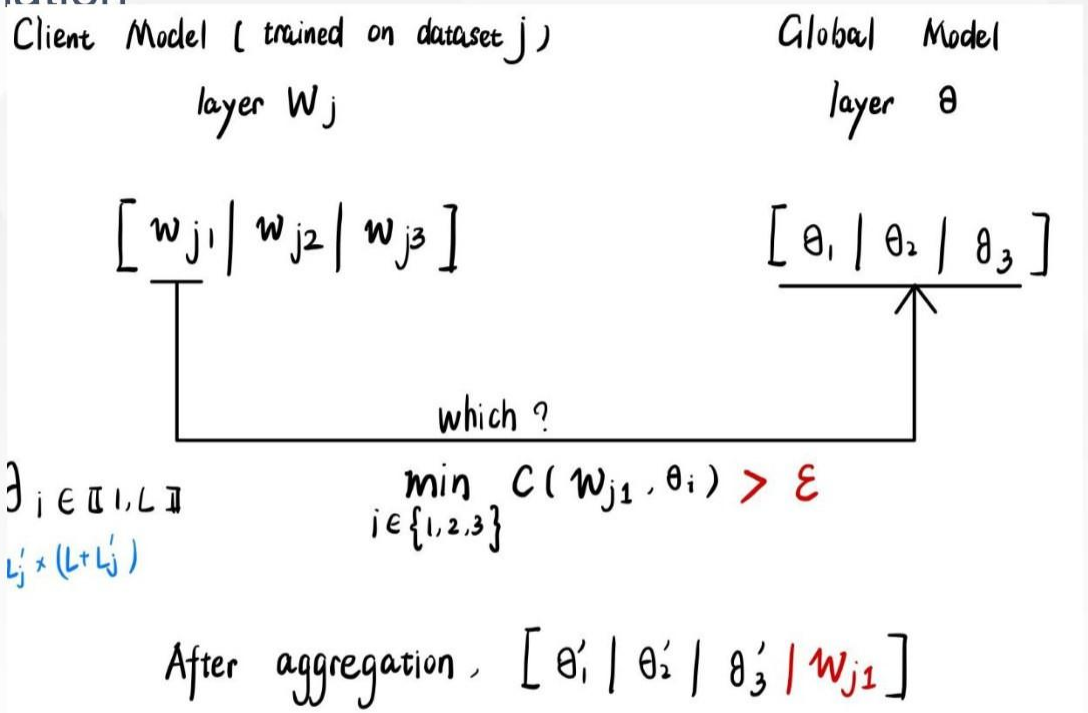
- Matched averaging formulation



$$\min_{\{\pi_{li}^j\}} \sum_{i=1}^{L} \sum_{j,l} \min_{\theta_i} \pi_{li}^j c(w_{jl}, \theta_i) \text{ s.t. } \sum_i \pi_{li}^j = 1 \; \forall \, j, l; \; \sum_l \pi_{li}^j = 1 \; \forall \, i, j.$$

FedMA algorithm

- Matched averaging formulation



$$\min_{\{\pi_{li}^{j'}\}_{l,i}} \sum_{i=1}^{L+L_{j'}} \sum_{j=1}^{L_{j'}} \pi_{li}^{j'} C_{li}^{j'} \text{ s.t. } \sum_{i} \pi_{li}^{j'} = 1 \; \forall\, l; \; \sum_{l} \pi_{li}^{j} \in \{0,1\} \; \forall\, i, \text{ where}$$

$$C_{li}^{j'} = \begin{cases} c(w_{j'l}, \theta_i), & i \leq L \\ \epsilon + f(i), & L < i \leq L + L_{j'}. \end{cases}$$

FedMA 补充：神经元配对的小例子

客户端神经元 w 与全局神经元 θ 的距离

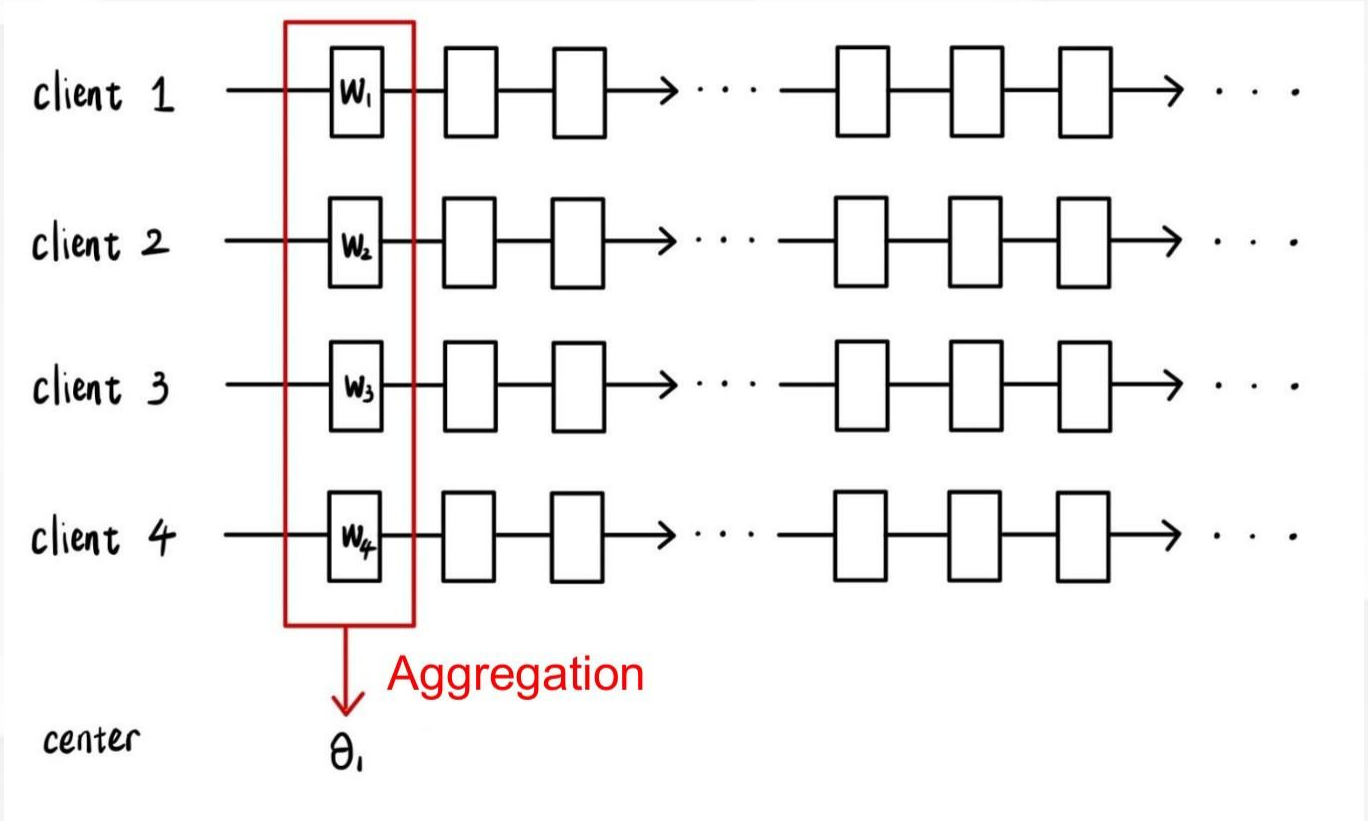
	θ_1	θ_2	θ_3	结果
w_1	0.9	**0.2**	1.4	$w_1 \leftrightarrow \theta_2$
w_2	**0.3**	1.1	1.2	$w_2 \leftrightarrow \theta_1$
w_3	1.6	1.5	1.7	**都 > ε = 1.0 → 新增 θ_4**

数值为示意。

- 每个客户端神经元找"最像"的全局神经元，每个只能配一个：这是**指派问题**（可用匈牙利算法）
- 配上的：先对齐、再平均（w_1 与 θ_2 平均，w_2 与 θ_1 平均）
- 谁都不像的（w_3）：说明学到了新特征，**加为新神经元**，全局层变宽
- 新增代价 ε + f(i) 随层宽 i 递增：层越宽，越难再加，防止无限膨胀

FedMA algorithm

- FedMA (https://github.com/IBM/FedMA)



FedMA algorithm

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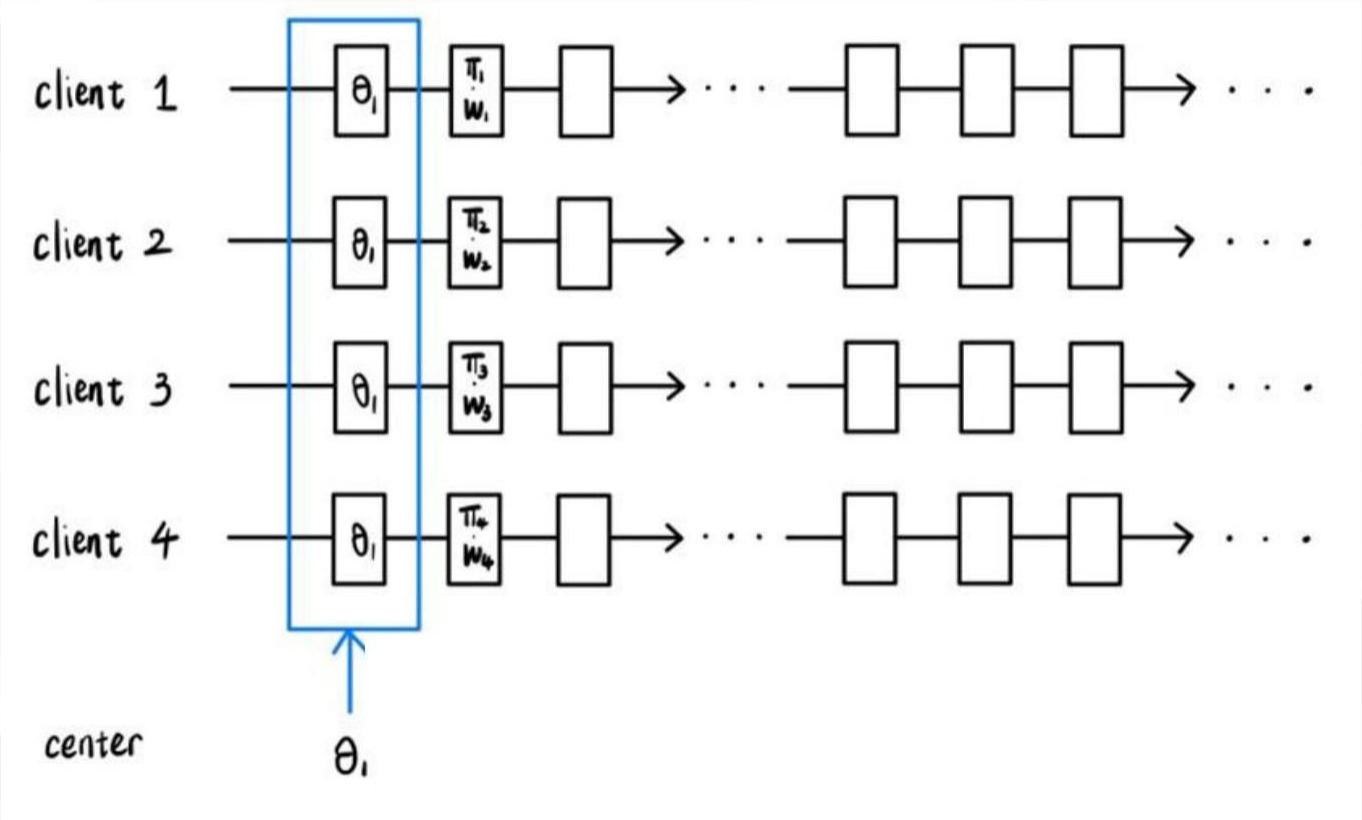

FedMA algorithm

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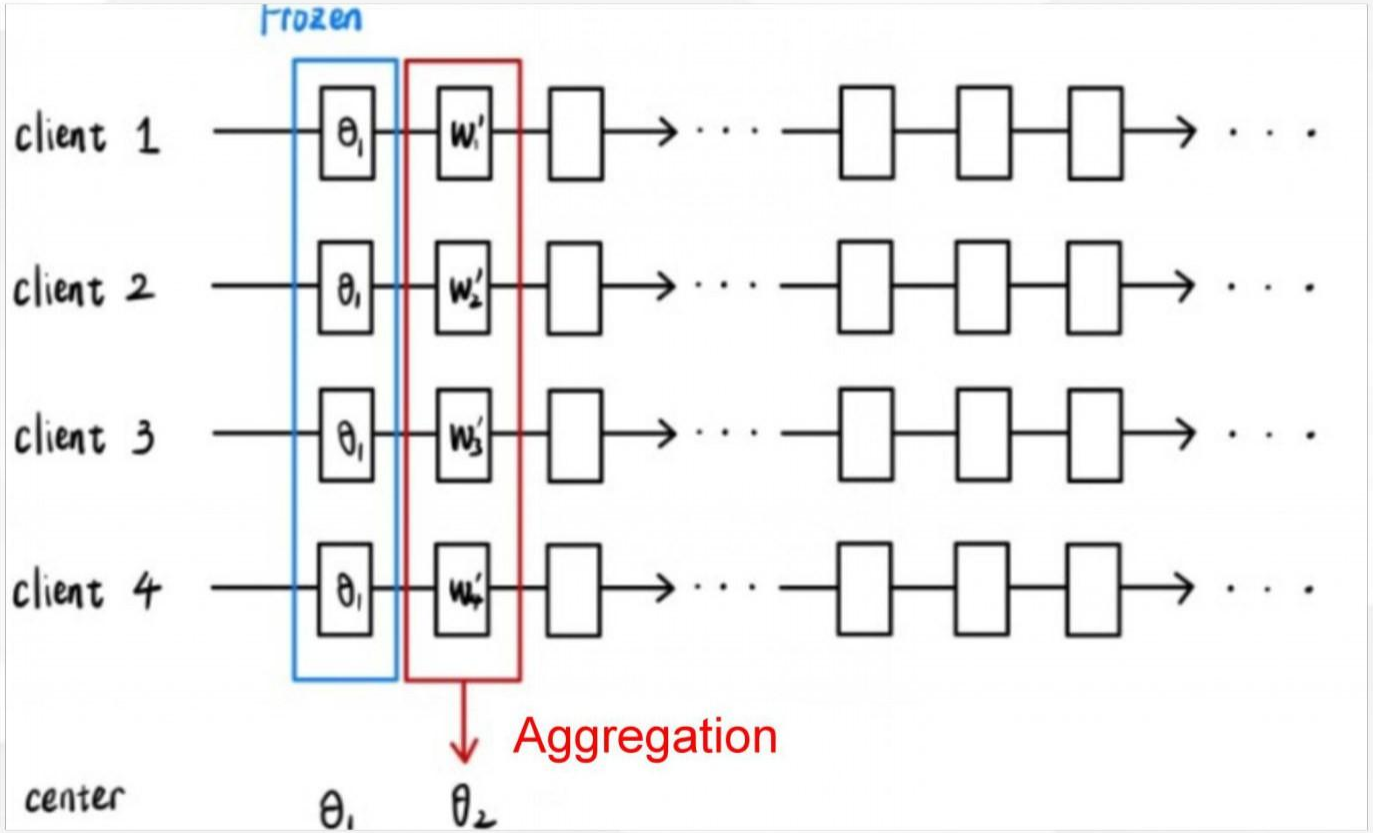
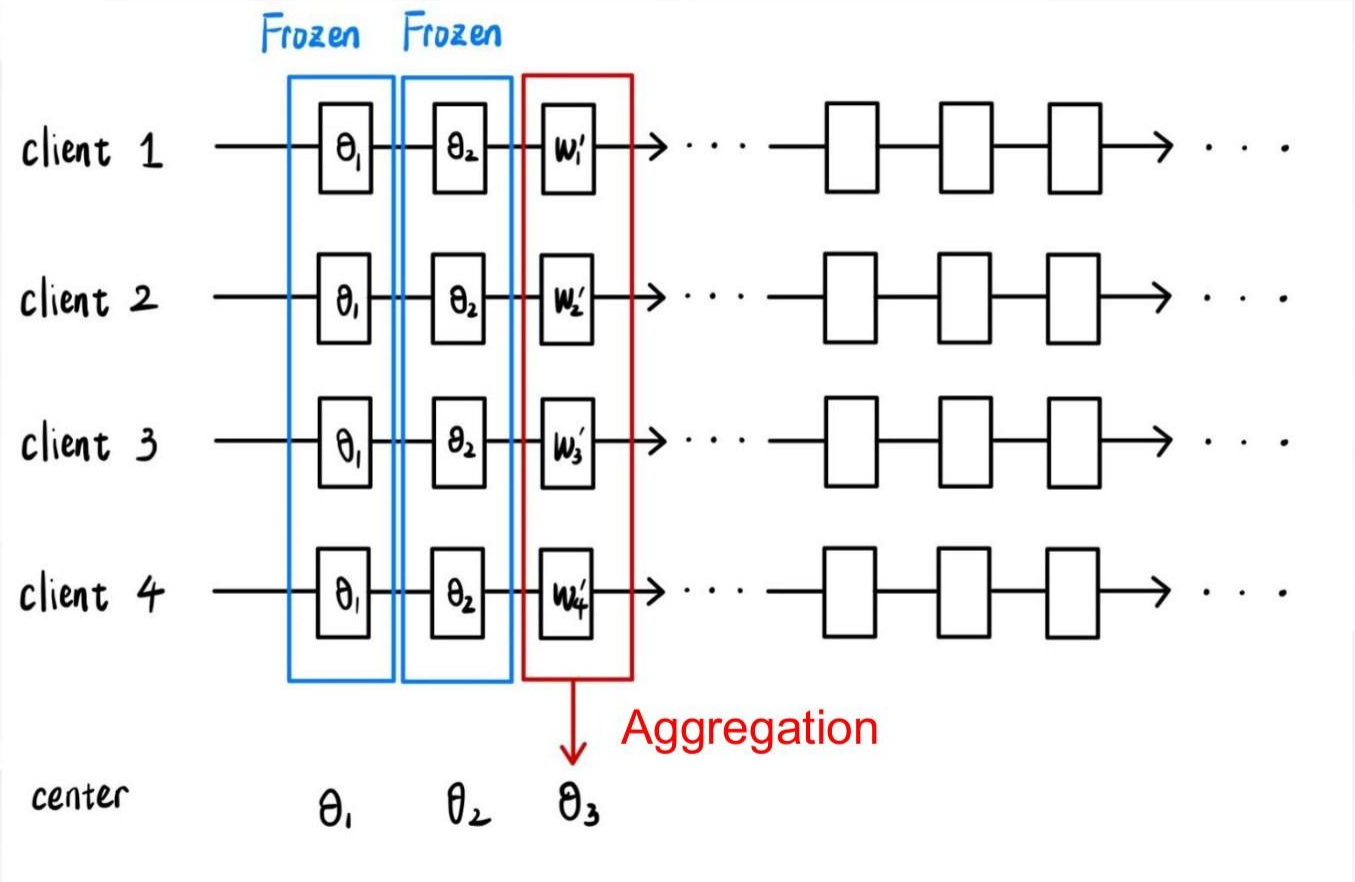


FedMA algorithm

- FedMA (https://github.com/IBM/FedMA)



FedMA algorithm

- FedMA (https://github.com/IBM/FedMA)

Algorithm 1: Federated Matched Averaging (FedMA)

Input : local weights of N-layer architectures $\{W_{j,1}, \ldots, W_{j,N}\}_{j=1}^{J}$ from J clients

Output: global weights $\{W_1, \ldots, W_N\}$

$n = 1$;

while $n \leq N$ **do**

  if $n < N$ **then**

    $\{\Pi_j\}_{j=1}^{J} = \text{BBP-MAP}(\{W_{j,n}\}_{j=1}^{J})$; // call BBP-MAP to solve Eq. 2

    $W_n = \frac{1}{J} \sum_j W_{j,n} \Pi_j^T$;

  else

    $W_n = \sum_{k=1}^{K} \sum_j p_{jk} W_{jl,n}$ where p_k is fraction of data points with label k on worker j;

  end

  for $j \in \{1, \ldots, J\}$ **do**

    $W_{j,n+1} \leftarrow \Pi_j W_{j,n+1}$; // permutate the next-layer weights

    Train $\{W_{j,n+1}, \ldots, W_{j,L}\}$ with W_n frozen;

  end

  $n = n + 1$;

end

FedMA 小结：一层一层地匹配

1. 各客户端训练好；服务器只**匹配平均第 1 层**，下发
2. 客户端**冻结第 1 层**，继续训练后面各层
3. 服务器匹配平均第 2 层……逐层推进
4. 输出层没有置换问题，按各家**各类数据比例**加权

为什么逐层：每层的 Π 互相耦合，一次全求误差会层层累积（前作 PFNM 的问题）

代价：通信轮数 ≈ 层数；模型变宽；不支持 BatchNorm，残差结构难处理

思想延续：模型合并（如 Git Re-Basin）也是先对齐神经元再合并

03

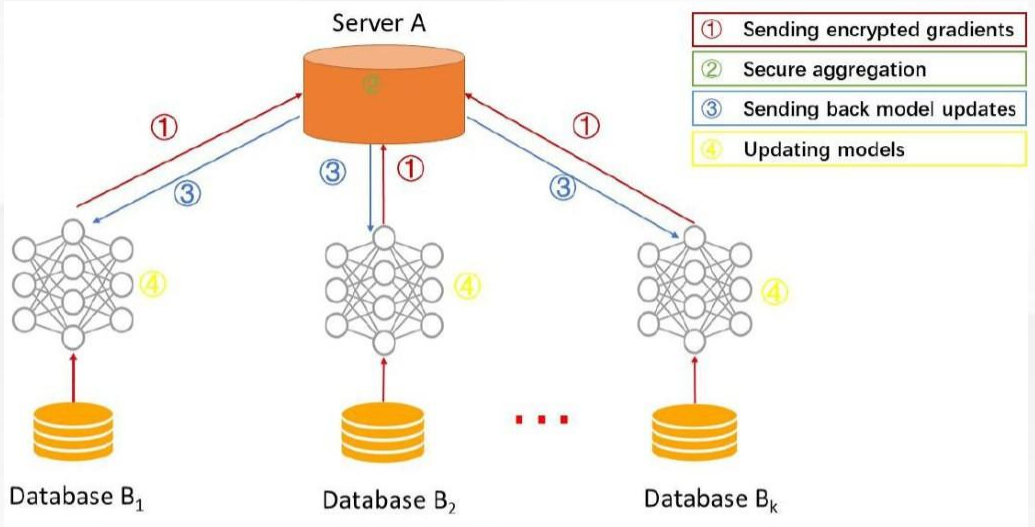
Advanced algorithms

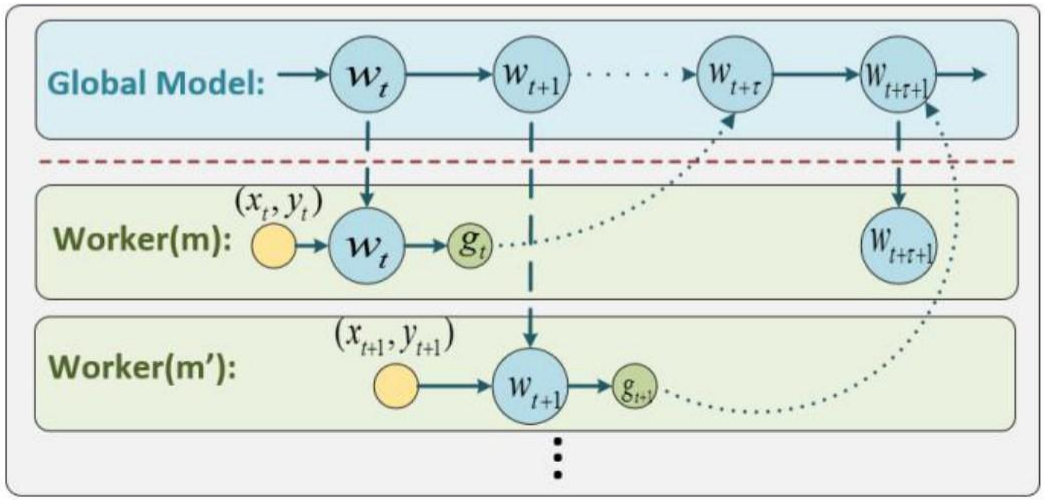
- SageFlow
- FLChain
- FAIR

SageFlow

Sageflow algorithm

- Stragglers: slow devices
  - **Keep waiting**: slow down the overall process
  - **Drop out**: important data missing
  - **Asynchronous**(staleness): + adversaries?
- Attackers: malicious attacks launched by adversaries
  - **untargeted attacks**: model poisoning, data poisoning
  - **targeted/backdoor attacks**: misclassify the targeted subtasks
  - Robust Federated Averaging & Multi-Krum
- Large portion of adversaries
  - Straggler: increase attack ratio





Sageflow ① 两个问题，叠加更糟 系统异构 + 安全 • 改 ④ 聚合

掉队者（straggler）：慢设备

- **一直等**：所有人陪着等，整体变慢
- **直接丢**：丢掉的设备数据分布不同 → 模型有偏
- **异步**：慢设备基于旧模型算的更新已**过时**（staleness），直接加进来会拉坏模型

攻击者（adversary）

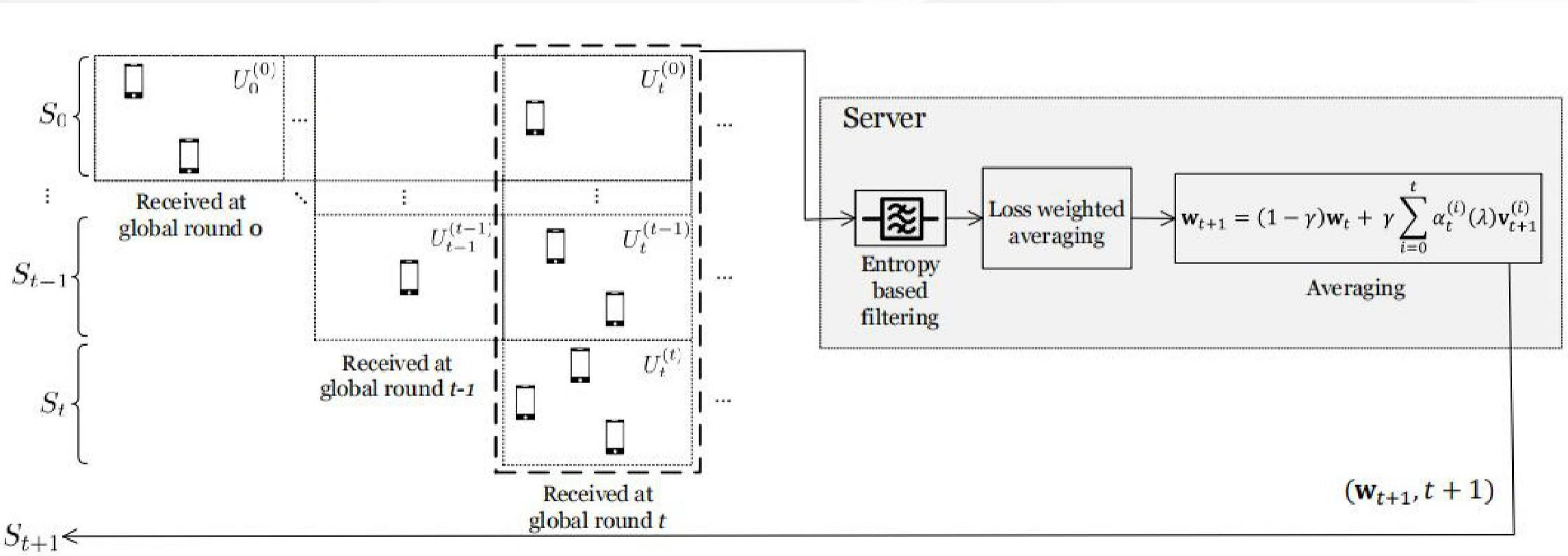
- **模型投毒**：篡改上传参数（如 × −0.1）
- **数据投毒**：翻转标签后正常训练
- **后门**：带特定触发图案时被分到指定类

叠加：正常设备有的没按时到 → 本轮到达者中攻击者**占比升高** → 依赖"攻击者是少数"的鲁棒聚合（RFA、Multi-Krum）失效。

Jungwuk Park, Dong-Jun Han, Minseok Choi, Jaekyun Moon, Sageflow: Robust Federated Learning against Both Stragglers and Adversaries, NeurIPS 2021

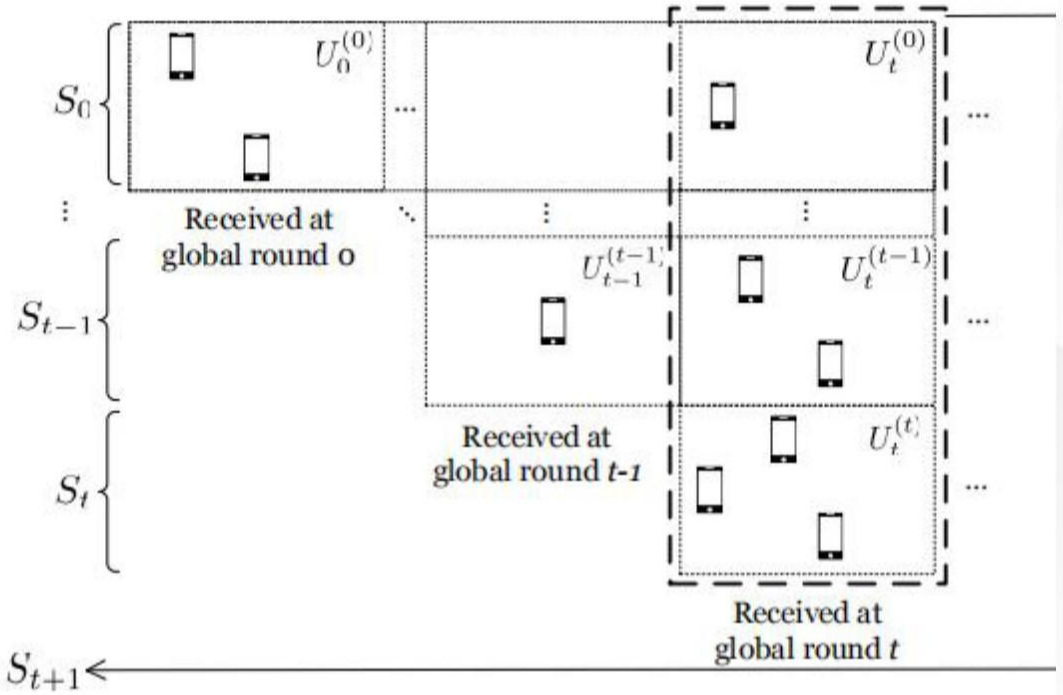
Sageflow algorithm

- Staleness-aware grouping
- Entropy-based filtering + Loss-weighted averaging



Sageflow algorithm

- Staleness-aware grouping
  - perform **periodic** global aggregation(fixed time deadline)
  - allow stragglers to be aggregated in **later rounds**
  - group with **same staleness** -> group representative model
  - aggregate according to **staleness**



Staleness group Number of data samples

$$\mathbf{v}_{t+1}^{(i)} = \sum_{k \in U_t^{(i)}} \frac{m_k}{\sum_{k \in U_t^{(i)}} m_k} \mathbf{w}_i(k)$$

$$\mathbf{w}_{t+1} = (1-\gamma)\mathbf{w}_t + \gamma \sum_{i=0}^{t} \alpha_t^{(i)}(\lambda)\mathbf{v}_{t+1}^{(i)}$$

$$\sum_{i=0}^{t} \alpha_t^{(i)}(\lambda)\mathbf{v}_{t+1}^{(i)}$$

$$\alpha_t^{(i)}(\lambda) \propto \frac{\sum_{k \in U_t^{(i)}} m_k}{(t-i+1)^{\lambda}}$$

Staleness function

Sageflow 补充：分组加权的数值例子

当前第 **t = 5** 轮，取 λ = 1。本轮收到 4 份更新：

组（基于第 i 轮模型）	迟到轮数 t−i	样本数	样本数 / $(t-i+1)^{\lambda}$	组权重 α
i = 5（2 台，准时）	0	300	300 / 1 = 300	**0.6**
i = 4（1 台）	1	200	200 / 2 = 100	**0.2**
i = 3（1 台）	2	300	300 / 3 = 100	**0.2**

数值为示意。i = 3 组样本数和 i = 5 组一样多，但迟到 2 轮，权重只有其 1/3。

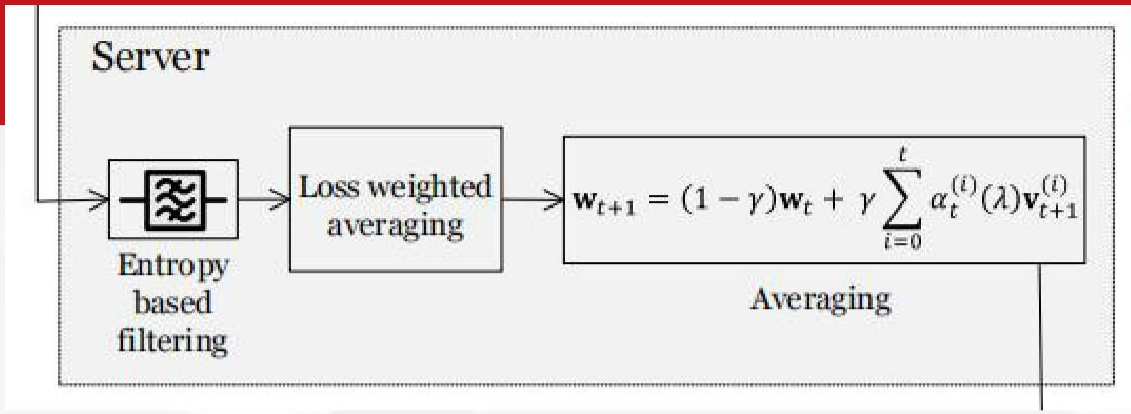
1. 组内：按样本数平均，得到组代表 $\mathbf{v}^{(i)}$
2. 组间：按上表权重 α 加权
3. 与旧模型混合：

$$\mathbf{w}_{t+1} = (1 - \gamma)\mathbf{w}_t + \gamma \Sigma_i \alpha^{(i)}\mathbf{v}^{(i)}$$

> 慢设备的更新**不丢弃**，只是按迟到程度降权；λ 越大惩罚越重。

Sageflow algorithm

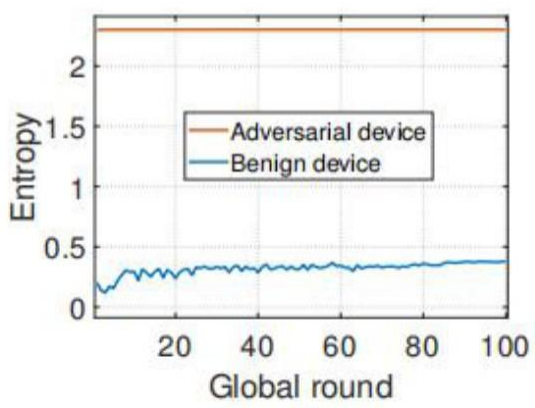
- Entropy-based filtering
  - **Public data** on server
  - **Filter out** high **entropy** models (loss)
  - For model poisoning



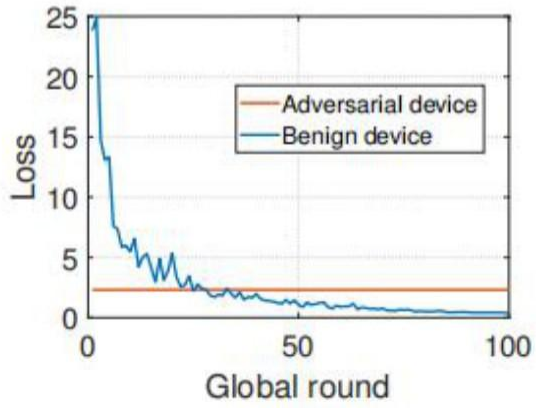
$$E_{x_{pub,j}}(k) = -\sum_{q=1}^{Q} P_{x_{pub,j}}^{(q)}(k) \log P_{x_{pub,j}}^{(q)}(k)$$

$$E(k) = \frac{1}{n_{pub}} \sum_{j=1}^{n_{pub}} E_{x_{pub,j}}(k)$$

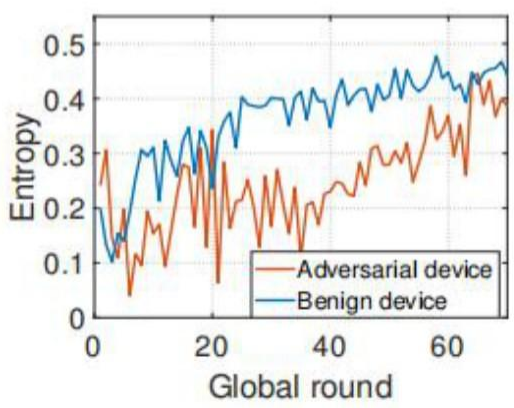
Shannon entropy



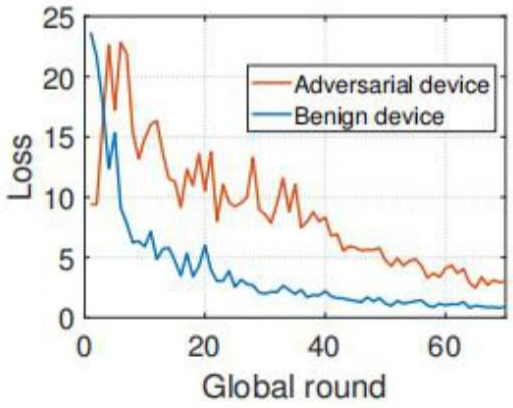
(a) Model poison, entropy



(b) Model poison, loss



(c) Data poison, entropy



(d) Data poison, loss

Sageflow 补充：熵怎么区分"乱猜"的模型

同一个公共样本，三分类的预测

模型	预测分布	熵 $-\Sigma\, p \ln p$
正常模型	[0.90, 0.05, 0.05]	≈ 0.39
被篡改参数的模型	[0.34, 0.33, 0.33]	≈ 1.10（最大值 ln3）

对所有公共样本取平均得 E(k)；**$E(k) > E_{th}$ 的模型直接丢弃。**

为什么还需要第三个模块

攻击	熵能区分?	损失能区分?
模型投毒（改参数）	**能**	不能
数据投毒（改标签）	不能	**能**

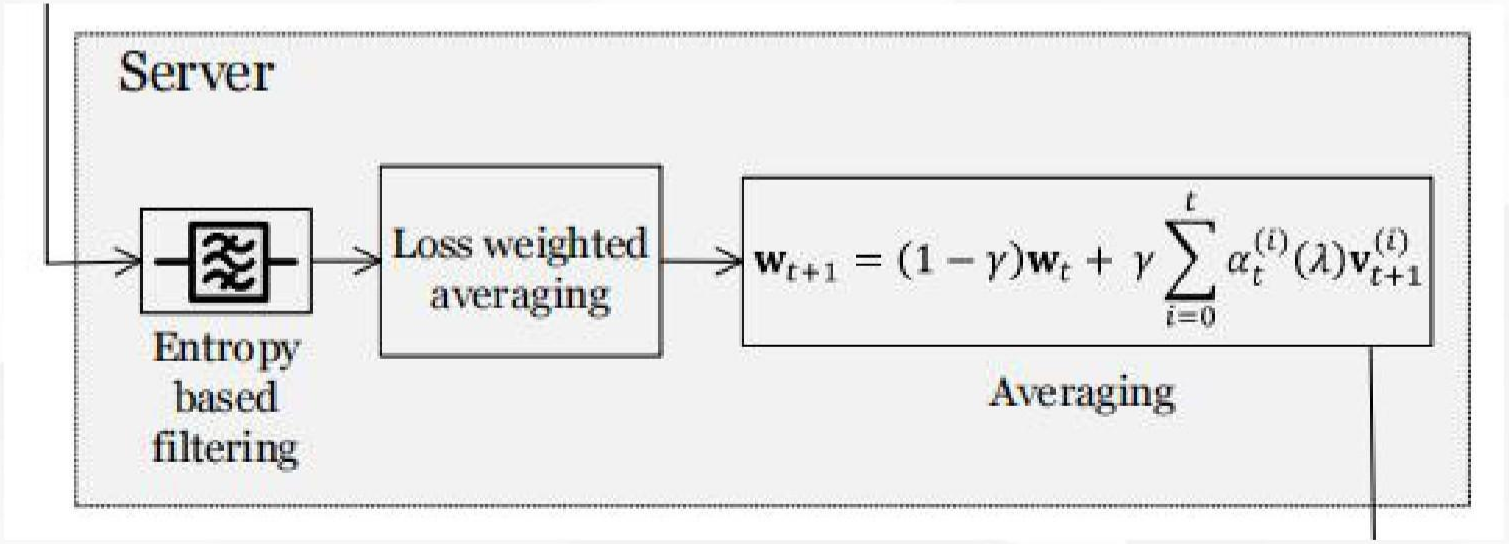
> 熵过滤只管模型投毒；数据投毒交给"按损失加权"。

Sageflow algorithm

- Loss-weighted averaging
  - Aggregation weight according to local models' **measured qualities**
  - Measure by **loss** on public data
  - data-poisoned model -> small weight + less impact
  - For data poisoning & scaled backdoor

$$\beta_t^{(k)}(\delta) \propto \frac{m_k}{\{F_{pub}(\mathbf{w}_t(k))\}^{\delta}} \quad \text{and} \quad \sum_{k \in S_t} \beta_t^{(k)}(\delta) = 1.$$

$$\mathbf{w}_{t+1} = \sum_{k \in S_t} \beta_t^{(k)}(\delta) \mathbf{w}_t(k)$$



Sageflow 补充：按损失加权的数值例子

$$\beta^{(k)} \propto m_k / F_{pub}(w(k))^{\delta}$$

三台设备样本数相同，取 δ = 1：

设备	公共数据上的损失	1 / 损失	权重 β
正常 A	0.5	2.00	**0.48**
正常 B	0.6	1.67	**0.40**
数据投毒 C	2.0	0.50	**0.12**

数值为示意。

- 损失越大，权重越小：被投毒的模型自动降权
- **软降权**而不是一刀切，误伤正常设备的代价小

三个模块合起来，每轮：

1. 等待 T_d，按过时程度分组
2. 组内：**熵过滤** → **按损失加权**平均
3. 组间：按过时程度加权，再与旧模型混合

Sageflow algorithm

- Time complexity

Public data | Model parameters | Clients number

$$\mathcal{O}(n_{pub}|w|K)$$

Algorithm 1 Proposed Sageflow Algorithm

Input: Initialized model $\mathbf{w}_0$, **Output:** Final global model $\mathbf{w}_T$

Process at the Server

1: **for** each global round $t = 0, 1, ..., T-1$ **do**
2: Choose S_t and send the current model and the global round $(\mathbf{w}_t, t)$ to the devices
3: Wait for T_d and then:
4: **for** $i = 0, 1, ..., t$ **do**
5: $U_t^{(i)}(E_{th}) = \{k \in U_t^{(i)} | E(k) < E_{th}\}$ // Entropy-based filtering in each group
6: $\mathbf{v}_{t+1}^{(i)} = \sum_{k \in U_t^{(i)}(E_{th})} \beta_t^{(k)}(\delta)\mathbf{w}_i(k)$ // Loss-weighted averaging in each group (with same staleness)
7: **end for**
8: $\mathbf{w}_{t+1} = (1-\gamma)\mathbf{w}_t + \gamma \sum_{i=0}^{t} \alpha_t^{(i)}(\lambda)\mathbf{v}_{t+1}^{(i)}$ // Averaging of representative models (with different staleness)
9: **end for**

Process at the Device: Device k receives $(\mathbf{w}_t, t)$ from the server and performs local updates to obtain $\mathbf{w}_t(k)$. Then each benign device k sends $(\mathbf{w}_t(k), t)$ to the server, while a malicious adversary sends a poisoned model depending on the type of attack.

Sageflow algorithm

- Theoretical Analysis
- Convergence analysis
  - Assumption 1: μ-strongly convex + L-smooth
  - Assumption 2: unbiased estimation
- Theoretical bound

$$F(x) \leq F(y) + \nabla F(x)^T (x-y) - \frac{\mu}{2}\|x-y\|^2$$

$$F(x) \geq F(y) + \nabla F(x)^T (x-y) - \frac{L}{2}\|x-y\|^2$$

$$\mathbb{E}\|\nabla F_k(\mathbf{w}_t^i(k), \xi_t^i(k)) - \nabla F(\mathbf{w}_t^i(k))\|^2 \leq \rho_1$$

$$\mathbb{E}[F(\mathbf{w}_T) - F(\mathbf{w}^*)] \leq \nu^T [F(\mathbf{w}_0) - F(\mathbf{w}^*)] + (1-\nu^T) Z(\lambda, E_{th}, \delta)$$

Convergence speed

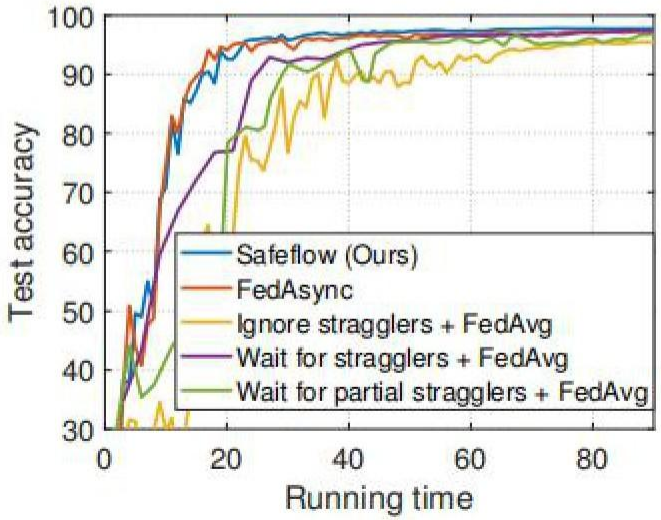
Error

Sageflow algorithm

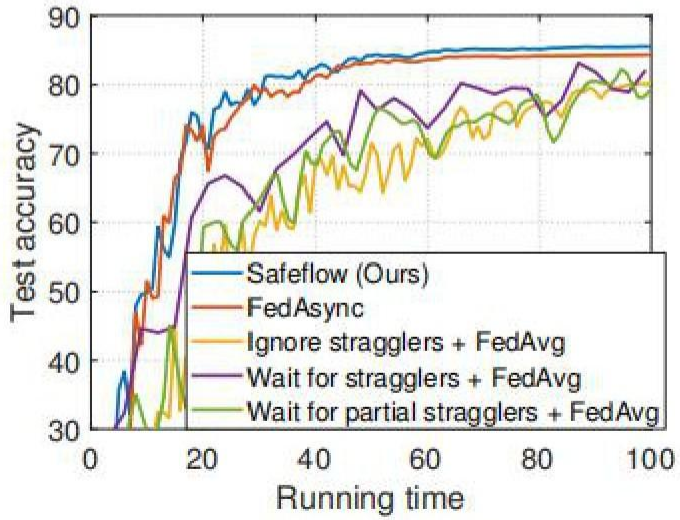
- Datasets: MNIST, FMNIST, CIFAR10
  - 2% as public data
- Models: CNN(2conv+2fc), CNN(2conv+1fc), CNN(VGG-11)
  - ignore batchnorm
- FL setting: 100 clients, two classes for each client, 5 local epochs, batch size of 10

Sageflow algorithm

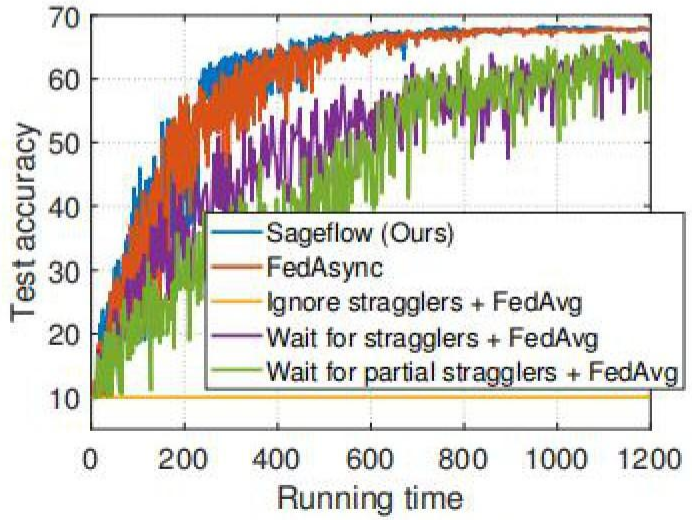
- Only stragglers: 10% participants
- Baselines: FedAvg(waiting, ignoring, waiting 50%), FedAsync
- Settings: uniform delay of [0,1,2] global rounds
- Ignoring lose significant data converges to a **suboptimal point**
- Waiting(all, 50%) requires the **largest running time** until convergence



(a) MNIST



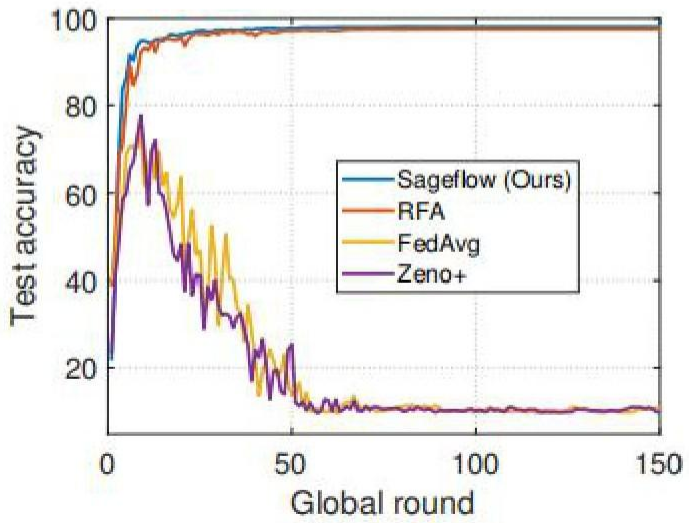
(b) FMNIST



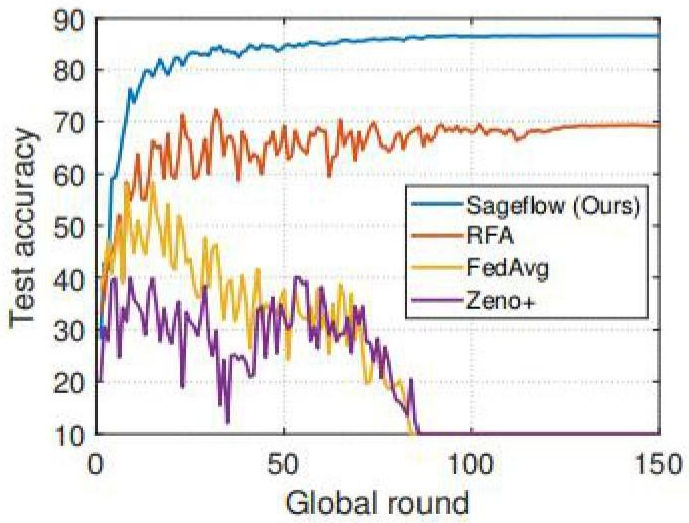
(c) CIFAR10

Sageflow algorithm

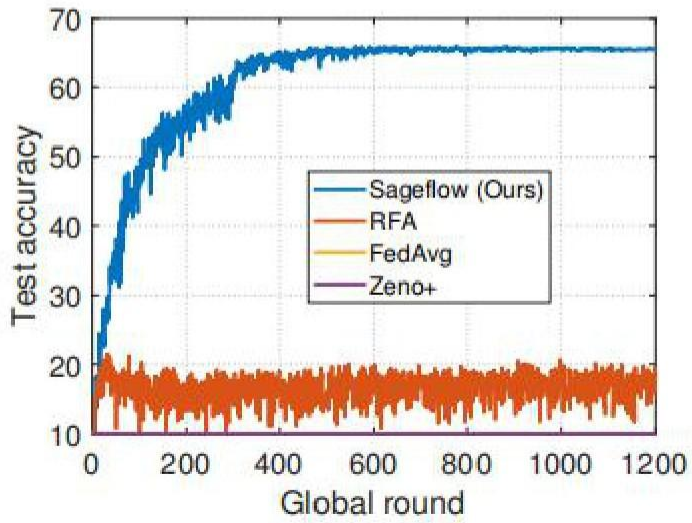
- Only adversaries: 20% participants
- Baselines: RFA, FedAvg, synchronized Zeno+, Multi-Krum
- Attacks: model(-0.1w), data(label-flipping), backdoor(model replacement, pixel-pattern attack)
- FedAvg **does not work well** on all datasets
- RFA: **complex model** led to worse performance
- Zeno+: bad on poisoning **but** good for backdoor
- Sageflow: **slow down** posioning



(a) MNIST



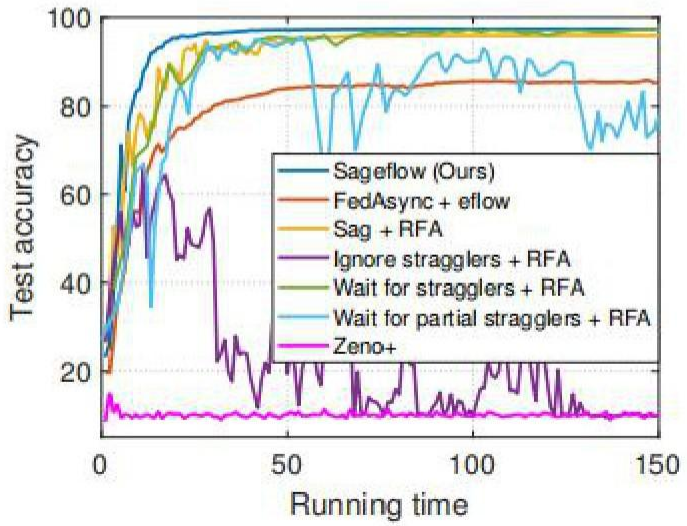
(b) FMNIST



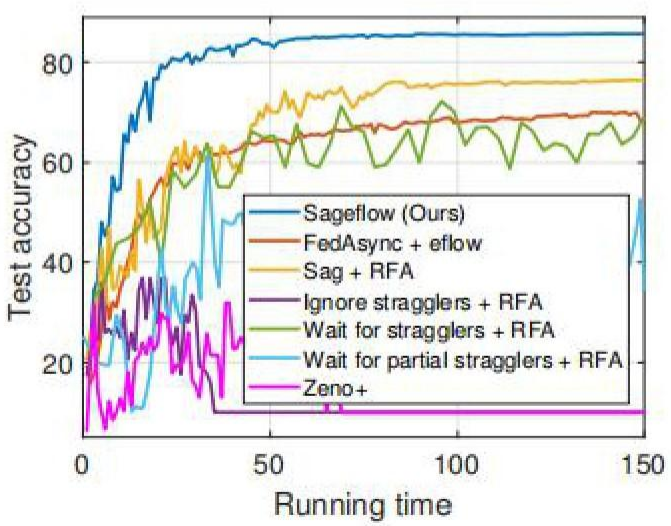
(c) CIFAR10

Sageflow algorithm

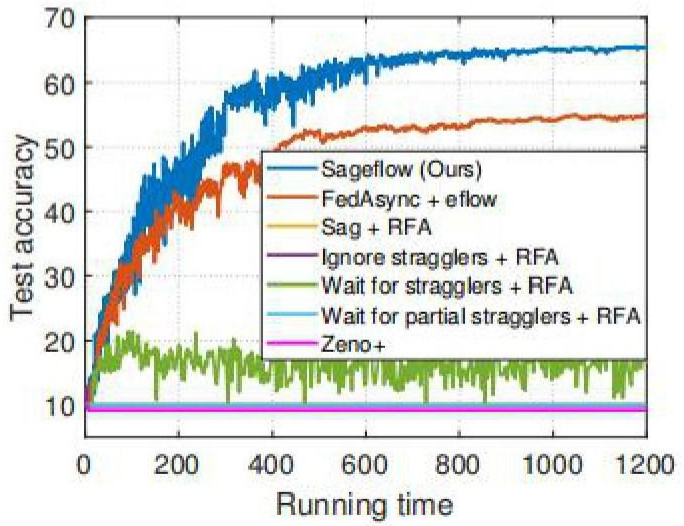
- Stragglers + adversaries: 20%(model/data), 10%(backdoor) participants
- Baselines: asynchronized Zeno+, Multi-Krum
- Zeno+: **does not perform well**(ignore staleness & entropy)
- Waiting + RFA: suffer from **straggler**
- Ignoring/Sag + RFA: poor(**high attack ratio**)
- eflow/RFA + FedAsync: poor(**one-by-one arrivals**)



(a) MNIST



(b) FMNIST



(c) CIFAR10

Sageflow algorithm

- Sageflow: robust FL scheme handle both **stragglers and adversaries**
  - staleness-aware grouping: stragglers
  - entropy-based filtering: model poisoning
  - loss-weighted averaging: data poisoning + backdoor
- Theoretical convergence analysis
- Extensive experimental results
- Future issues: Sageflow + secure aggregation

Sageflow 小结

模块	对付什么	做法
按过时程度分组聚合	掉队者	固定时间聚合；迟到的下次再交，按迟到轮数降权
熵过滤	模型投毒	公共数据上预测熵过高 → 丢弃
按损失加权	数据投毒、后门	公共数据上损失越大 → 权重越小

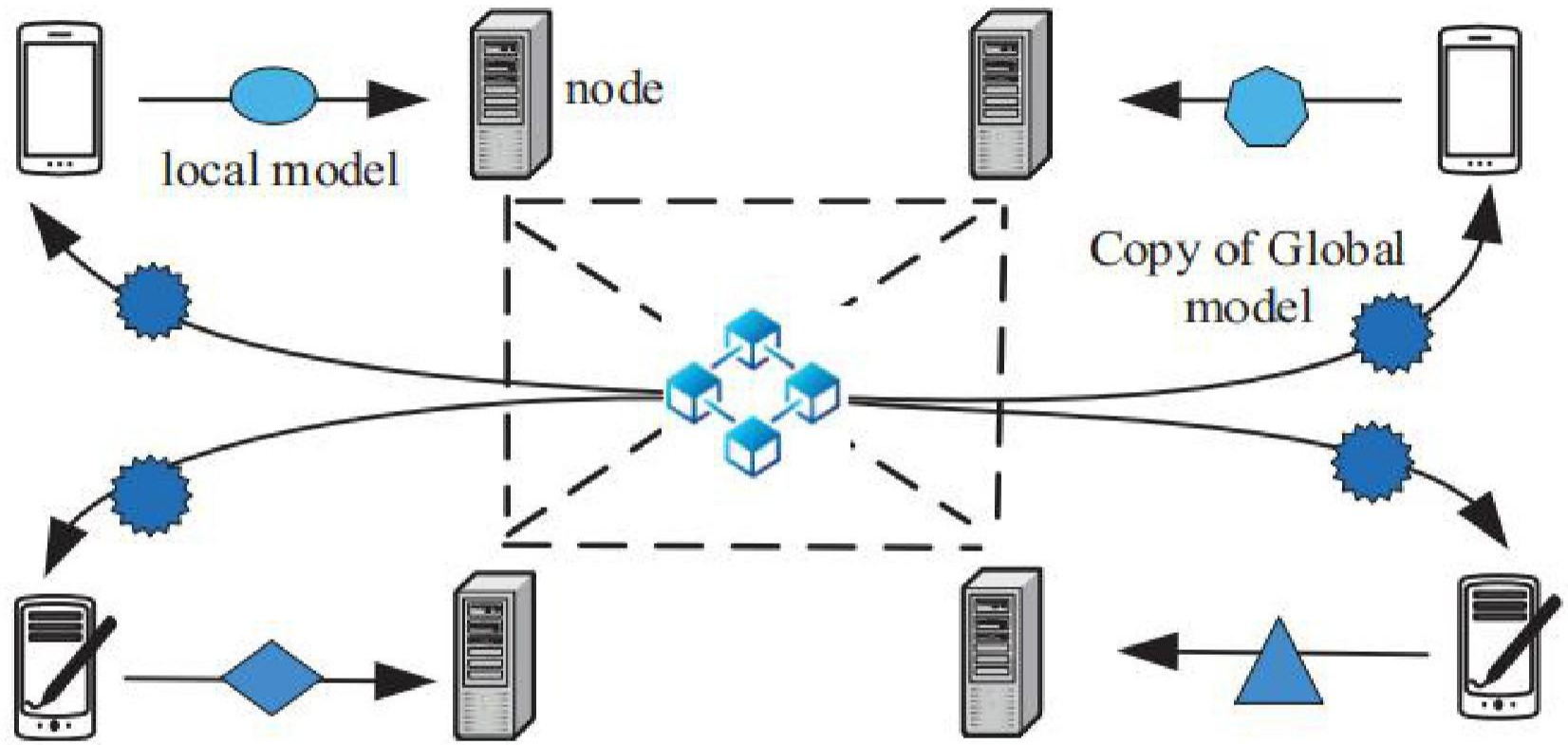
实验设置	结论
只有掉队者	直接丢 → 收敛到较差的点；全等 → 最慢；Sageflow 又快又准
只有攻击者	FedAvg 崩溃；RFA 在复杂模型上差；Zeno+ 防后门、不防投毒；Sageflow 都能压住
两者都有	基线都明显变差；消融实验说明三个模块各有用处

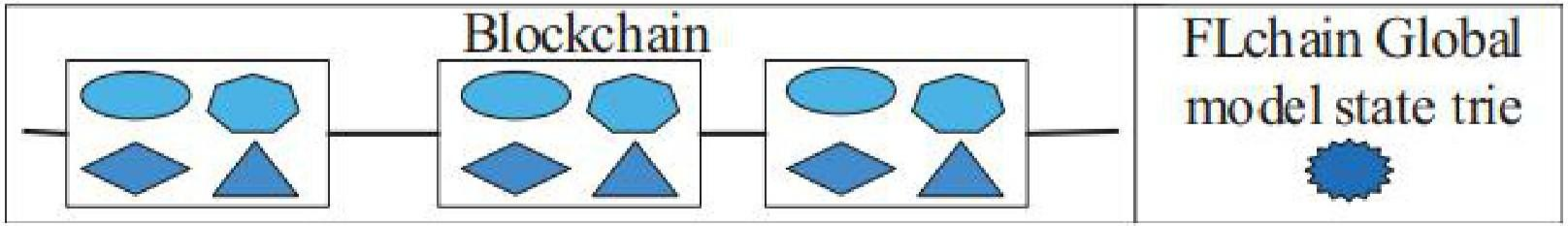
> **局限**：依赖服务器公共数据；服务器要看到每个模型，与"安全聚合"冲突。

FLChain

FL + Blockchain

- complete dependency on the reliability of a central server for storage and computation of the global model update
- Any malicious activity leads to flawed global model update which is detrimental for accuracy of subsequent local model updates





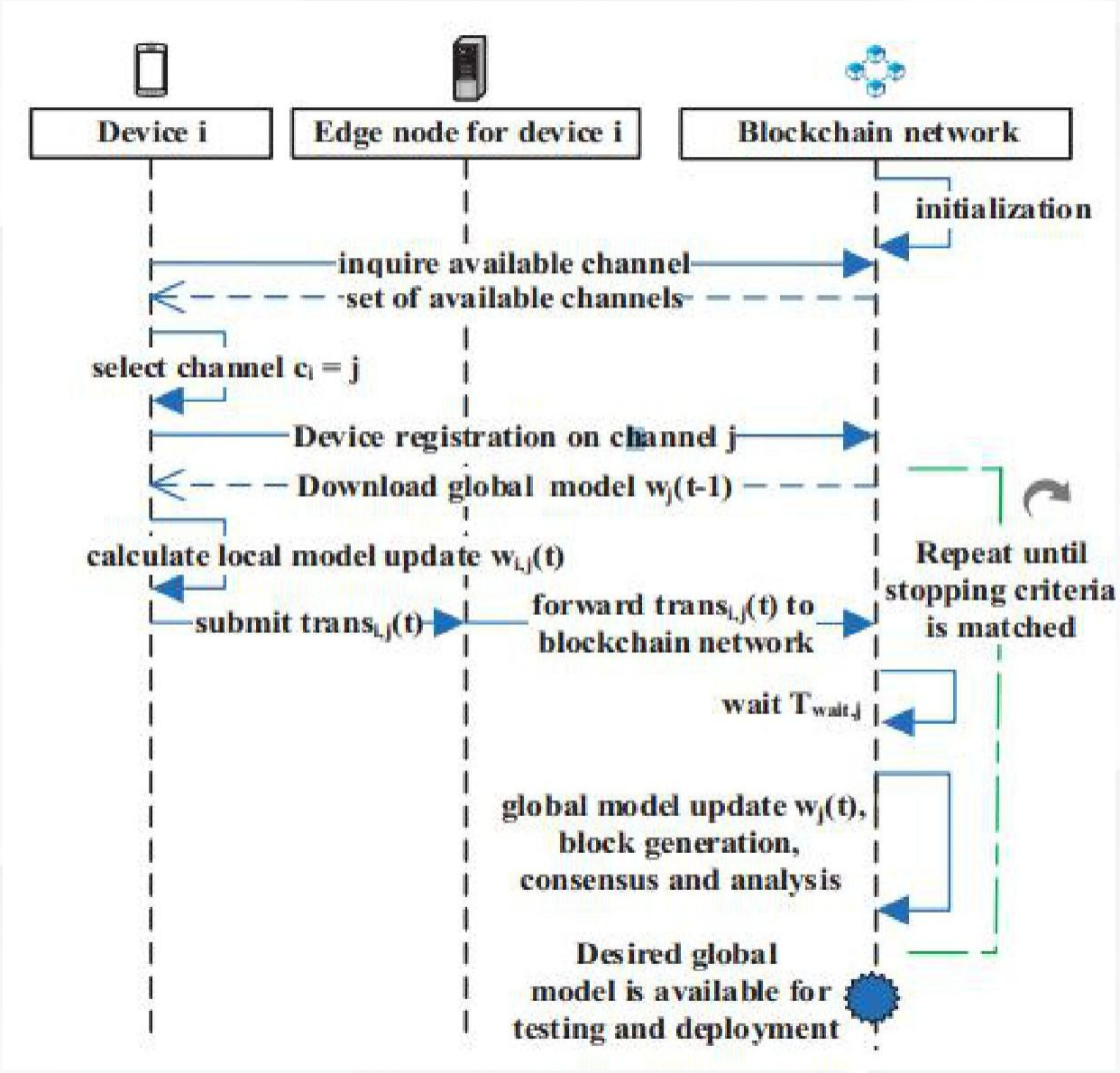
FL + Blockchain

- Initialization
- Channel Inquiry
- Channel Selection
- Device Registration
- Local Model Update
- Transaction Pool
- Global Model Update
- Consensus Protocol
- Analysis

Algorithm 1 : FLchain operation for channel j

Setup channel j for global model M_j
initialization: $t = 0$; $w_j(0), w_{i,j}(0) \in [0, w_{j,max}]$;
for all $i \in D_j$ **do in parallel**
    Inquire available channels
    Select channel $c_i = j \in C \triangleq \{1, 2, 3, \ldots, C_n\}$
    Register device i to channel j
end for
while $\|w_j(t) - w_j(t-1)\|_2 \leq \varepsilon_{threshold,j}$ **do**
    for all $i \in D_j$ **do in parallel**
        Download $w_j(t)$ from channel j to device i
        $t \leftarrow t + 1$; $w^0_{i,j}(t) = w_j(t)$;
        for $v = 1, ..., V$ **do**
            $w^v_{i,j}(t) = w^{v-1}_{i,j}(t) - \eta \nabla \Phi$, and Eq. (6)
        end for
        $w_{i,j}(t) = w^V_{i,j}(t)$, Generate $trans_{i,j}(t)$ and forward to blockchain network
        Wait for notification from channel j
    end for
    Calculate $w_j(t)$ using Eq. (7)
    global model state trie updation, block generation and consensus
end while

FL + Blockchain



FLChain 小结：中心服务器 vs 区块链

	中心服务器（FedAvg）	区块链（FLChain）
全局模型存在哪	一台服务器	链上，多个节点共同维护
信任假设	必须信任服务器	不依赖单一可信方
被篡改的风险	服务器出错或作恶 → 全局模型坏掉	需经共识确认，可追溯、难篡改
速度与能耗	快	共识带来延迟和能耗

> 联邦学习需要很多轮、要求快，与区块链共识的开销存在矛盾。

04

Other algorithms

- 联邦学习效率
- 联邦学习安全
- 联邦学习异构性
- 其他论文

实验室近年的联邦学习工作

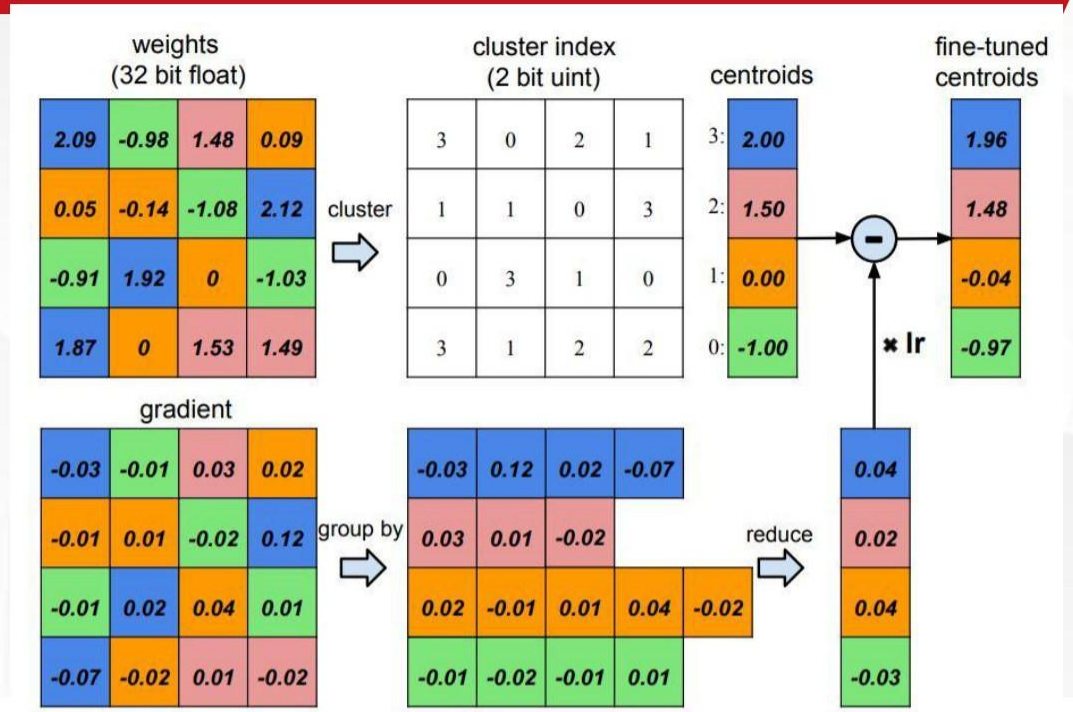
工作	方向	怎么做	效果
pFedEff IEEE TCDS 2023	效率：通信	掩码剪枝：去掉绝对值小的参数；上传/下载时去掉对损失贡献小的更新	通信减少约 94%，准确率下降 < 6%
自动流水线并行 IEEE JSAC 2023	效率：调度	GBDT 预测子任务耗时 → 遗传算法决定放在哪台设备 → 贪心排时间表	延迟低于模型/流水线并行，能用小内存设备
Siren ACM SoCC 2021	安全：拜占庭	客户端主动报警：收到的全局模型比自己还差就报警；服务器据此排除	恶意客户端占 80% 仍能防御；被攻击后能恢复
FedALA AAAI 2023	异构：个性化	接收全局模型时不直接覆盖，学一组**逐元素权重**决定每个参数取多少全局、留多少本地	多数据集准确率最高；通信与 FedAvg 相同
FedVB 2023	异构：系统	多分类拆成多个二分类分支，每家只训练、只上传自己有标签的分支	收敛更快、准确率更高

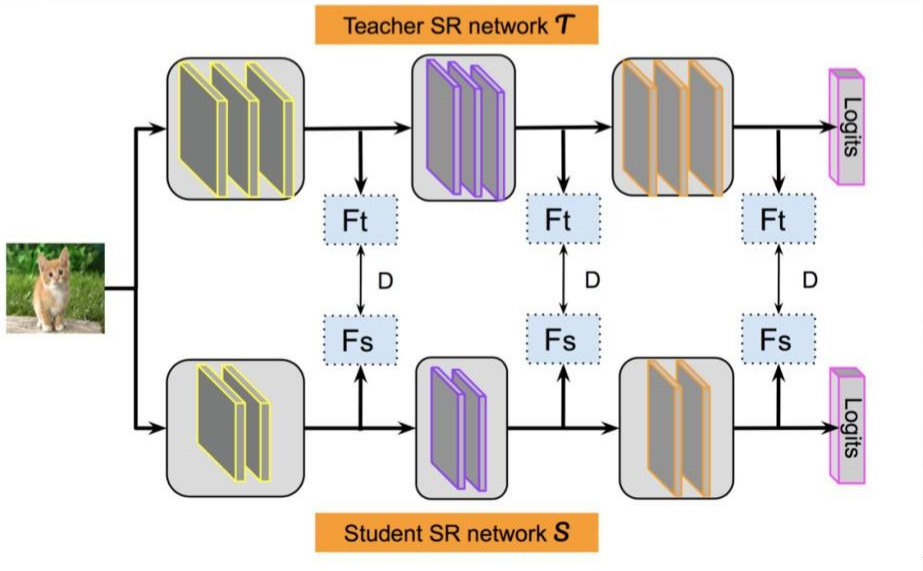
联邦学习效率

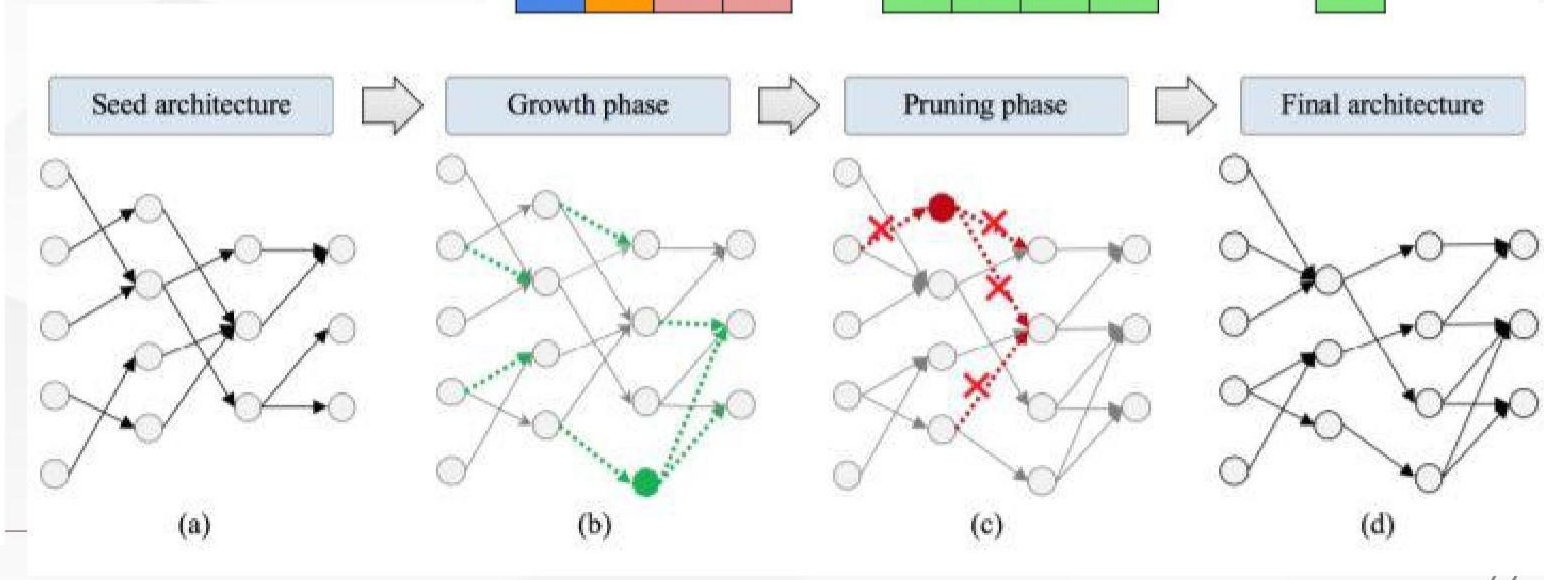
- 数据传输量大 - 模型压缩
- 网络环境复杂 - 传输调度

联邦学习效率-模型压缩

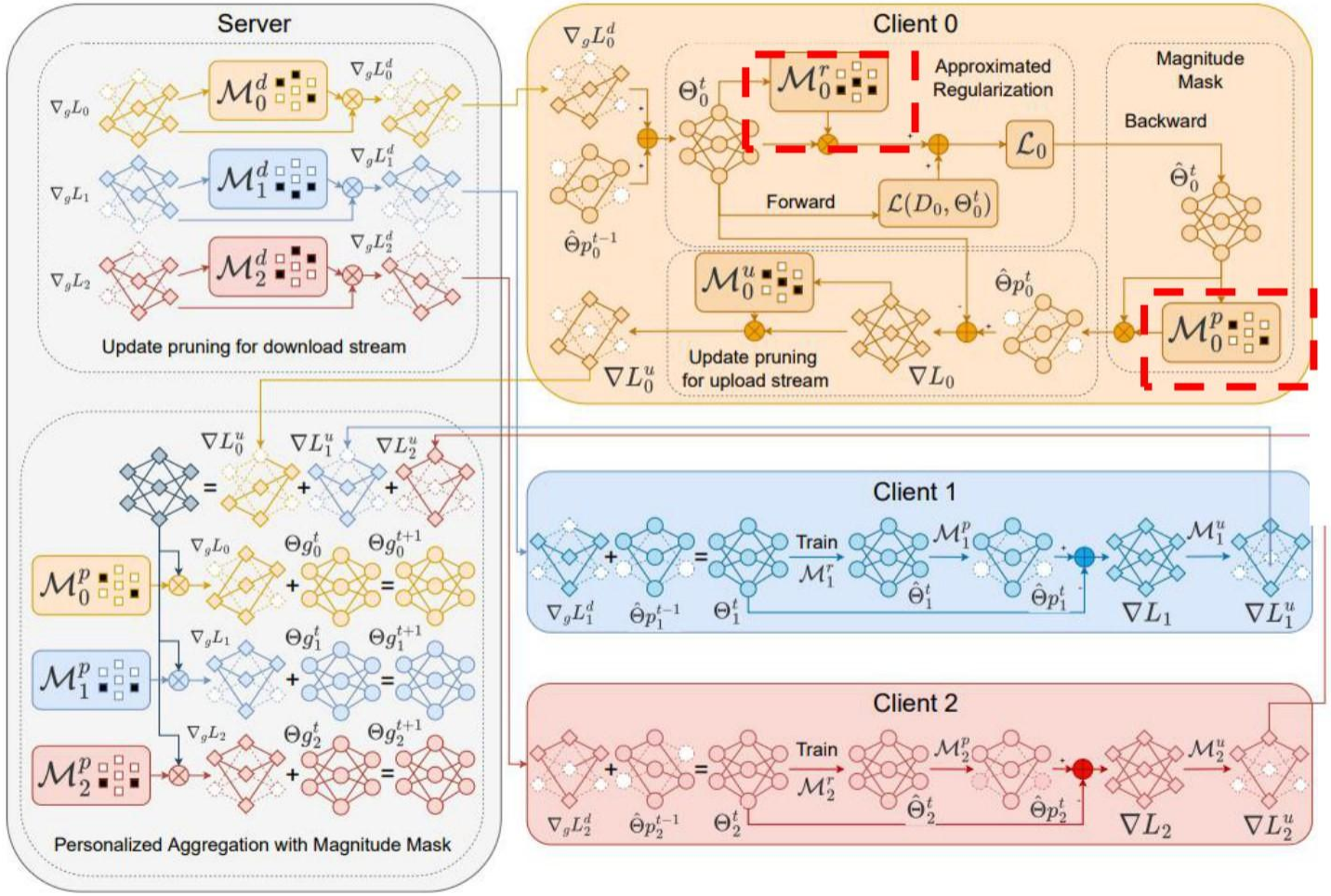
- 通信效率
  - 智能算法本身通信量大，通信开销大
- 模型压缩：**模型剪枝，模型量化，知识蒸馏**
  - 减少数据通信、计算量
  - 构造准确但小的神经网络模型







联邦学习效率



绝对值剪枝

$$\mathcal{H}(w) = |w|.$$

损失函数正则化

定位对推理结果影响较低的参数

$$\mathcal{L}_i = l(D_i, \Theta_i^t) + \lambda_1 \cdot ||\Theta_i^t \odot \mathcal{M}_i^r||_1.$$

训练后模型剪枝

过滤对推理结果影响较低的参数

$$\hat{\Theta}p_i^t \leftarrow \hat{\Theta}_i^t \odot \mathcal{M}_i^p.$$

Hongjian Shi, Jianqing Zhang, Shuming Fan, Ruhui Ma, Haibing Guan, pFedEff: An Efficient and Personalized Federated Cognitive Learning Framework in Multi-agent Systems, IEEE Transactions on Cognitive and Development Systems, 2023.

联邦学习效率

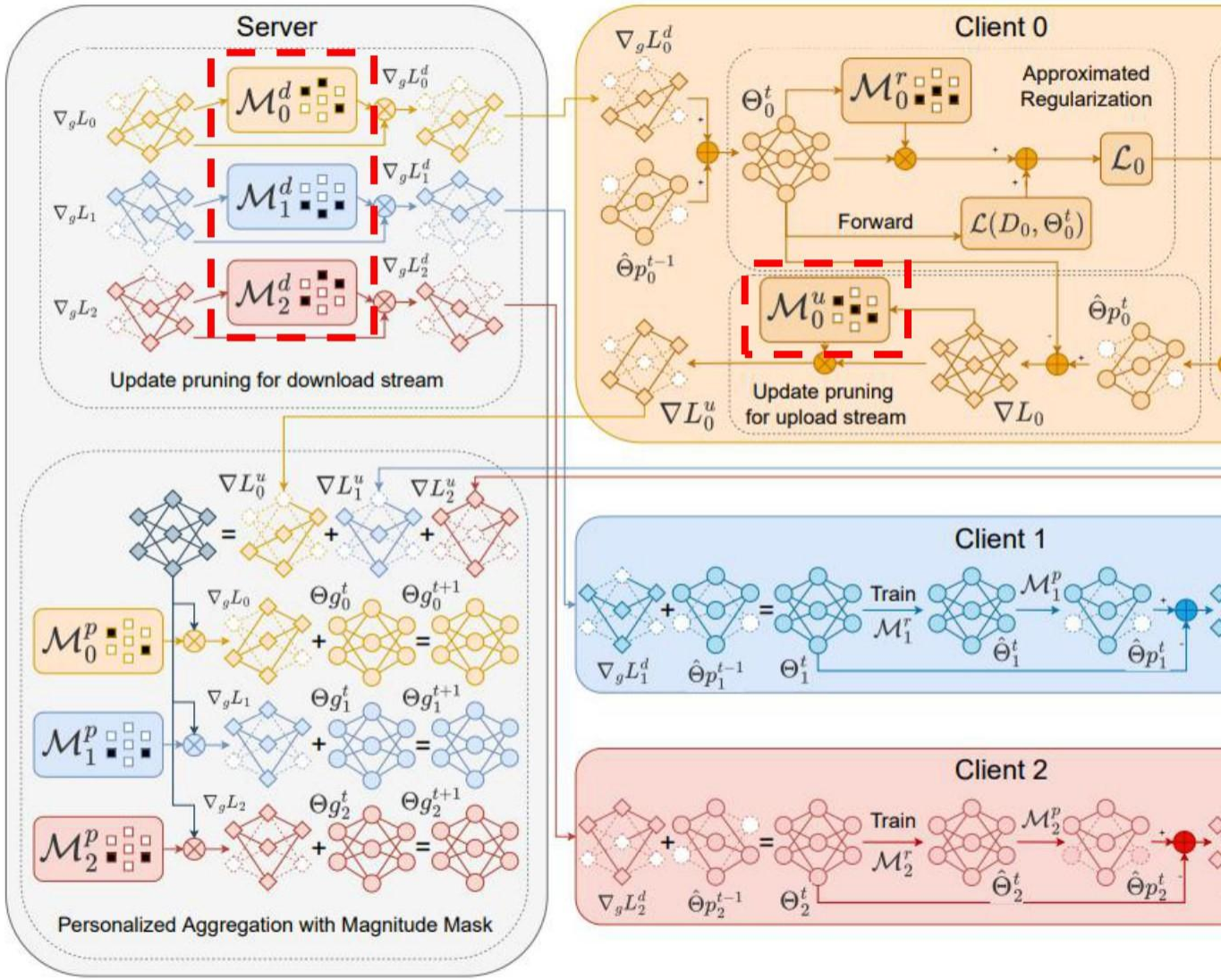
- 重要性剪枝

$$\begin{aligned}\mathcal{I}(w) &= |\mathcal{L}(D, \Theta_{w\to 0}) - \mathcal{L}(D, \Theta)| \\ &= |\frac{\partial \mathcal{L}(D,\Theta)}{\partial w} \cdot (0 - w) + o(w^2)| \\ &= |\frac{\partial \mathcal{L}(D,\Theta)}{\partial w} \cdot w|.\end{aligned}$$

- 上传/下载传输剪枝
- 过滤更新量较小的上传/下载模型更新

$$\nabla_g \mathcal{L}_i^u \leftarrow \nabla \mathcal{L}_i \odot \mathcal{M}_i^u.$$

$$\nabla \mathcal{L}_i^d = (\sum_{j=1}^{M} k_j \cdot \nabla_g \mathcal{L}_i) \odot \mathcal{M}_i^d$$



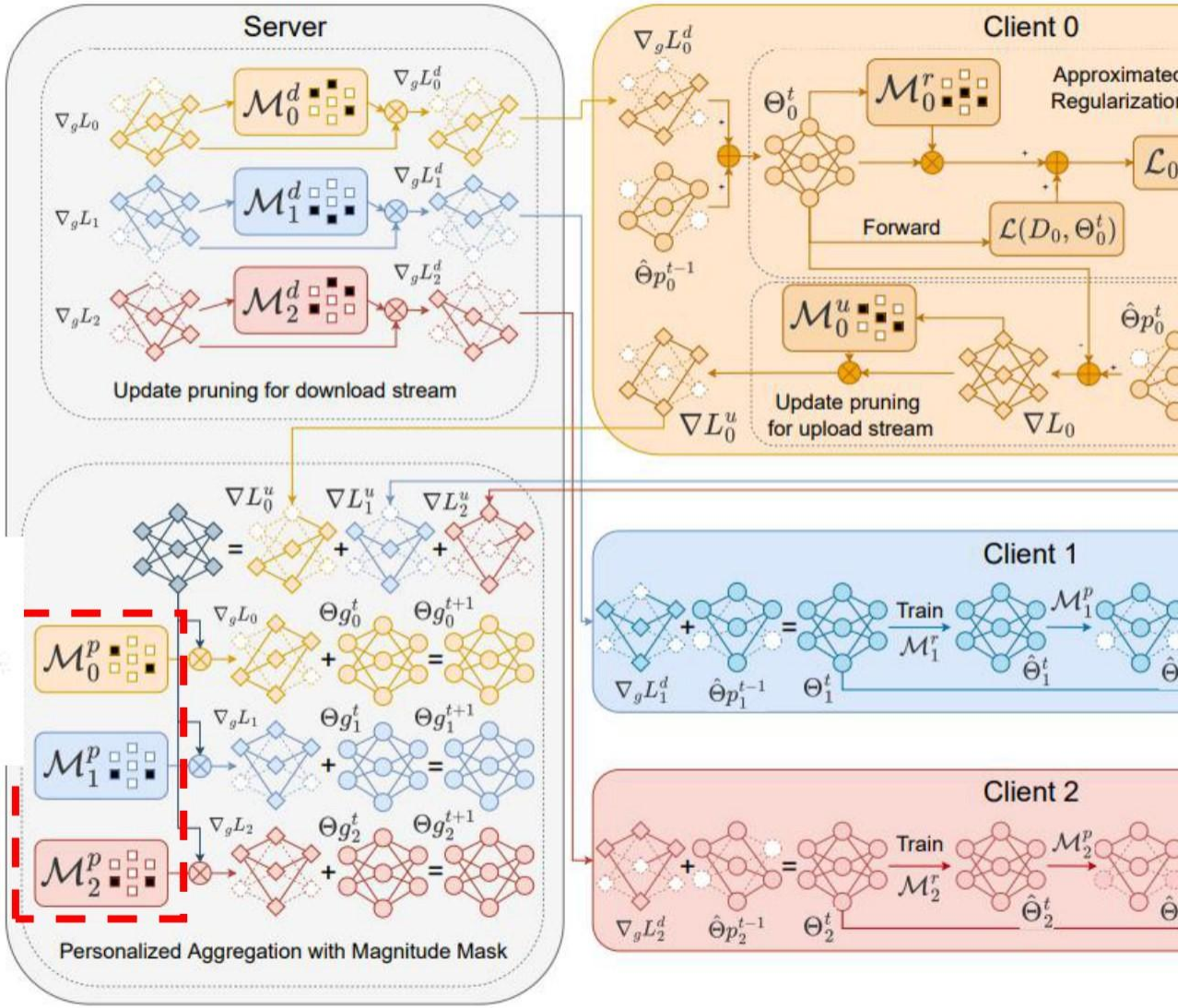
Hongjian Shi, Jianqing Zhang, Shuming Fan, Ruhui Ma, Haibing Guan, pFedEff: An Efficient and Personalized Federated Cognitive Learning Framework in Multi-agent Systems, IEEE Transactions on Cognitive and Development Systems, 2023.

联邦学习效率

- 个性化模型聚合
- 只对有效更新进行聚合

$$\Theta g_i^{t+1} \leftarrow \lambda_2 \cdot \Theta g_i^t +$$

$$(1-\lambda_2) \cdot \mathcal{M}_i^p \odot \sum_{j=1}^{M} k_j \cdot (\Theta g_j^t - \eta \cdot \nabla_g \mathcal{L}_j^u),$$



Hongjian Shi, Jianqing Zhang, Shuming Fan, Ruhui Ma, Haibing Guan, pFedEff: An Efficient and Personalized Federated Cognitive Learning Framework in Multi-agent Systems, IEEE Transactions on Cognitive and Development Systems, 2023.

联邦学习效率-模型压缩

- 在大幅减少通信量（~94%）的情况下
- 尽可能保障了训练准确率（<6%）

			Dataset&Data distributions									
			MNIST					FMNIST				
			IID	DIR75	DIR50	DIR25	PAT	IID	DIR75	DIR50	DIR25	PAT
Metrics & Baselines	Acc.	FedAvg	**0.9913**	0.9907	0.9881	0.9833	0.9576	0.9142	0.9021	0.9029	0.8923	0.7795
		FedPer	0.9687	0.9569	0.9512	0.9291	0.8778	0.8490	0.8373	0.8439	0.8230	0.7334
		ADP-50	0.9900	0.9899	0.9882	0.9848	0.9593	0.9016	0.9022	0.9047	0.8920	0.8006
		Hermes-50	0.9902	0.9899	0.9883	0.9852	0.9596	**0.9152**	0.9030	0.9031	0.8930	0.7803
		SplitFed	0.9911	**0.9909**	0.9882	0.9834	0.9545	0.9134	0.9022	0.9050	0.8919	0.7680
		FCCL	0.9820	0.9808	0.9821	**0.9866**	**0.9981**	0.8831	0.9037	0.9122	**0.9254**	**0.9949**
		LotteryFL-75	0.9700	0.9700	0.9750	0.9800	0.9775	**0.9152**	**0.9275**	**0.9125**	0.9250	0.9825
		LotteryFL-50	0.9700	0.9725	0.9700	0.9775	0.9875	0.8625	0.9075	0.9075	0.9225	0.9875
		LotteryFL-25	0.9350	0.9400	0.9350	0.9525	0.9600	0.7625	0.7675	0.7075	0.7850	0.8600
		pFedEff-75	0.9908	0.9873	**0.9888**	0.9814	0.9729	0.9108	0.8887	0.8961	0.8989	0.9293
		pFedEff-50	0.9897	0.9843	0.9862	0.9786	0.9715	0.9086	0.8774	0.8867	0.8894	0.9585
		pFedEff-25	0.9863	0.9723	0.9745	0.9601	0.9745	0.9018	0.8427	0.8544	0.8557	0.9500
	Conv.	FedAvg	198	198	192	197	192	196	200	200	198	200
		FedPer	196	197	199	200	197	196	196	199	197	200
		ADP-50	148	187	**179**	162	197	174	197	198	200	192
		Hermes-50	**90**	160	193	199	198	178	184	198	177	198
		SplitFed	192	198	198	189	199	196	189	194	190	193
		FCCL	79	187	167	132	184	151	127	191	189	51
		LotteryFL-75	131	178	198	**125**	98	**70**	109	93	152	41
		LotteryFL-50	197	**153**	183	168	149	159	182	157	165	81
		LotteryFL-25	187	189	181	159	172	199	**0**	**4**	**8**	97
		pFedEff-75	197	193	196	194	**2**	200	187	195	198	**4**
		pFedEff-50	199	199	195	193	4	190	191	199	196	6
		pFedEff-25	195	200	200	200	19	184	200	199	199	16
	Comm. Vol.	FedAvg	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
		FedPer	0.0895	0.0895	0.0895	0.0895	0.0895	0.0895	0.0895	0.0895	0.0895	0.0895
		ADP-50	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
		Hermes-50	1.0037	1.0037	1.0037	1.0037	1.0037	1.0037	1.0037	1.0037	1.0037	1.0037
		SplitFed	0.1071	0.1071	0.1071	0.1071	0.1071	0.1071	0.1071	0.1071	0.1071	0.1071
		FCCL	0.1031	0.1031	0.1031	0.1031	0.1031	0.1031	0.1031	0.1031	0.1031	0.1031
		LotteryFL-75	0.7350	0.7351	0.7351	0.7351	0.7350	0.7350	0.7351	0.7351	0.7351	0.7350
		LotteryFL-50	0.4876	0.4877	0.4877	0.4877	0.4876	0.4876	0.4877	0.4877	0.4877	0.4876
		LotteryFL-25	0.2377	0.2378	0.2378	0.2378	0.2377	0.3362	0.2430	0.2575	0.2523	0.2377
		pFedEff-75	0.5277	0.5625	0.5625	0.5625	0.5149	0.5548	0.5625	0.5625	0.5625	0.5623
		pFedEff-50	0.2398	0.2500	0.2500	0.2500	0.2407	0.2500	0.2500	0.2500	0.2500	0.2500
		pFedEff-25	**0.0596**	**0.0625**	**0.0625**	**0.0625**	**0.0625**	**0.0625**	**0.0625**	**0.0625**	**0.0625**	**0.0625**

Hongjian Shi, Jianqing Zhang, Shuming Fan, Ruhui Ma, Haibing Guan, pFedEff: An Efficient and Personalized Federated Cognitive Learning Framework in Multi-agent Systems, IEEE Transactions on Cognitive and Development Systems, 2023.

联邦学习效率-模型压缩

- 不同绝对值mask下不同压缩率结果
- pre-training mask和post-training mask能够有效减少通信量，但是对准确率影响较大

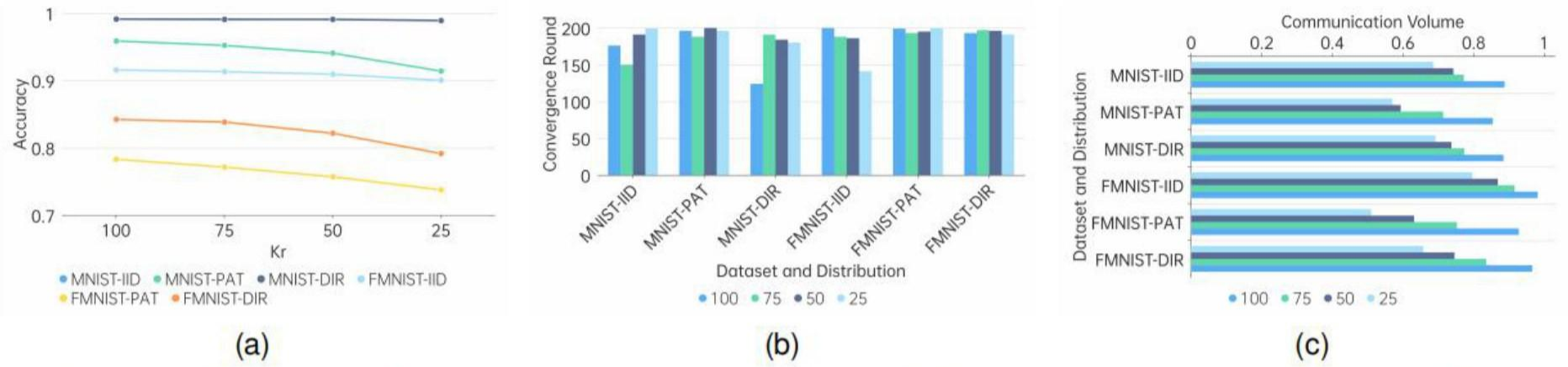


Fig. 5. The metrics when using different K_i^r for different distributions on different datasets. (a) Accuracy; (b) Convergence rate; (c) Communication Volume. We can observe that K_i^r can decrease the communication volume but introduce minor accuracy loss.

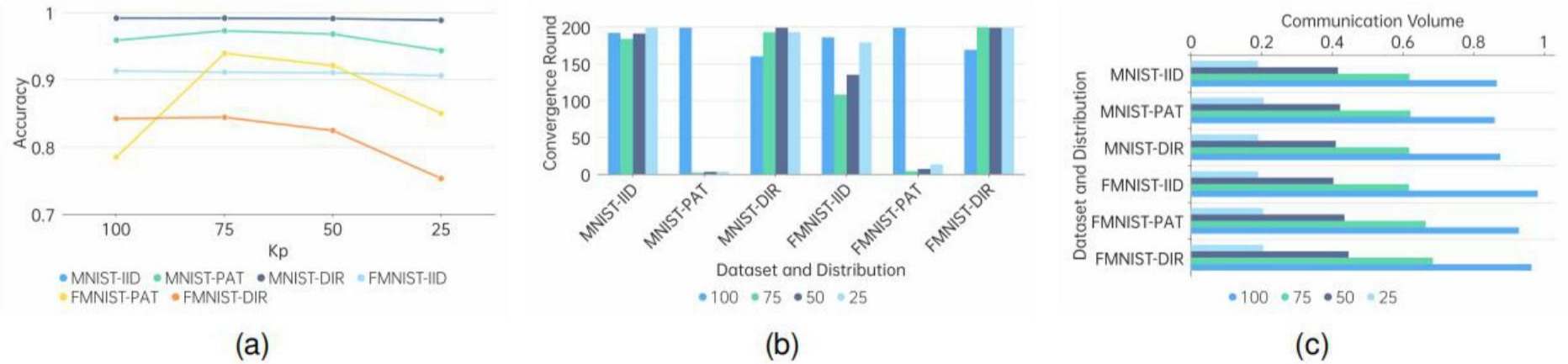
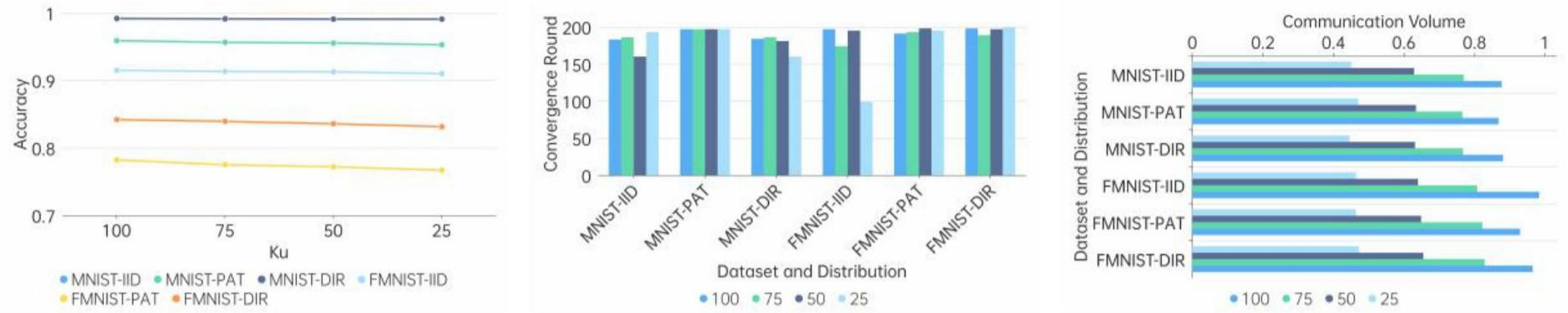


Fig. 6. The metrics when using different K_i^p for different distributions on different datasets. (a) Accuracy; (b) Convergence rate; (c) Communication Volume. We can observe that K_i^p significantly reduces the communication volume but introduce relatively large accuracy loss.

Hongjian Shi, Jianqing Zhang, Shuming Fan, Ruhui Ma, Haibing Guan, pFedEff: An Efficient and Personalized Federated Cognitive Learning Framework in Multi-agent Systems, IEEE Transactions on Cognitive and Development Systems, 2023.

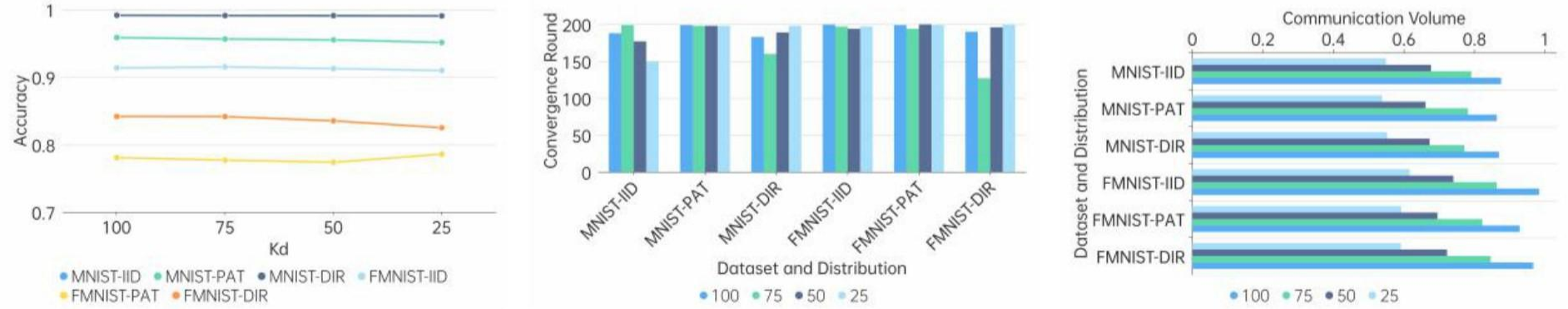
联邦学习效率-模型压缩

- 不同重要性mask下不同压缩率结果
- upload mask和download mask虽然通信量减少有限，但是能够有效维持准确率



(a) (b) (c)

Fig. 7. The metrics when using different K_i^u for different distributions on different datasets. (a) Accuracy; (b) Convergence rate; (c) Communication Volume. We can observe that K_i^u can keep the training accuracy while decreasing the communication volume.



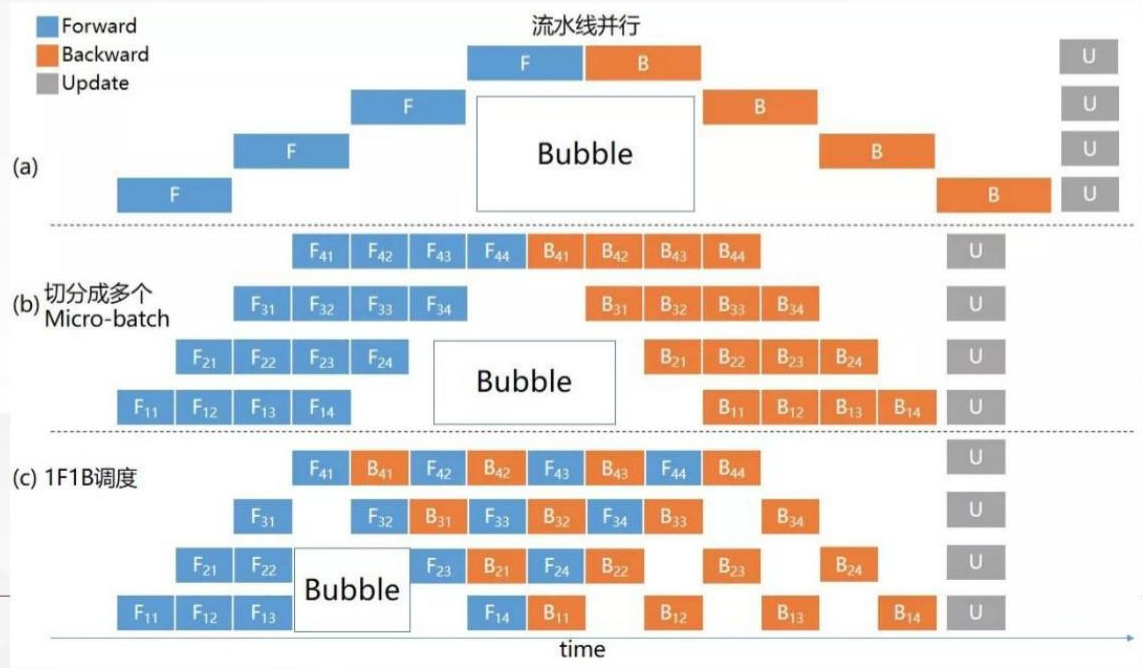
(a) (b) (c)

Fig. 8. The metrics when using different K_i^d for different distributions on different datasets. (a) Accuracy; (b) Convergence rate; (c) Communication Volume. We can observe that K_i^d can keep the training accuracy while decreasing the communication volume.

Hongjian Shi, Jianqing Zhang, Shuming Fan, Ruhui Ma, Haibing Guan, pFedEff: An Efficient and Personalized Federated Cognitive Learning Framework in Multi-agent Systems, IEEE Transactions on Cognitive and Development Systems, 2023.

联邦学习效率-传输调度

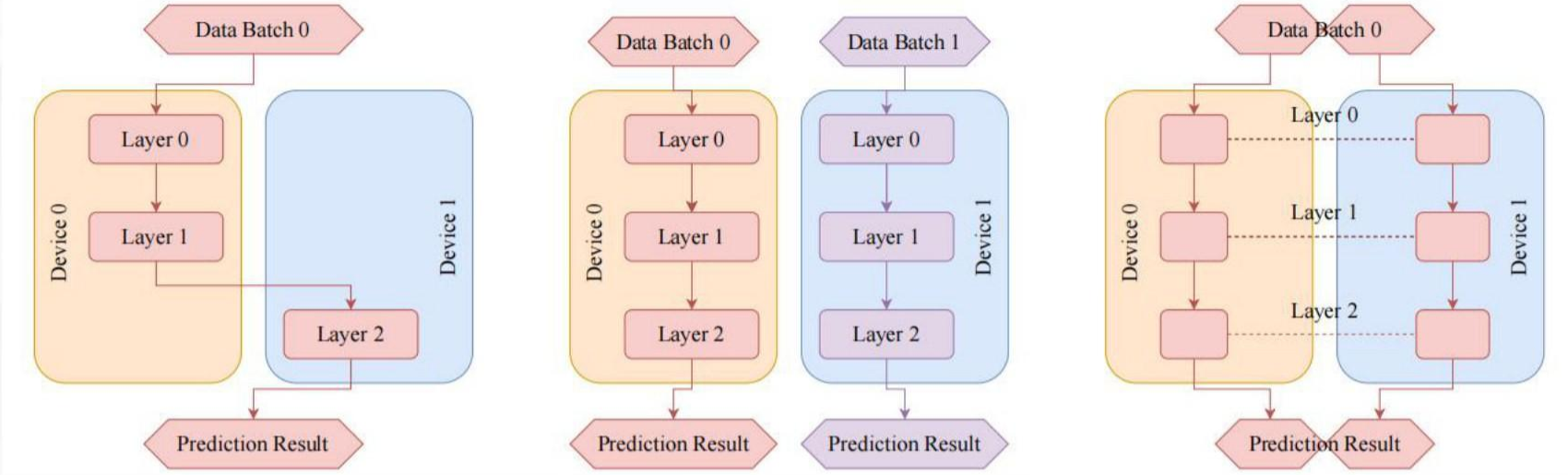
- 任务调度
  - 任务复杂，资源丰富，需要调度计算子任务
- 并行算法：
  - 计算and计算、计算and通信、通信and通信
  - 数据并行、模型并行、算子并行、优化器并行

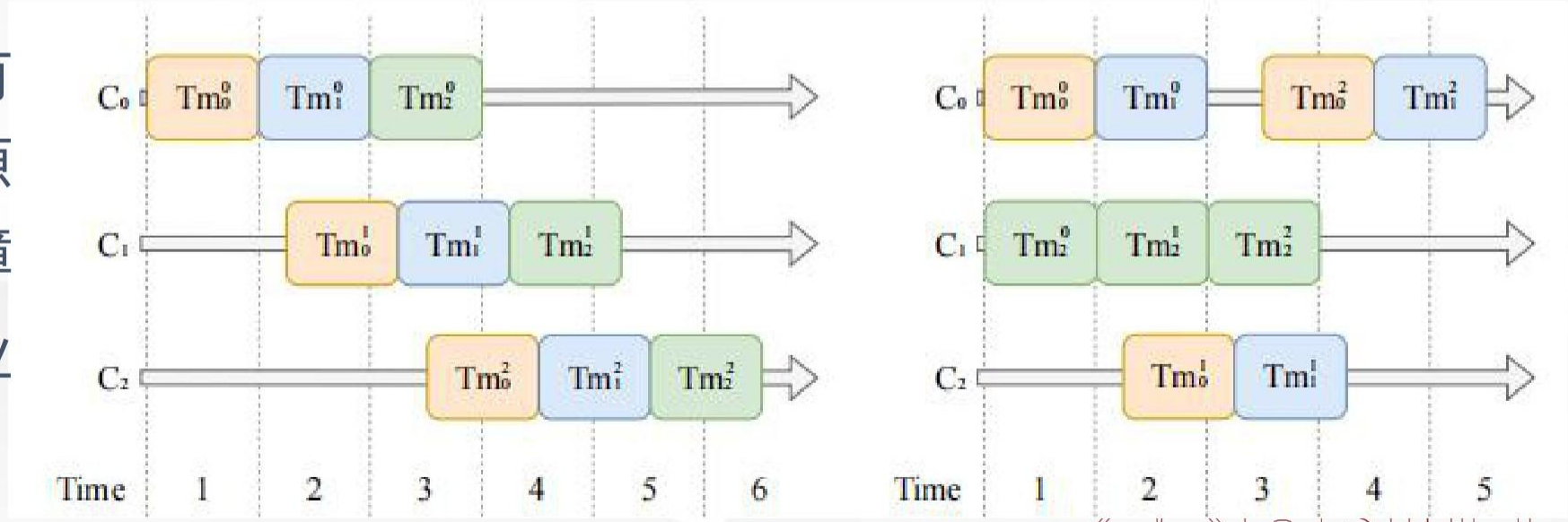


Parallel Strategy	Affiliation
Pangu[2]	Huawei
Pipedream[3]	Microsoft
Gpipe[4]	Google
Fairscale[5]	Facebook
Megatron-LM[6]	NVIDIA
DeepSpeed[7]	Microsoft
Colossal-AI[8]	HPC-AI
FlexFlow[9]	Stanford
OneFlow[10]	OneFlow

联邦学习效率-传输调度

- 并行算法
  - 模型：减少空间占用
  - 数据：减少处理延迟
  - 算子：提高处理速度
- 存在的问题
  - 单一并行算法效率有限，存在大量的资源浪费，可用性无保障
  - 多并行算法需要专业设计

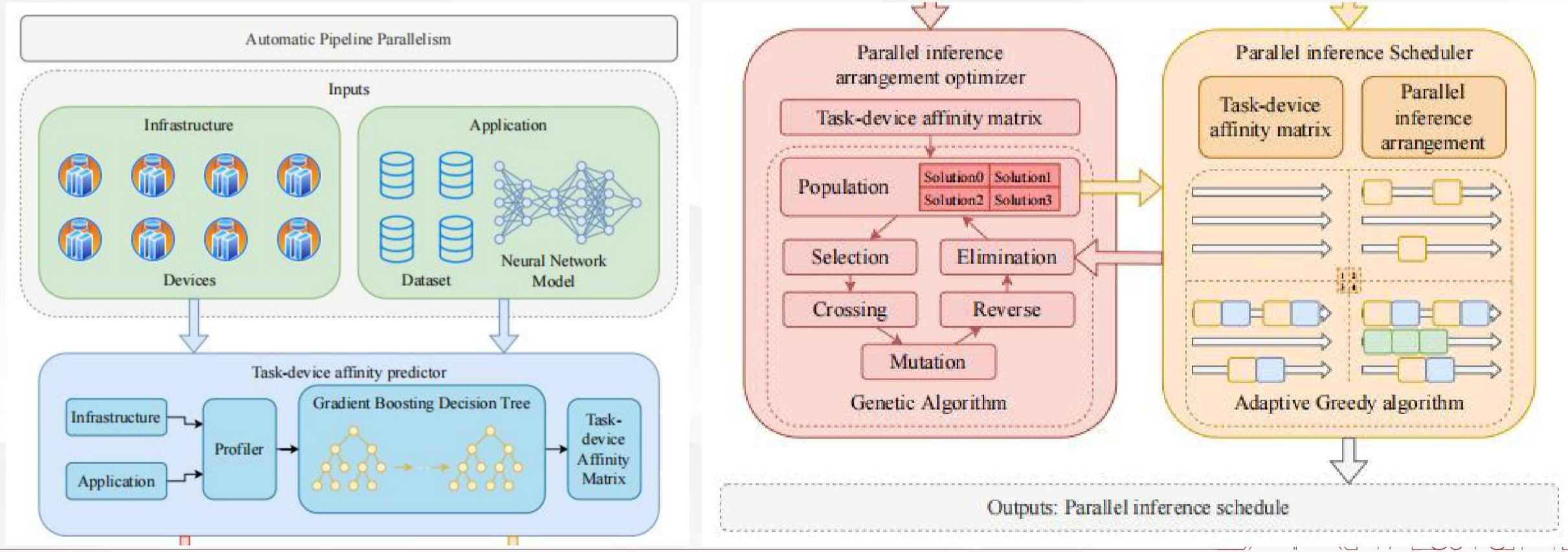




Hongjian Shi, Weichu Zheng, Zifei Liu, Ruhui Ma, Haibing Guan, Automatic Pipeline Parallelism: A Parallel Inference Framework for Deep Learning Applications in 6G Mobile Communication Systems, in IEEE Journal on Selected Areas in Communications，2023.

联邦学习效率-传输调度

- 先采用GBDT预测各任务在各设备上的运行时间，来减少profiling的用时
- 而后采用遗传算法决定任务位置，确认各个任务所在设备，以减少任务处理延迟
- 遗传算法中会用贪心算法排列各设备上的任务，得到适应度函数，并得到实际可运行的任务时间表



Hongjian Shi, Weichu Zheng, Zifei Liu, Ruhui Ma, Haibing Guan, Automatic Pipeline Parallelism: A Parallel Inference Framework for Deep Learning Applications in 6G Mobile Communication Systems, in IEEE Journal on Selected Areas in Communications，2023.

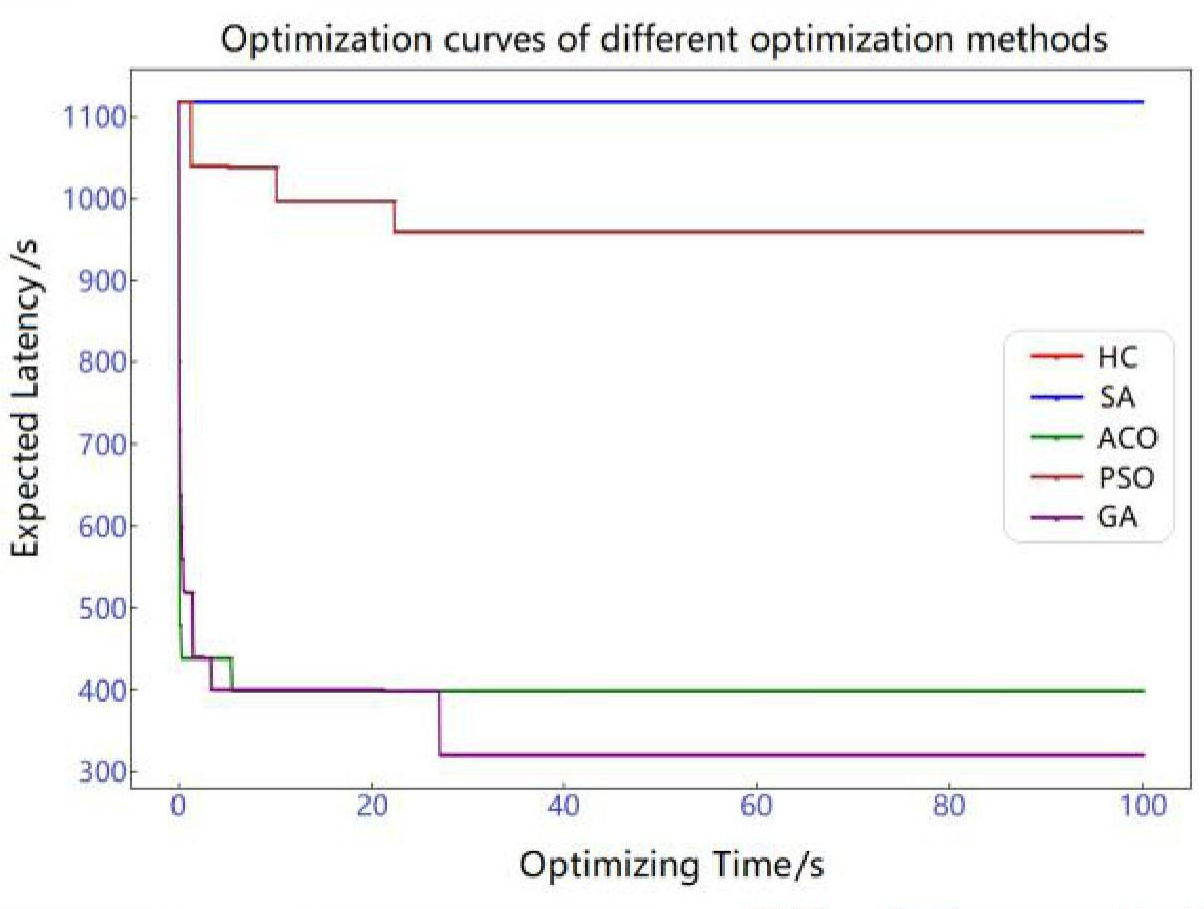
联邦学习效率

- 整体算法能够提高设备利用率
- 减少延迟提高吞吐量
- 同时能提高可用性

Method	Latency(s)	Throu.(/s)	Dev. Util.(%)	Rel.(%)
Standalone	6.1053	1310.34	12.2831	0
Model#8	9.6717	827.15	8.3009	0
Data	0.9468	8449.12	83.9462	100
Pipeline#8	2.9603	2702.43	27.0113	0
AP^2#5	1.3739	5822.89	49.9417	100

Method	Latency(s)	Throu.(/s)	Dev. Util.(%)	Rel.(%)
Standalone	6.9814	1432.37	9.8450	0
Model#5	11.9720	835.29	6.2150	0
Data	0.9734	10273.53	82.3580	100
Pipeline#5	1.7692	5652.16	39.2073	89
AP^2#5	1.7178	5821.57	45.4050	100

- 采用遗传算法的效果和收敛性较好



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联邦学习效率

- GBDT预测方法准确率明显由于其他预测方法
- 并且由于是线下训练，训练时间不会对线上推理产生负面影响
- 主要是由于GBDT对于表格数据和feature较少的数据有着十分优越的表现

Method	Test Accuracy	Train Accuracy	Training time(ms)
GBDT	**0.7176±0.0001**	**0.8058±0.0000**	107.48±0.64
FCNN-leakyrelu	0.6583±0.0003	0.7670±0.0000	6620.46±108.63
FCNN-relu [78]	0.6719±0.0002	0.7664±0.0000	6699.71±114.25
FCNN-tanh	0.6860±0.0000	0.7674±0.0000	6674.88±95.96
FCNN-sigmoid	0.7039±0.0000	0.7624±0.0000	6574.73±106.43
ABDT [79]	0.7114±0.0000	0.8001±0.0000	70.20±1.06
RF [80]	0.7096±0.0011	0.7738±0.0004	9.56±0.57
SVR [81]	0.7166±0.0000	0.7399±0.0000	**7.97±0.52**

n_estimators	Train Accuracy	Test Accuracy	Training Time(ms)
100	0.7410±0.0000	0.7172±0.0000	**48.39±0.72**
120	0.7634±0.0000	0.7251±0.0000	58.17±0.57
140	0.7783±0.0000	**0.7266±0.0000**	67.46±0.79
160	0.7882±0.0000	0.7238±0.0000	76.44±0.68
180	0.7978±0.0000	0.7209±0.0000	86.48±1.20
200	0.8058±0.0000	0.7176±0.0001	95.27±0.77
220	0.8120±0.0000	0.7132±0.0001	106.06±4.01
240	0.8219±0.0025	0.7064±0.0029	115.31±3.25
260	0.8273±0.0020	0.7020±0.0025	125.07±4.59
280	**0.8309±0.0000**	0.6992±0.0036	132.34±1.07

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联邦学习安全

- 拜占庭攻击 - 双端协同预警
- 不可信第三方 - 区块链辅助聚合

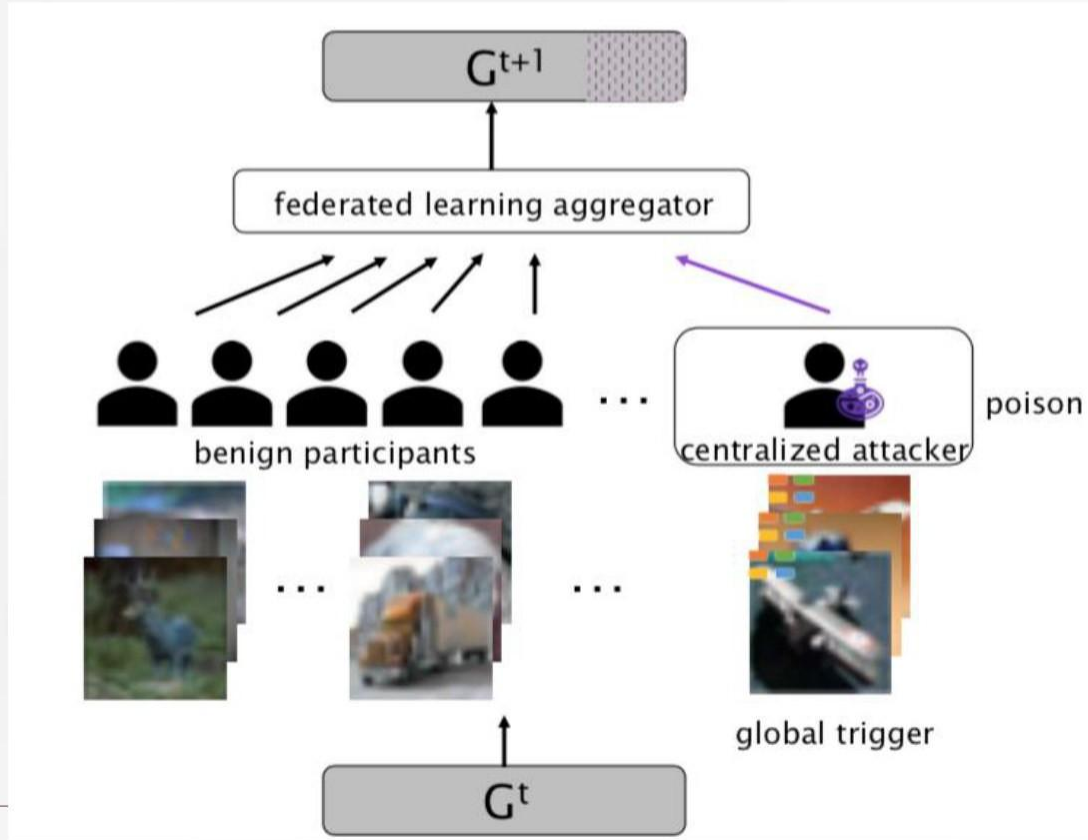
联邦学习安全---拜占庭攻击

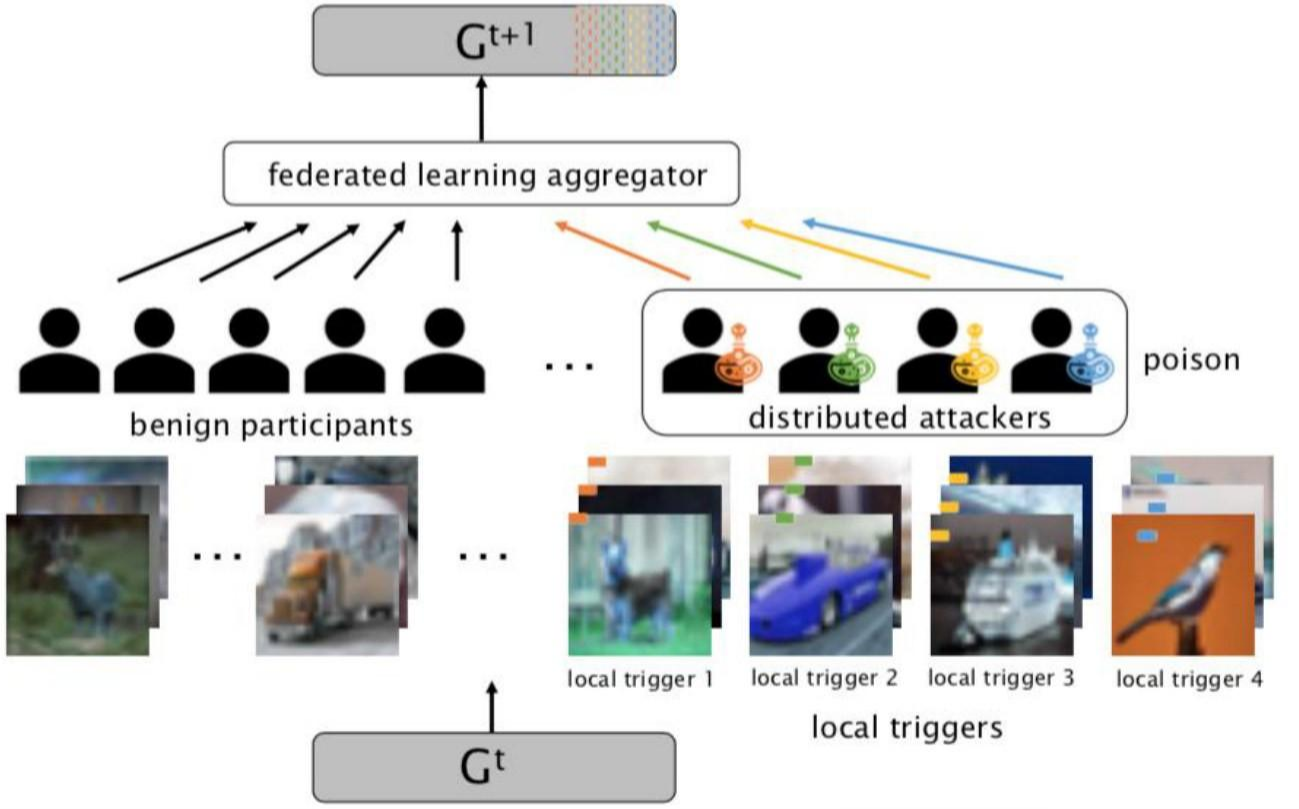
计算安全

- 拜占庭攻击，系统中存在恶意节点通过传输错误数据破坏训练模型

异常更新判断：

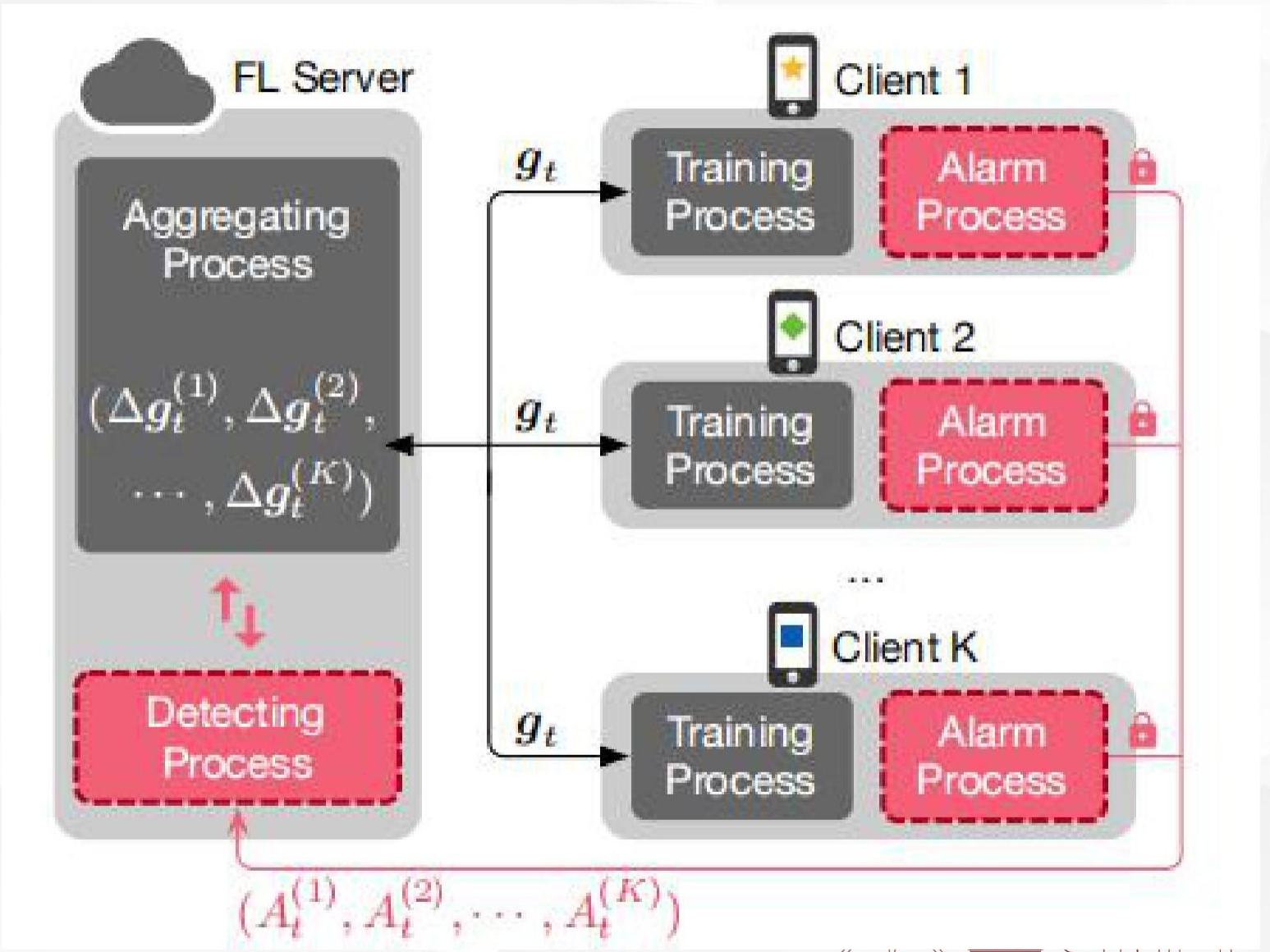
- 判断模型更新中的离群点
- 剔除离群点或是抑制离群点贡献





联邦学习安全---拜占庭攻击

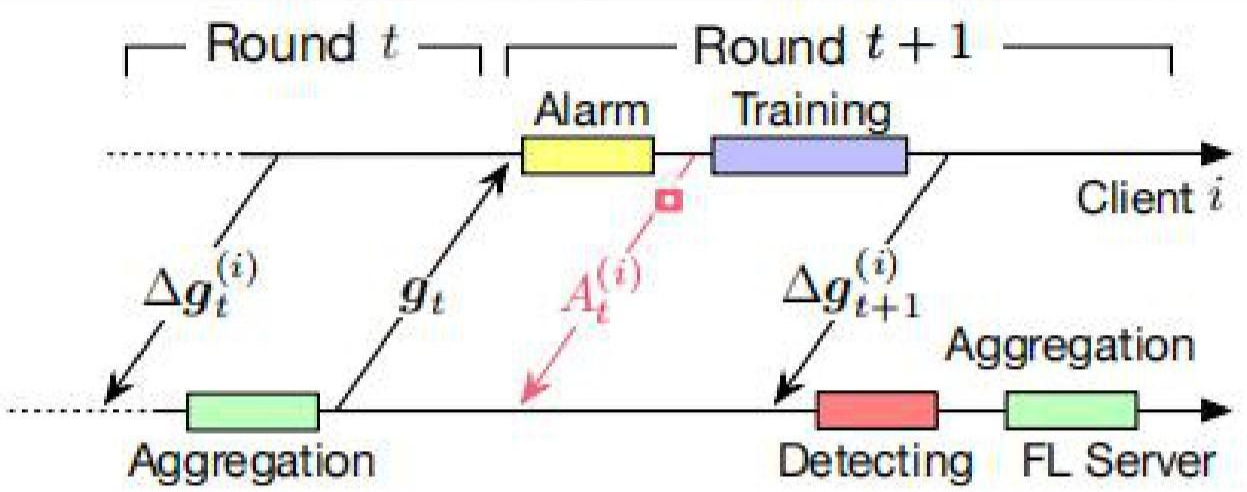
- 利用客户端信息辅助排除恶意节点
- 客户端侧拥有**示警机制**，可以和服务器侧的决策流程互相配合
- 服务器侧使用基于准确度筛查和权重分析的**决策流程**
- 系统提供两种辅助机制，分别为**惩罚机制**和**奖励机制**

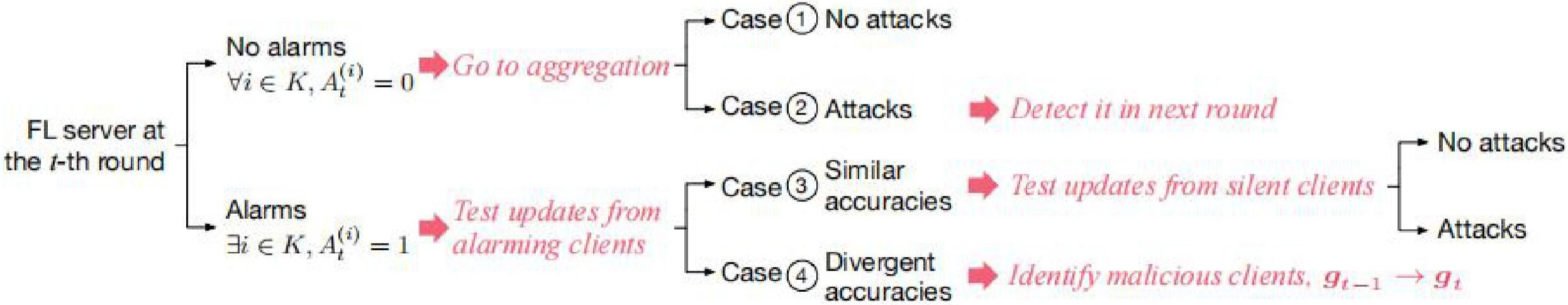


Hanxi Guo, Hao Wang, Tao Song, Yang Hua, Zhangcheng Lv, Xiulang Jin, Zhengui Xue, Ruhui Ma, Haibing Guan, Siren: Byzantine-robust Federated Learning via Proactive Alarming, Proceedings of ACM Symposium on Cloud Computing, pp. 47--60, 2021.

联邦学习安全---拜占庭攻击

- 客户端侧结构与方法流程
  - 根据全局模型和本地模型**区别**进行判断
- 服务器侧结构与方法流程
  - 根据客户端**告警**判断是否存在恶意攻击





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联邦学习安全---拜占庭攻击

- 在sign-flipping, label-flipping和targeted model poisoning攻击下可以有效收敛到一个较优准确率
- 也可以看出针对targeted model poisoning时，Siren稳定性较好

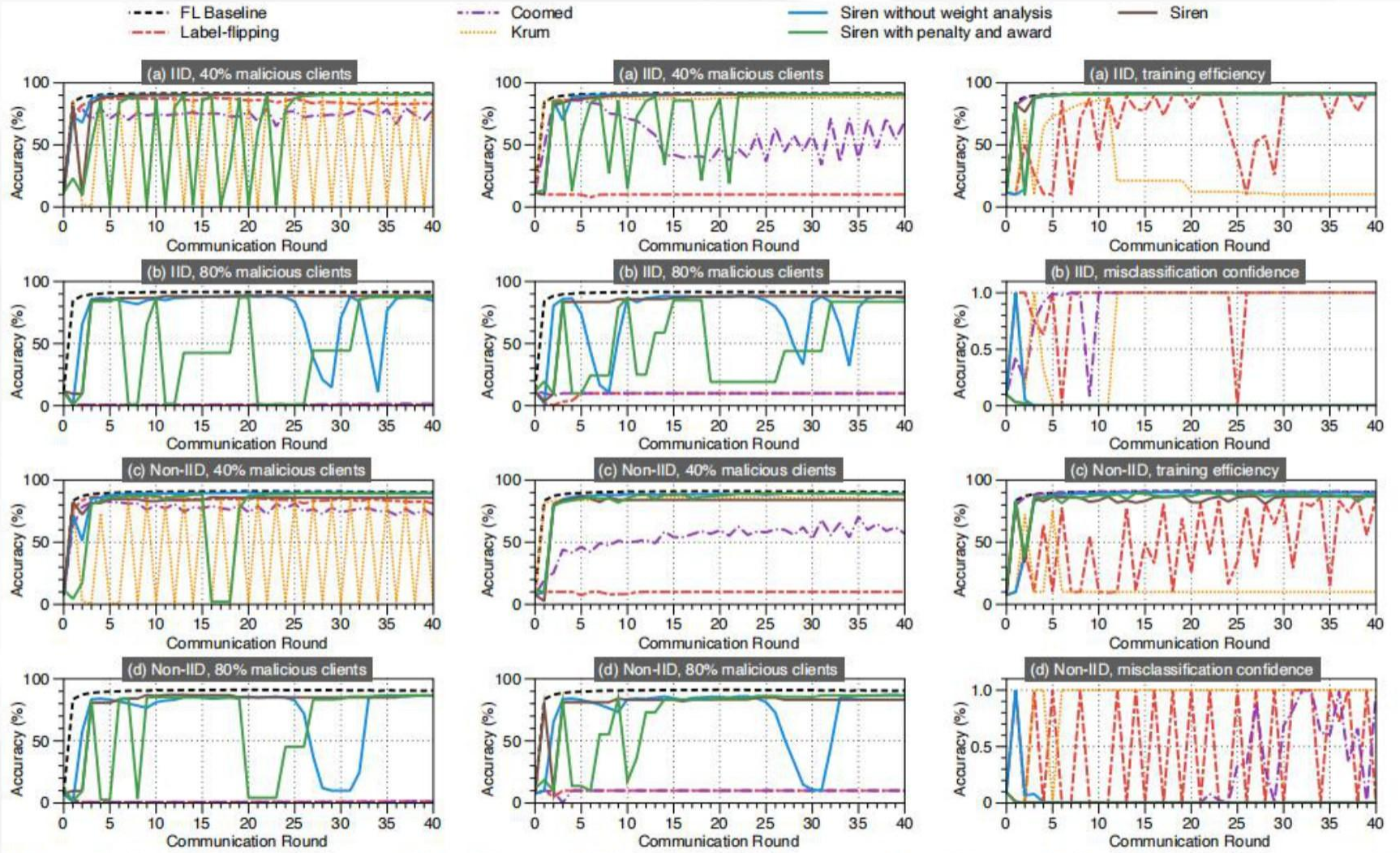

Figure 5: Training efficiency under label-flipping attack when $|K| = 10$.

Figure 6: Training efficiency under sign-flipping attack when $|K| = 10$.

Figure 7: Training efficiency and misclassification confidence under targeted model poisoning, $|K| = 10$.

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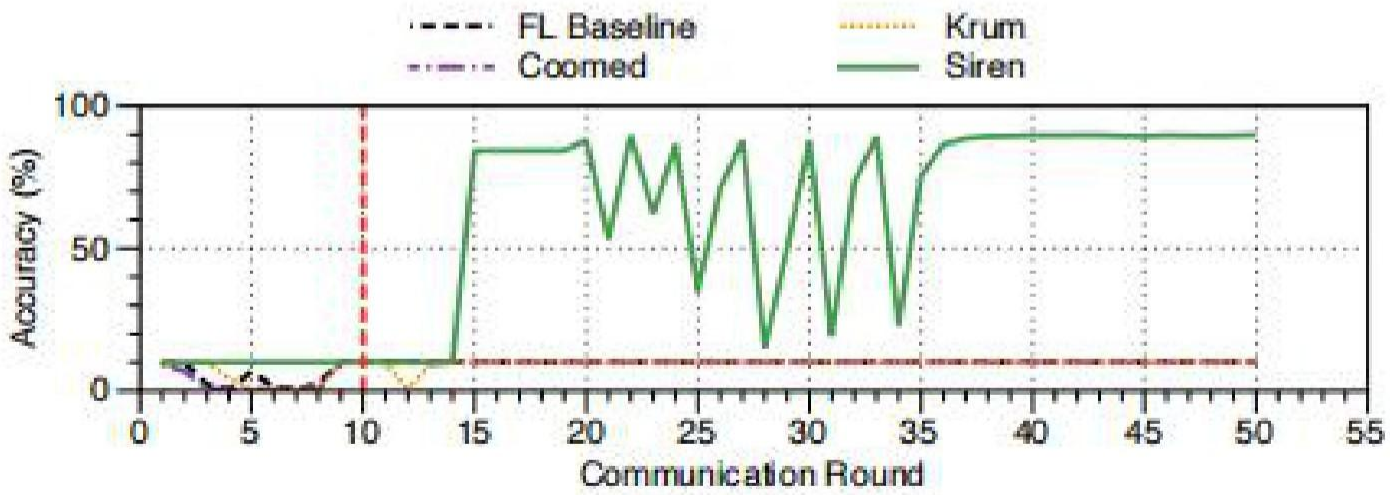
联邦学习安全---拜占庭攻击

- Siren可以在大比例恶意客户端的情况下（80%）拥有较强的防御能力
- Siren可以在系统已经遭受攻击的情况下恢复原有系统的训练准确率



Attack Type	Defense Methods	Malicious Proportion	Accuracy
None	None	0%	55.33%
Sign-flipping	None	40%	10.01%
		80%	10.01%
	Krum[1]	40%	41.53%
	Coomed	40%	34.25%
		80%	9.99%
	SIREN	40%	**51.58%**
		80%	**45.50%**
Label-flipping	None	40%	34.70%
		80%	11.68%
	Krum[1]	40%	44.72%
	Coomed	40%	37.21%
		80%	9.60%
	SIREN	40%	**49.82%**
		80%	**43.52%**

[1] Krum cannot work properly when malicious clients' proportion reaches 80%.

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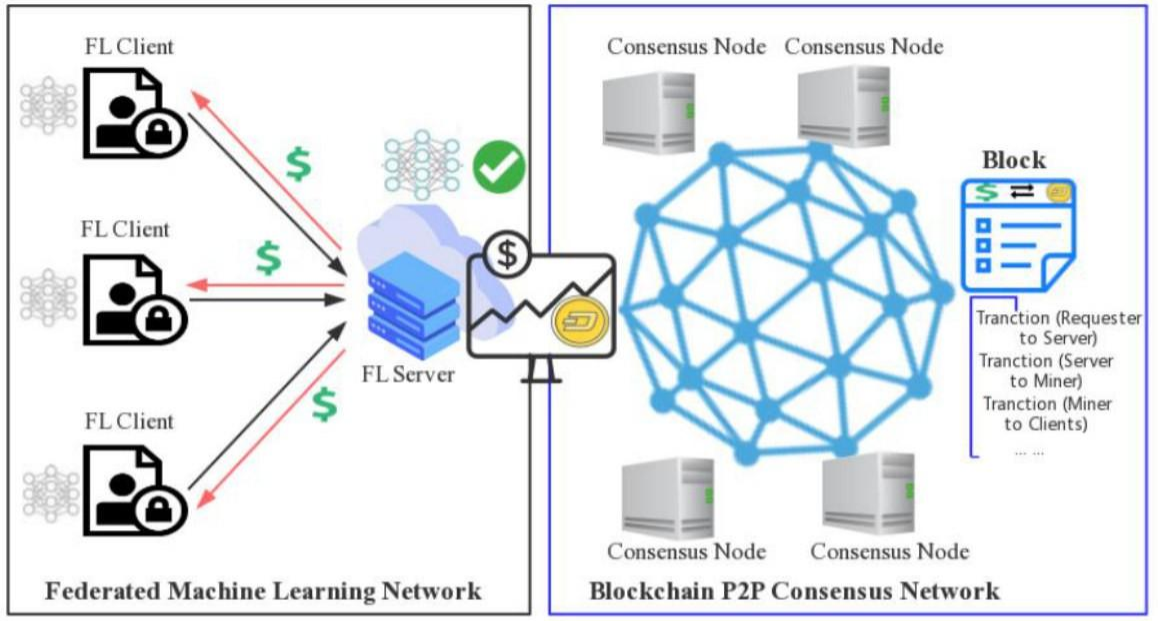
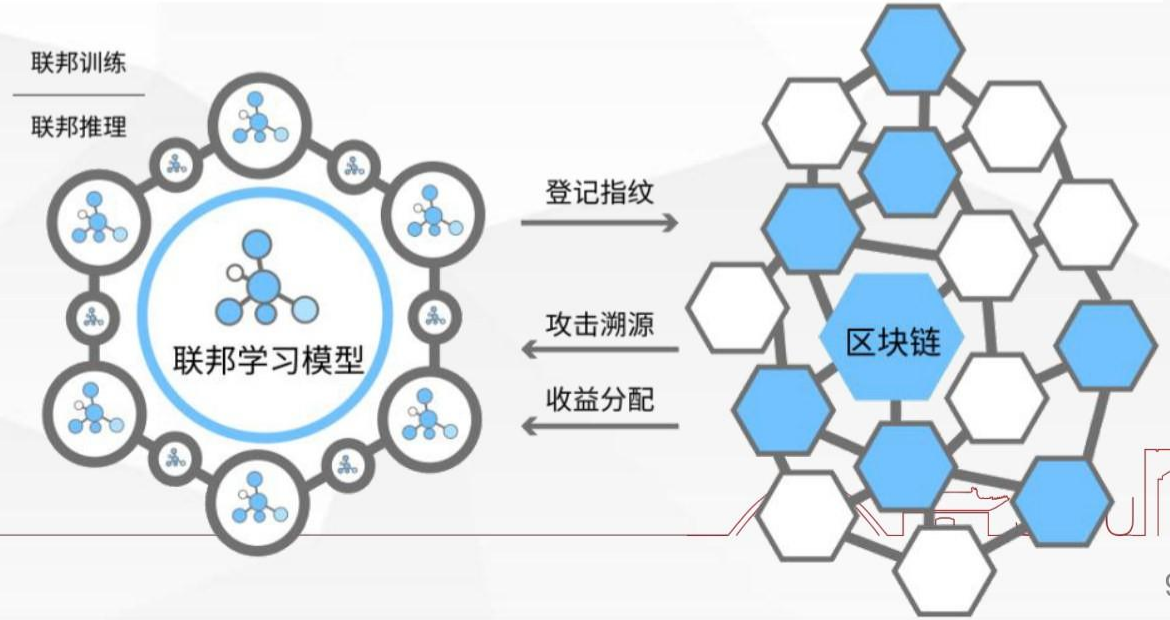
联邦学习安全---不可信第三方

- 通信安全
  - 无可信第三方服务器，易受攻击
- 联邦学习+区块链
  - 用区块链替换原有server保存全局模型，可以提高全局模型的防篡改能力
  - 区块链本身的共识机制和联邦学习的聚合权重有很多相似之处



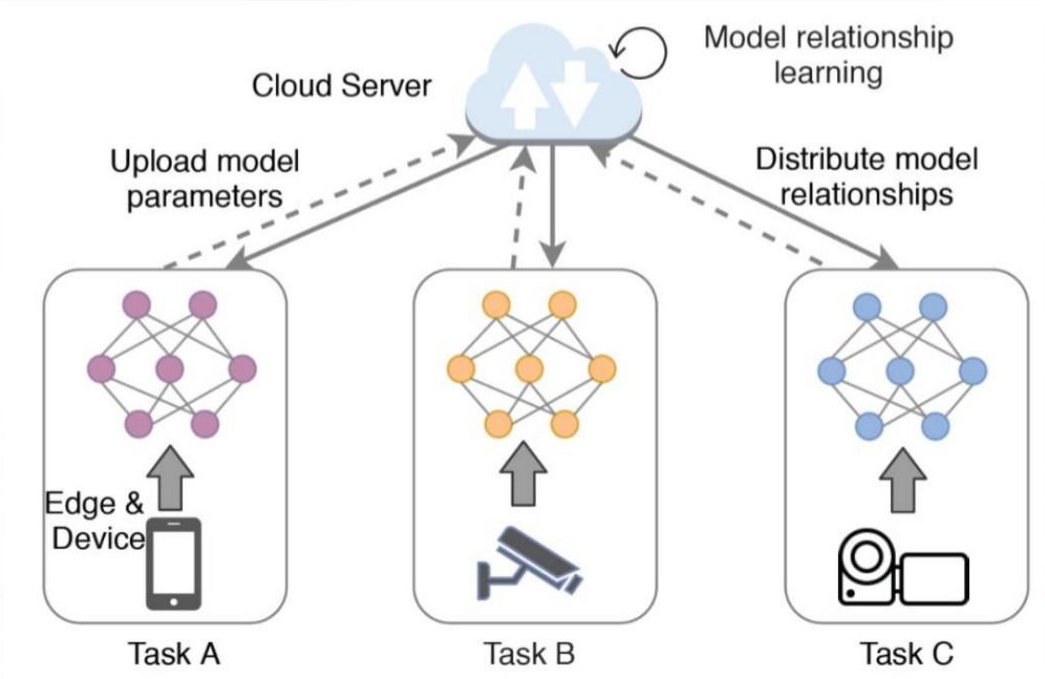


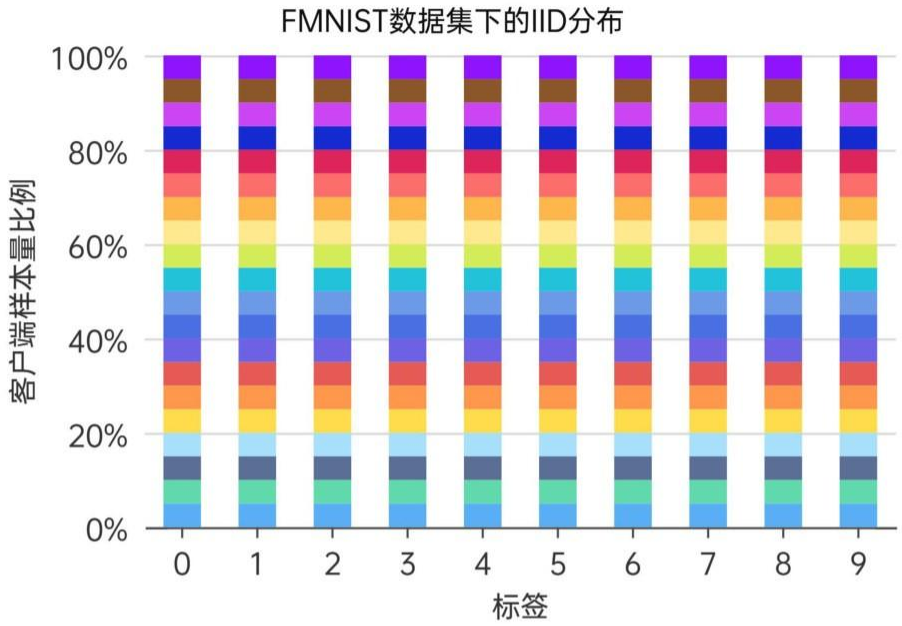
联邦学习异构性

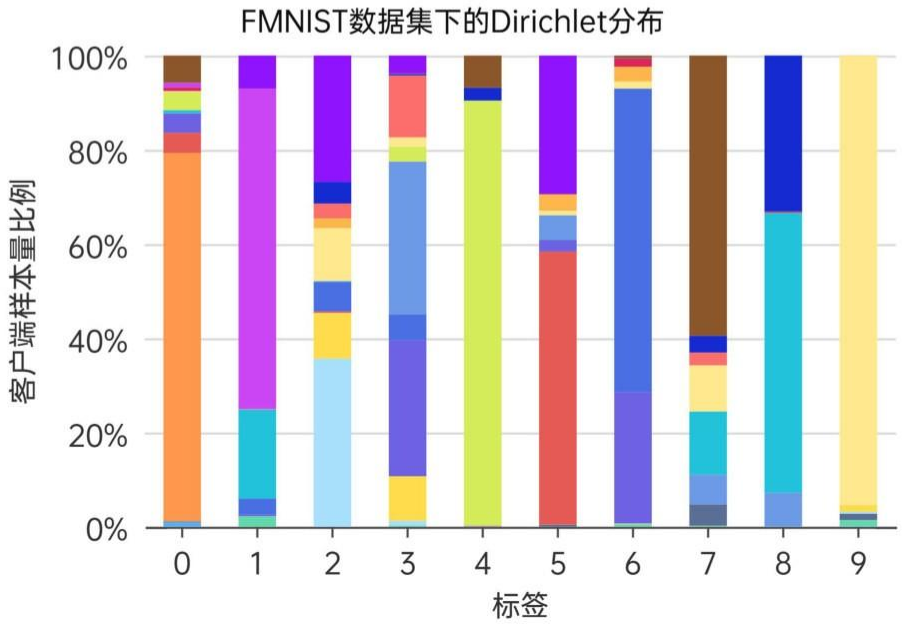
- 统计异构性 - 非聚合联邦学习
- 系统异构性 - 适应性训练调整

联邦学习异构性---统计异构性

- 统计异构
  - 各客户端数据分布不同，模型不准确
- 跨域联邦学习：
  - 模型之间交换知识，不单纯进行聚合
  - 模型在本地测试集上的效果会更好







联邦学习异构性---统计异构性

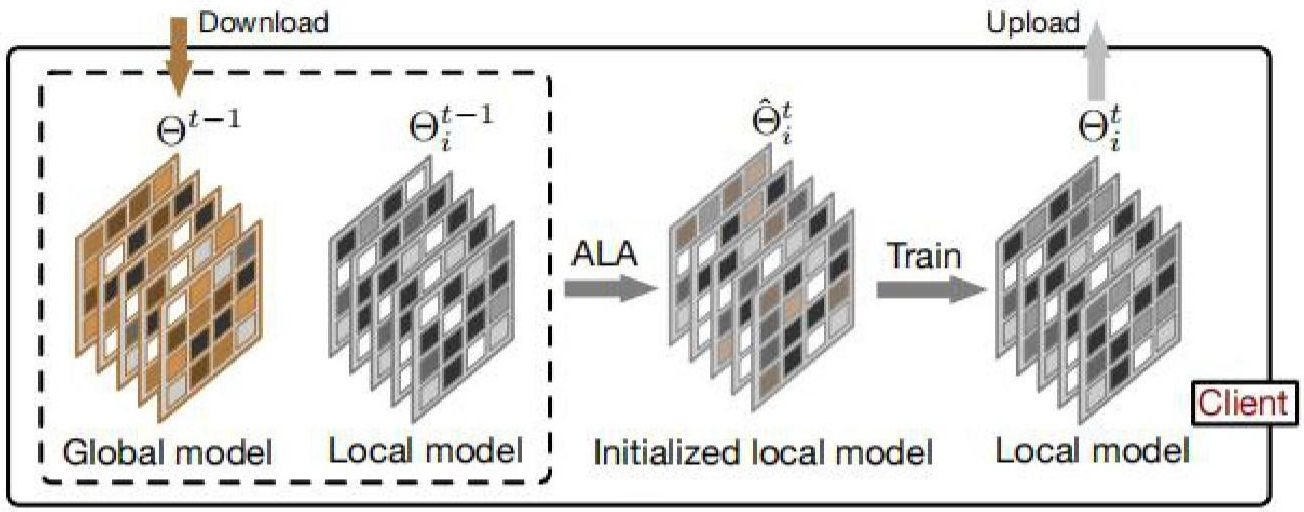
- 引入个性化mask做element-wise的更新

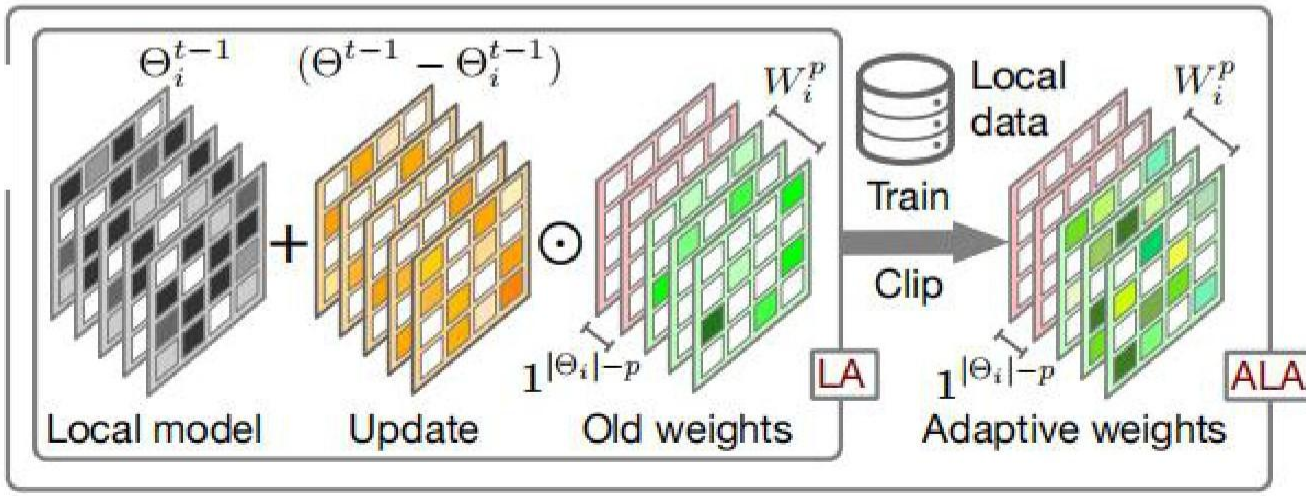
$$\hat{\Theta}_i^t := \Theta_i^{t-1} \odot W_{i,1} + \Theta^{t-1} \odot W_{i,2},$$
$$s.t. \quad w_1^q + w_2^q = 1, \forall \text{ valid } q$$

$$\hat{\Theta}_i^t := \Theta_i^{t-1} + (\Theta^{t-1} - \Theta_i^{t-1}) \odot W_i,$$

$$\hat{\Theta}_i^t := \Theta_i^{t-1} + (\Theta^{t-1} - \Theta_i^{t-1}) \odot [\mathbf{1}^{|\Theta_i|-p}; W_i^p],$$

$$W_i^p \leftarrow W_i^p - \eta \nabla_{W_i^p} \mathcal{L}(\hat{\Theta}_i^t, D_i^{s,t}; \Theta^{t-1}),$$





Jianqing Zhang, Yang Hua, Hao Wang, Tao Song, Zhengui Xue, Ruhui Ma, Haibing Guan, FedALA: Adaptive Local Aggregation for Personalized Federated Learning, Proceedings of the 37th AAAI Conference on Artificial Intelligence, 2022.

联邦学习异构性---统计异构性

- 在多个数据集多种设定上均取得了最高的准确率

Table 2: The test accuracy (%) in the pathological heterogeneous setting and practical heterogeneous setting.

Settings	Pathological heterogeneous setting			Practical heterogeneous setting				
Methods	MNIST	Cifar10	Cifar100	Cifar10	Cifar100	TINY	TINY*	AG News
FedAvg	97.93±0.05	55.09±0.83	25.98±0.13	59.16±0.47	31.89±0.47	19.46±0.20	19.45±0.13	79.57±0.17
FedProx	98.01±0.09	55.06±0.75	25.94±0.16	59.21±0.40	31.99±0.41	19.37±0.22	19.27±0.23	79.35±0.23
FedAvg-C	99.79±0.00	92.13±0.03	66.17±0.03	90.34±0.01	51.80±0.02	30.67±0.08	36.94±0.10	95.89±0.25
FedProx-C	99.80±0.04	92.12±0.03	66.07±0.08	90.33±0.01	51.84±0.07	30.77±0.13	38.78±0.52	96.10±0.22
Per-FedAvg	99.63±0.02	89.63±0.23	56.80±0.26	87.74±0.19	44.28±0.33	25.07±0.07	21.81±0.54	93.27±0.25
FedRep	99.77±0.03	91.93±0.14	67.56±0.31	90.40±0.24	52.39±0.35	37.27±0.20	39.95±0.61	96.28±0.14
pFedMe	99.75±0.02	90.11±0.10	58.20±0.14	88.09±0.32	47.34±0.46	26.93±0.19	33.44±0.33	91.41±0.22
Ditto	99.81±0.00	92.39±0.06	67.23±0.07	90.59±0.01	52.87±0.64	32.15±0.04	35.92±0.43	95.45±0.17
FedAMP	99.76±0.02	90.79±0.16	64.34±0.37	88.70±0.18	47.69±0.49	27.99±0.11	29.11±0.15	94.18±0.09
FedPHP	99.73±0.00	90.01±0.00	63.09±0.04	88.92±0.02	50.52±0.16	35.69±3.26	29.90±0.51	94.38±0.12
FedFomo	99.83±0.00	91.85±0.02	62.49±0.22	88.06±0.02	45.39±0.45	26.33±0.22	26.84±0.11	95.84±0.15
APPLE	99.75±0.01	90.97±0.05	65.80±0.08	89.37±0.11	53.22±0.20	35.04±0.47	39.93±0.52	95.63±0.21
PartialFed	99.86±0.01	89.60±0.13	61.39±0.12	87.38±0.08	48.81±0.20	35.26±0.18	37.50±0.16	85.20±0.16
FedALA	**99.88±0.01**	**92.44±0.02**	**67.83±0.06**	**90.67±0.03**	**55.92±0.03**	**40.54±0.02**	**41.94±0.05**	**96.52±0.08**

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联邦学习异构性---统计异构性

FedALA对在不同统计异构性和扩展性的情况都展开了测试，在多种设定下其效果均比传统方法要好。

	Heterogeneity			Scalability		Applicability of ALA			
Datasets	Tiny-ImageNet		AG News	Cifar100		Tiny-ImageNet		Cifar100	
Methods	$Dir(0.01)$	$Dir(0.5)$	$Dir(1)$	50 clients	100 clients	Acc.	Imps.	Acc.	Imps.
FedAvg	15.70±0.46	21.14±0.47	87.12±0.19	31.90±0.27	31.95±0.37	40.54±0.17	21.08	55.92±0.15	24.03
FedProx	15.66±0.36	21.22±0.47	87.21±0.13	31.94±0.30	31.97±0.24	40.53±0.26	21.16	56.18±0.65	24.19
FedAvg-C	49.88±0.11	16.21±0.05	91.38±0.21	49.82±0.11	47.90±0.12	—	—	—	—
FedProx-C	49.84±0.02	16.36±0.19	92.03±0.19	49.79±0.14	48.02±0.02	—	—	—	—
Per-FedAvg	39.39±0.30	16.36±0.13	87.08±0.26	44.31±0.20	36.07±0.24	30.90±0.28	5.83	48.68±0.36	4.40
FedRep	55.43±0.15	16.74±0.09	92.25±0.20	47.41±0.18	44.61±0.20	37.89±0.31	0.62	53.02±0.11	0.63
pFedMe	41.45±0.14	17.48±0.61	87.08±0.18	48.36±0.64	46.45±0.18	27.30±0.24	0.37	47.91±0.21	0.57
Ditto	50.62±0.02	18.98±0.05	91.89±0.17	54.22±0.04	52.89±0.22	40.75±0.06	8.60	56.33±0.07	3.46
FedAMP	48.42±0.06	12.48±0.21	83.35±0.05	44.39±0.35	40.43±0.17	28.18±0.20	0.19	48.03±0.23	0.34
FedPHP	48.63±0.02	21.09±0.07	90.52±0.19	52.44±0.16	49.70±0.31	40.16±0.24	4.47	54.28±0.21	3.76
FedFomo	46.36±0.54	11.59±0.11	91.20±0.18	42.56±0.33	38.91±0.08	—	—	—	—
APPLE	48.04±0.10	24.28±0.21	84.10±0.18	55.06±0.20	52.81±0.29	—	—	—	—
PartialFed	49.38±0.02	24.20±0.10	91.01±0.28	48.95±0.07	39.31±0.01	35.40±0.02	0.14	48.99±0.05	0.18
FedALA	**55.75±0.02**	**27.85±0.06**	**92.45±0.10**	**55.61±0.02**	**54.68±0.57**	—	—	—	—

Jianqing Zhang, Yang Hua, Hao Wang, Tao Song, Zhengui Xue, Ruhui Ma, Haibing Guan, FedALA: Adaptive Local Aggregation for Personalized Federated Learning, Proceedings of the 37th AAAI Conference on Artificial Intelligence, 2022.

联邦学习异构性---统计异构性

- 同时算法相比其他baseline，减少了一定的时间，提高了整体的计算效率和训练效率
- 其通信量和FedAvg相当，并没有引入额外的传输

	Computation		Communication
	Total time	Time/iter.	Param./iter.
FedAvg	365 min	1.59 min	$2 * \Sigma$
FedProx	325 min	1.99 min	$2 * \Sigma$
FedAvg-C	607 min	24.28 min	$2 * \Sigma$
FedProx-C	711 min	28.44 min	$2 * \Sigma$
Per-FedAvg	121 min	3.56 min	$2 * \Sigma$
FedRep	471 min	4.09 min	$2 * \alpha_f * \Sigma$
pFedMe	1157 min	10.24 min	$2 * \Sigma$
Ditto	318 min	11.78 min	$2 * \Sigma$
FedAMP	92 min	1.53 min	$2 * \Sigma$
FedPHP	264 min	4.06 min	$2 * \Sigma$
FedFomo	193 min	2.72 min	$(1 + M) * \Sigma$
APPLE	132 min	2.93 min	$(1 + M) * \Sigma$
PartialFed	693 min	2.13 min	$2 * \Sigma$
FedALA	7+116 min	1.93 min	$2 * \Sigma$

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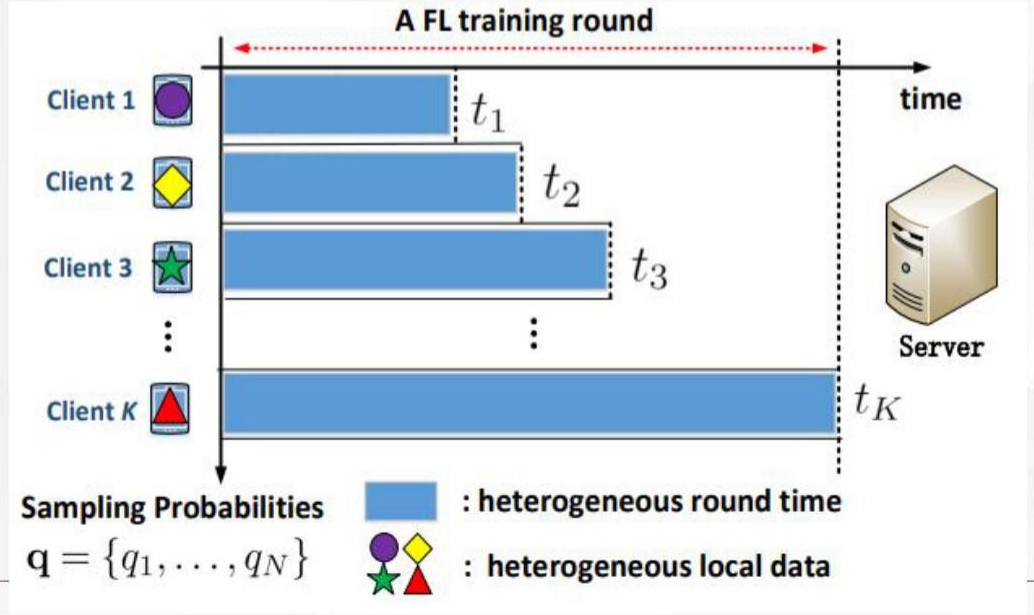
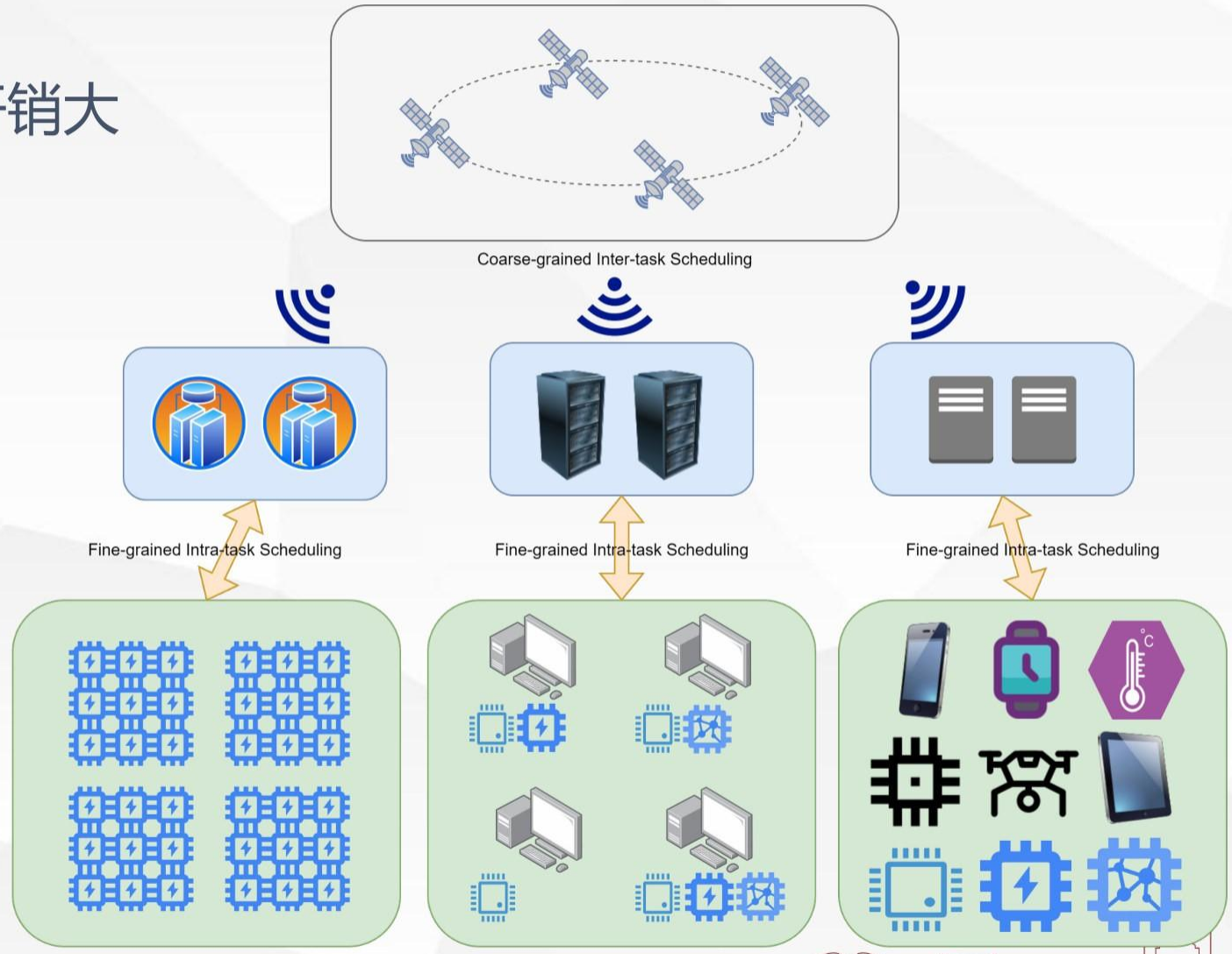
联邦学习异构性---系统异构性

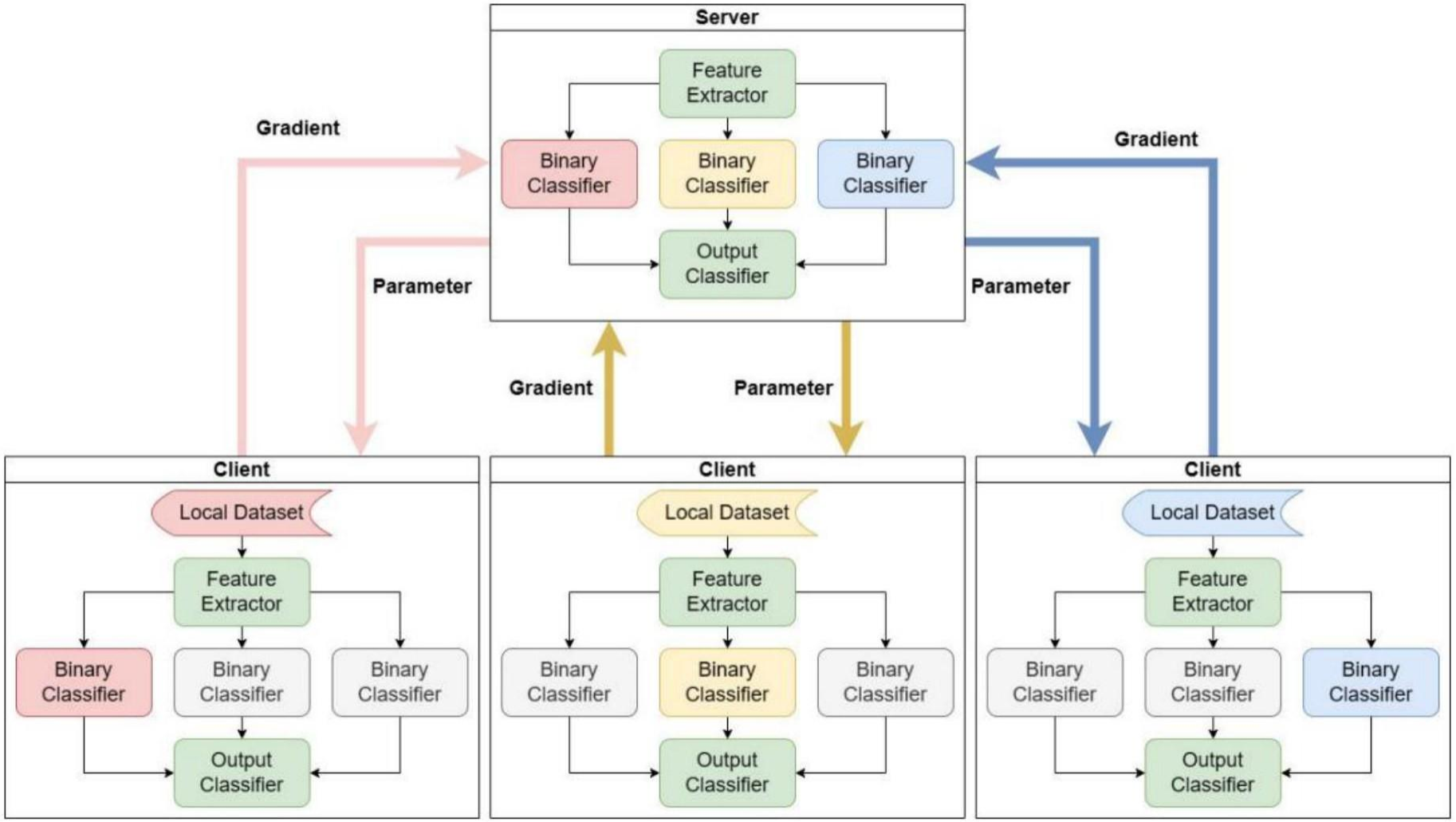
- 系统异构
  - 各客户端计算能力不同，通信开销大
- 适应性训练调整
  - 采用不同模型不同训练参数
  - 采用异步或者回合制聚合





联邦学习异构性---系统异构性

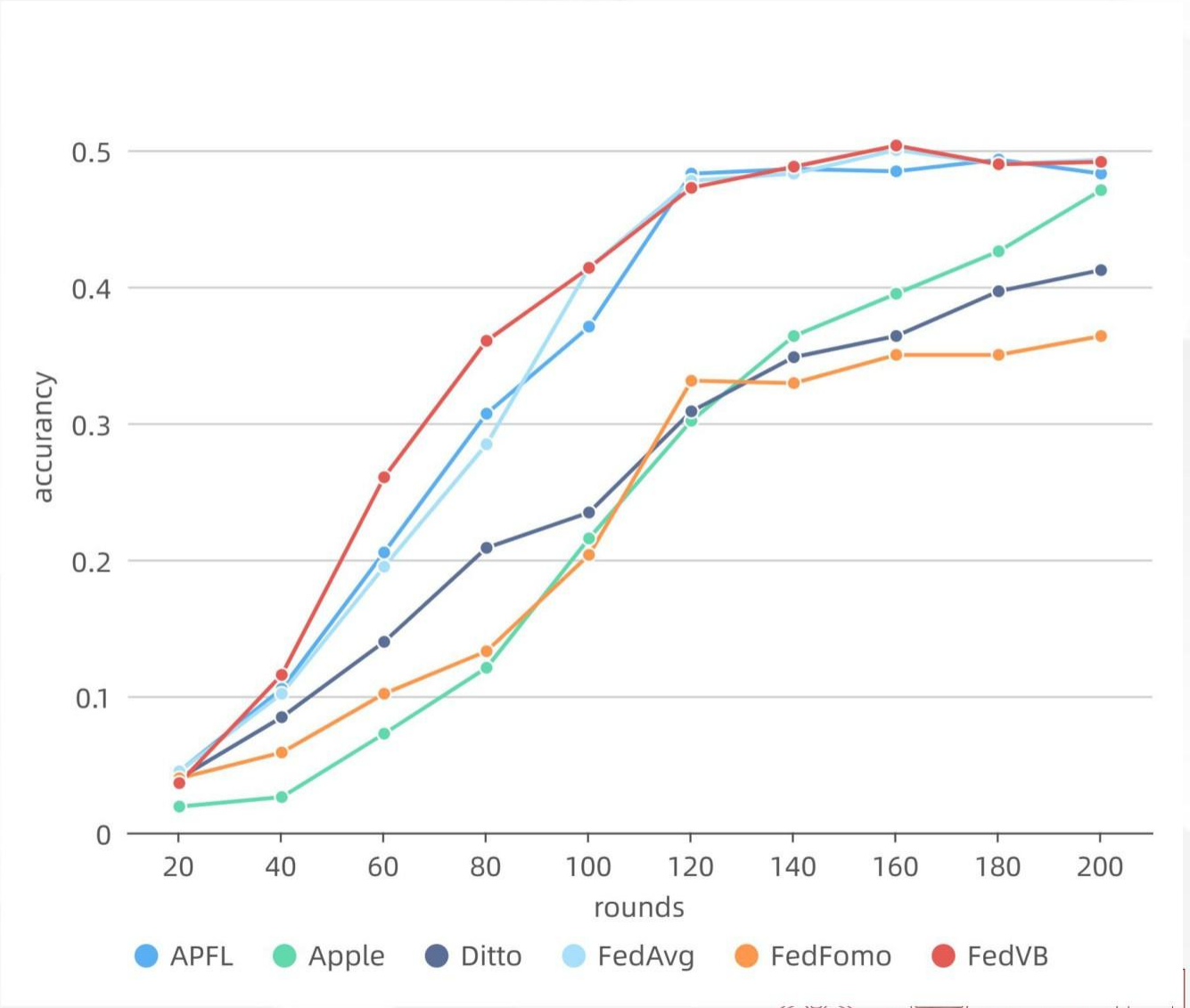
- 将多分类模型分解成多个二分类模型
- 训练时只对**本地标签对应的分支**进行训练
- 传输时只传输**经过训练的模型分支**
- 聚合时只聚合**传输上来的对应分支**



联邦学习异构性---系统异构性

- FedVB相比其他baseline收敛更快，准确率更高
- 多分支模型相比传统模型能够在准确率上有所提升

Algorithm	Simple classifier	Multi-branch Network
FedAvg	0.3983	0.5000
APFL	0.4276	0.5000
Apple	0.3983	0.4707
Ditto	0.3137	0.4259
FedBN	0.3983	0.5017
FedMTL	0.3224	0.5017
FedProx	0.4000	0.5017
pFedMe	0.0259	0.0379
FedFomo	0.8897	0.8500
LOCAL	0.8638	0.8621
PerAvg	0.6000	0.7483



Yunhui Wang, Weichu Zheng, Zifei Liu, Jinyan Wang, Hongjian Shi, Minyu Gu, Ruhui Ma, and Haibing Guan, A Federated Network Intrusion Detection System with Multi-branch Network and Vertical blocking Aggregation, 2023.

总结：没有通吃的算法

算法	会议	针对的问题	改哪一步	代价
FedAvg	AISTATS 2017	基线	—	—
MOON	CVPR 2021	统计异构	② 本地损失（表示空间）	多两次前向，多存一个模型
FedDyn	ICLR 2021	统计异构	② 本地损失 + ④ 服务器状态	客户端存状态；通信不变
KT-pFL	NeurIPS 2021	统计/模型异构	③ 传预测而非参数	需要公共数据，N×N 矩阵
FedMA	ICLR 2020	排列不变性	④ 逐层匹配聚合	轮数 ≈ 层数，模型变宽
Sageflow	NeurIPS 2021	掉队者 + 攻击者	④ 分组、过滤、加权	需要公共数据，与安全聚合冲突

每个方法都在用某种代价——额外计算、存储、公共数据、超参数——换取对某类问题的鲁棒性。
读论文时先问：**针对什么问题？改了哪一步？代价是什么？**

课后动手（自愿）：用 PFLlib 观察 Non-IID 的影响

```
cd PFLlib/dataset
python generate_MNIST.py iid - -       # 先生成 IID
cd ../system
python main.py -data MNIST -m CNN -algo FedAvg -gr 50

cd ../dataset
python generate_MNIST.py noniid - dir  # 再生成 Non-IID
cd ../system
for a in FedAvg FedProx MOON FedDyn; do
  python main.py -data MNIST -m CNN -algo $a -gr 50
done
```

- MNIST + 小 CNN，笔记本 CPU 也能跑；轮数可再调小
- 重新生成数据会覆盖旧划分，按顺序跑
- 观察：IID 与 Non-IID 下 FedAvg 的准确率曲线差多少？
- Non-IID 下各算法的差别有多大？换更小的 α 呢？

> 不计分、不提交，作为理解今天内容的练习。

Q&A

饮 水 思 源　爱 国 荣 校